VALETON GP200

THE MISSING MANUAL

V1.8.2



Up to date
for firmware
1.8.0/1.8.2
and editor 1.8.1

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Vincenzo Pancotti

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INTRODUCTION

This mini-book was sparked off by a collection of posts I made to the Valeton GP200 users' Facebook group in 2025. It tries to expand greatly on the official manual, as well as correcting it in the (mercifully few) places where Valeton get it wrong about their own product.

My original posts included technical details about how I made the measurements and deductions that enabled the guides; I have edited most of these out, since the reader really only needs to know the end results.

If you find this useful and feel generous, go to https://buymeacoffee.com/vincenzopancotti to leave a donation. You can always find the latest version of this book for free there.

Disclaimer

While this book begins with a FAQ and the other chapters sometimes include suggestions for the reader, in the end there is only one rule: **What sounds good is good**. If you get the sound you want by doing the opposite of everything in here, more power to you!

Everything in this book has been tested with the **GP200R** running firmware **1.8.0**. The **GP200** and **GP200X** are identical hardware and firmware, so they will behave the same. While I expect that much of the content will also apply to the GP200JR and GP200LT, it is possible that these devices have minor differences.

READ ME BEFORE BUYING ONE

Reasons you might NOT want the GP200:

1. *If you need* really *accurate modelling of classic amps*. The models are fine and can sound very good on their own terms; but if you, say, own a real Fender Deluxe Reverb and want that exact sound, you'll likely be disappointed. The GP200 is also unable to run the newer "HQ" models offered in the Ampero 2 devices.

2. *If you need fine control over signal routing*. The GP200 only offers a single linear path - no parallel, no branching. Moreover, in order to save money it takes some shortcuts (not documented in the manual) about how the noise gate is routed, how stereo FX work, etc. So if you're very particular about crafting signal flow, as opposed to being happy with typical sequences of FX, you could get frustrated.

3. *If you need multiple effects of one type*. You can only run one reverb, one delay, one modulation at a time. No stacking delays, no stacking chorus. There is limited support for stacking two drive pedals.

4. *If you need lots of "off the wall" effects*. There is a bit crusher and a few (not great) pitch shift effects, but if you are into shoe gaze, ambient, avant-garde - again, might be limiting.

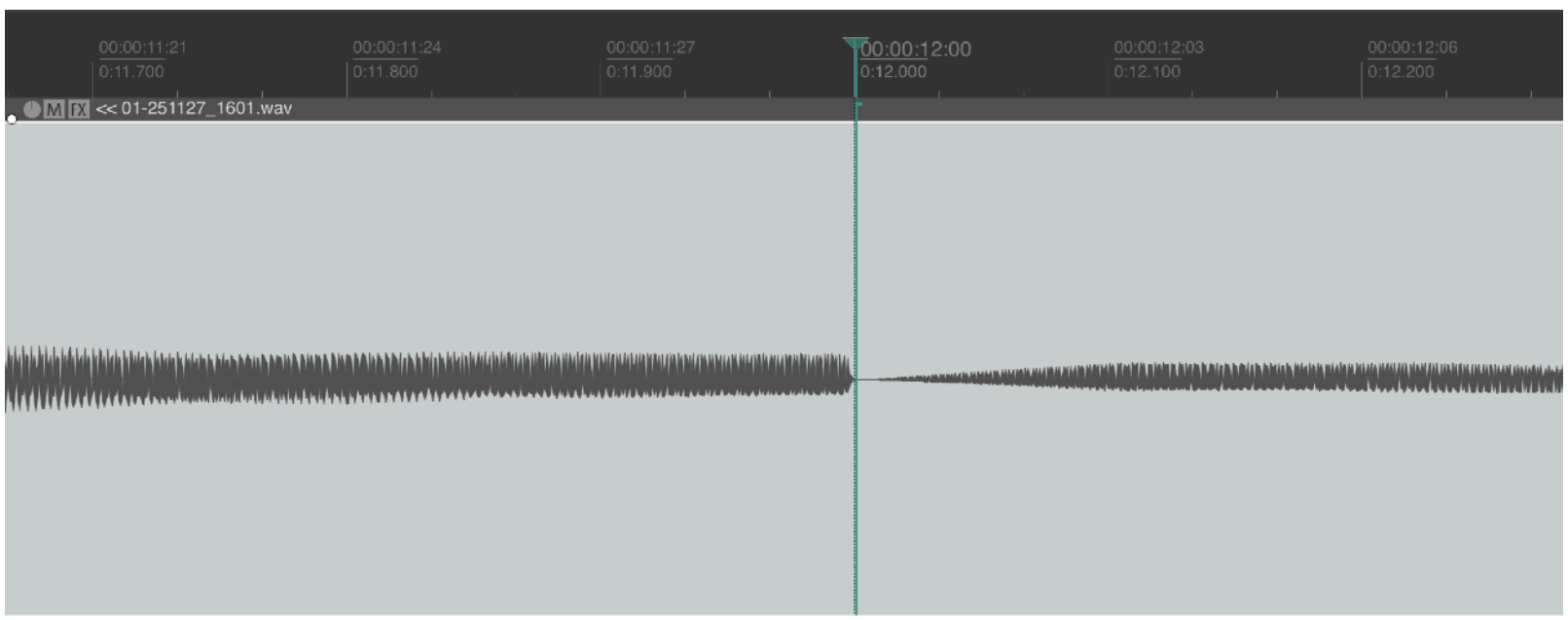
5. *If you need a convincing spring reverb*. GP200 has one, it's … let's just say it doesn't survive a close listen. imho.

6. *If you need comprehensive MIDI integration*. The GP200 has the capability to be amazing here, with proper MIDI sockets, and has a dedicated MIDI mode where it can act as a MIDI controller. But it has some surprising omissions: for example, you cannot tell it to send a specific MIDI command when it changes to a specific patch. So some automations are not possible.

7. *If you need a good looper*. The looper is not really usable in a live context for a number of reasons, at least not with the current firmware.

8. *If you need scene support*. The GP200 does not have true scene support (changing multiple effect parameters with a single stomp)

9. *If you need completely seamless patch switching with reverb/delay spillover*. On the GP200, a patch change results in 10ms of mute while the new patch is loaded and then - interestingly - about 90-100 ms of a fade in of the sound. Any reverb or delay trails from the old patch are cut off.



It's worth mentioning here that 1, 2, 3, 6 and 8 (and to some extent 4) are all better into the GP200's big brother, the Ampero 2 (original, Stomp and Stage). So if your budget stretches that far and you can live with fewer footswitches, consider those units.

FAQ
(OR "JUST TELL ME WHAT TO DO!")

Maybe you don't care about the technical details of why things are the way they are; after all, making music is the point. So this chapter attempts to give direct answers to commonly asked questions on the GP200 groups on Facebook or Reddit. Detailed justifications are left until later chapters.

All answers are targeted at giving decent results for the majority of players working in mainstream genres (rock, pop, metal, jazz).

HOW DO I SET MY INPUT LEVEL?

Leave it at 0dB unless you really, really know you need to set it otherwise.

HOW DO I SET MY GLOBAL EQ?

Ideally, you don't set it at all. The idea is that you set up your patches (without using Global EQ) using your normal rig, and then when you have to use a different rig, you can use Global EQ to compensate without having to touch the EQ block on every single patch.

For example, suppose you normally play by connecting an unbalanced output from the GP200 into the FX return of your tube amp, but when gigging you go direct into the PA. Then you can minimise the hassle of swapping like this:

1. Turn on **No CAB** for the L output.
2. Plug output L into the tube amp's effect return; or, if it doesn't have a loop, into the cleanest input you can get.
3. Dial in your patches to sound the way you like, without using Global EQ.
4. Unplug from tube amp and plug output R into PA.
5. Leaving Global EQ off initially, find the combination of CAB selection and High/Low filters that gets you closest to your tube amp sound
6. Then turn on Global EQ and use it to fine-tune the sound to get as close as possible.

Now you're set up, and you just need to turn the Global EQ on and off depending on whether you're using the tube amp or the PA, and you will have the most consistent sound.

The reason you should avoid using Global EQ to fine tune your sound is that it is global. The adjustments that seem perfect with one amp model, cabinet IR and effect selection may be completely inappropriate with a different patch where all of these are different. Or in other words, there is no magic "one size fits all" eq curve. Before reaching for the Global EQ, see if the EQ block can get you there on a per-patch basis.

If you find yourself (as is very common) wanting to filter out extreme lows and highs to remove rumble, mud, fizz and grain, then initially use the cut filters in the CAB block (which are the same as those in Global EQ). If that isn't enough, consider finding or creating a suitable custom IR, since this can have a steeper cutoff, rather than reaching for Global EQ. Another way to look at this is to think of the Global EQ as a tool for *identifying* a problem, which you then solve in a more flexible way: "Oh, this IR sounded even better with a lift at 4kHz.. maybe I can find one in the set that has that naturally, or push that band on the EQ block."

HOW DO I SET UP THE FX LOOP?

Set Send Level to 95 and Return Level to 100, then use regular guitar pedals in the loop. You did remember to leave your input level at 0dB, didn't you? Great! Now you can leave your pedals input levels exactly how you had them with your guitar. The settings have no effect on your tone when you're not actually using the loop, so you can dial this into every patch.

If you are not using the FX loop, then turn Return level down to 0 and move the return point to the end of the signal chain. This minimises residual noise from that input, since you can't turn it completely off.

HOW DO I TURN THE FX LOOP ON AND OFF WITH A FOOTSWITCH?

You can't[1]. Sorry. Complain to Valeton.

HOW DO I MAKE IT SEND MIDI PROGRAM CHANGE WHEN I CHANGE PATCHES?

You can't[2]. Sorry. Complain to Valeton.

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[1] At least, as of the time of writing (firmware 1.8.0)

[2] At least, as of the time of writing (firmware 1.8.0)

You can use MIDI mode (see official manual page 8) to turn the GP200 into a MIDI floor controller, but while in this mode you can't change patches. So you have to tap-dance: select the patch on the GP200, enter MIDI mode, tap a footswitch to send a PC, and then exit MIDI mode. Which is essentially useless for anything except between-song changes.

HOW DO I REMAP THE FOOTSWITCHES INSIDE TUNER, MIDI OR LOOPER MODES?

You can't[3]. Sorry. Complain to Valeton.

HOW DO I SET THE MASTER VOLUME?

If you are using the unbalanced (jack) outputs, you should ideally have it cranked all the way up. That makes the unbalanced outputs have the same level as the FX SEND, and having the strongest signal possible down those unbalanced outputs reduces noise pickup. It also means if you bypass all modules (manually or via the tuner) the unbalanced output is the same level as the input - potentially important if you use the GP200 like a pedalboard in front of a clean amp. If you are only using the XLR outputs, it doesn't matter as much, but louder is still better, other things being equal.

That said, if max volume is too hot for your monitoring device (some tube amp FX returns can be easily overloaded, especially if they have no trim pot or are post master volume) or desk then of course turn it down. Compared with the maximum level, 3 o'clock is about -3dB, 12 o'clock about -13dB, and 9 o'clock about -34dB.

The right level for your amp/PA may be far too loud for the headphone output. Suggestion: find the sensible settings for you and mark them with a dab of nail varnish/paint/sticker.

HOW DO I MAKE PATCHES SWITCH FASTER?

Since firmware 1.8.0 at least, you can make footswitches action faster simply by removing any secondary function from that footswitch. They will then action on press. The older behaviour of actioning on release still applies for any footswitch that also has a setting for HOLD or one of the combination presses (1+5, 5+6, 6+7, 7+8). Since by default pressing 7+8 together enters MIDI mode, this catches people out. If you don't use MIDI, turn that footswitch combination off - now the C and D foot switches will work like the A and B ones.

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[3] Can you guess?

The mysterious “FS Mode” setting in the computer editor is (at least in 1.8.0) now the exact same thing as “Bank Sel Mode” in Global settings on the GP200 itself, and controls how the bank and patch change functions interact. See the official manual for a description.

HOW DO I HAVE CAB SIMS ON THE XLRS BUT NOT THE UNBALANCED JACKS?

You can’t, and this isn’t something that can be fixed with software as they are fed from the same DAC. You can use **No CAB** mode in Global Settings to have cab sim disabled on the L output only.

One possible workaround, if you don’t plan on using the FX loop and only need mono, is to place the CAB block at the end of the signal chain, with the FX loop send point just before, and then use the FX loop send as your unbalanced output. This has the side effect of global EQ only applying to the XLRs, but this may work in your favour.

HOW DO I RUN A TRADITIONAL PEDALBOARD IN FRONT OF THE GP200?

If you are used to running a board into a clean amp (a so-called ‘pedal-platform) then there are a couple of considerations that may trip you up, one minor and one potentially more frustrating.

1. The last pedal in your chain likely has low output impedance, and so you may benefit from changing the **Input Mode** to **Line** instead of **E.GT** - but try both settings to see which gives you the strongest signal.

2. If you are using overdrive pedals for dirt and to increase signal level, then you will need to use an amp model (if you are using one) that can handle high signal levels without going into saturated mush. See Appendix 3 for details.

HOW DO I USE THE GROUND LIFT SWITCH?

This little button is often misunderstood, largely because the official manual skims over it with a few words. When in the “out” position the grounds on the XLRs *only* are disconnected. Two things follow from this:

1. If you aren’t using the XLR outputs, the ground lift switch does *absolutely nothing*. It is not a generic hum killer.

2. If you are using the XLR outputs, you must have another source of ground or you can make any hum *worse* - sometimes much worse! Again, ground lift is not a panacea.

So, the right question to ask is *when* to try using the ground lift switch? The answer is when your GP200 is connected to two (or more) other grounded devices: e.g. going to a PA system via the XLRs and simultaneously going via an unbalanced jack to the FX return of a tube amp onstage. This can cause a ground loop and annoying hum - if it does, lifting ground on the XLRs can fix it.

The standard PSU that ships with the GP200, and many higher-quality 3rd party replacement supplies, is a Class 2 PSU with no true ground. (If you have one with a UK plug, you may note that the earth pin is plastic.) If you replace it with a Class 1 earthed PSU to avoid the headphone hum gotcha, then that gives another path to ground and again, you may encounter a ground loop that the button can fix if you are connecting to a PA via the XLRs.

HOW DO I GET THE SOUND OF [INSERT EXPENSIVE PEDAL HERE]?

The GP200 is, for its price (and especially if you get a used one in good condition), a very capable all-in-one unit. People can and do gig professionally with it and nothing else. It is nevertheless a budget unit which has to share a single processor across all of the effects it is running; and so any single purpose dedicated pedal will always sound lusher/more detailed when in an A/B comparison. For example, you simply can't select a reverb that is as huge and expansive as the Strymon Big Sky.

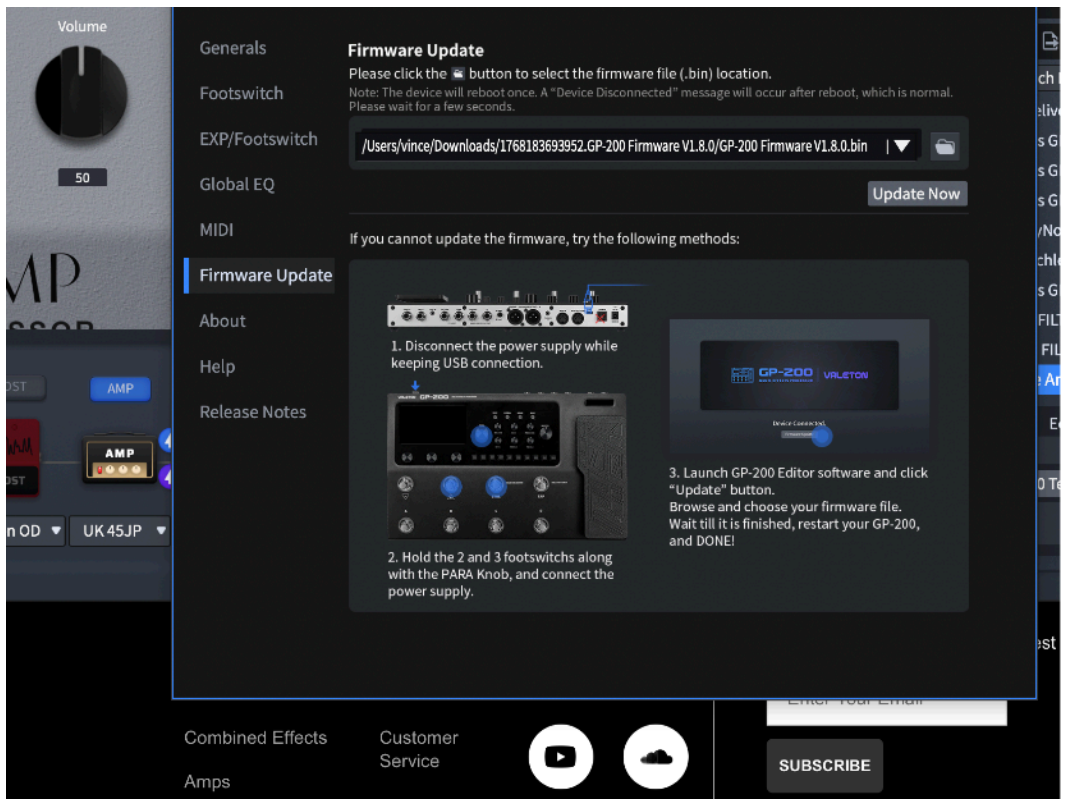
Sometimes it can help, if you have the blocks spare, to combine multiple effects and think of them as a single item. For example, combining Ambience in DLY with Arena in RVB to make a "bigger" reverb.

Rather than fretting about direct comparisons through headphones or studio monitors, concentrate on the overall feel you are after and test different options in the context of a live band mix, where subtle details will often be obscured by everything else that's happening. You will probably get close enough for the audience to be happy!

KNOWN ISSUES

HELP! SOMETHING WENT WRONG DURING A FIRMWARE UPDATE AND NOW THE DEVICE IS STUCK ON THE UPDATE SCREEN.

Firstly, try again by forcibly restarting the update, using the process that is shown in the editor program:



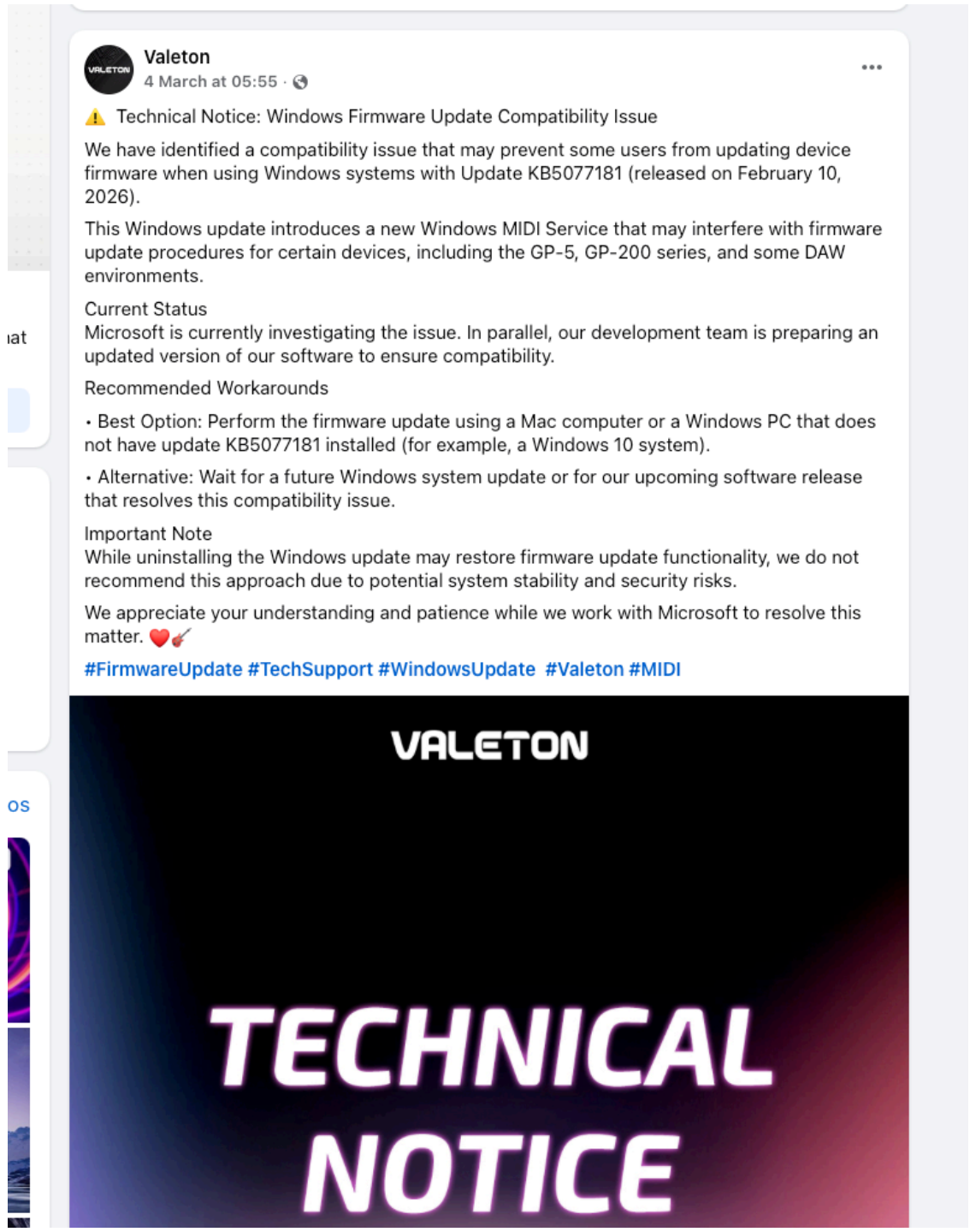
In March 2026 Valeton issued a technical notice that on some Windows 11 PCs, the Windows update KB5077181 changed the MIDI subsystem in a way that disrupts communication with the editor software. So, if you can, try borrowing a Windows 10 PC or a Mac to complete the update.

HELP! I'M ON WINDOWS 11 AND IMPORTING NAMS/IRS/ PATCHES STOPPED WORKING.

In March 2026 Valeton issued a technical notice that on some Windows 11 PCs, the Windows update KB5077181 changed the MIDI subsystem in a way that disrupts communication with the editor software. So, if you can, try borrowing a Windows 10 PC or a Mac to perform the import.

If you only have Windows 11 available, consult this post and thread for a workaround that has been endorsed by Valeton:

https://www.facebook.com/groups/1080579552678581/permalink/2222355018501023/



Valeton's March 2026 technical notice

HELP! THE EDITOR CRASHES WHEN I TRY TO LOAD A NAM.

Update to firmware and software v1.8 to fix this issue. If you still have problems, exit the software, delete the entire folder `C:\ProgramData\HTCache`, and restart.

HELP! THE EDITOR CRASHES WHEN I TRY TO CHANGE MIDI SETTINGS.

This is a known bug on MacOS. You can change all MIDI settings directly on the unit itself via Global Settings, and that is the workaround.

HELP! I LOADED AN IR AND IT'S INCREDIBLY QUIET.

When you load an IR, the editor software actually sends a pulse through it and recaptures a 1024-sample version to send to the device. The recaptured version has more or less the same level as the original. However, what you may not know is that most normal guitar/bass cab capture IRs would, if processed as-is, actually result in a much louder output signal than input signal[4]. The IR loader in your DAW will usually invisibly compensate for this. The IR playback block in the GP200 is quite simple, and applies a blanket -18dB correction factor. This is almost always good enough for guitar and bass cab captures, but completely wrong for very simple artificial IRs used to recreate an EQ curve.

If you load a synthetic EQ curve IR, it's very likely it will be quiet - you haven't done anything wrong, but will have to compensate elsewhere.

If a regular guitar/cab IR is very quiet, then the creator has already applied compensation and you need to undo it. Try normalising the file to 0dFS in a DAW or audio editor.

HELP! I DOWNLOADED A PATCH AND IT'S WAY LOUDER/ QUIETER THAN THE ONES I ALREADY HAVE.

This is perfectly normal, since the creator of the patch will have a different rig from yours and thus will have dialled in different levels. Start by resetting both the **Patch Volume** and the **Volume** of **CAB** to 50, and then see if you are close enough to get the level you want by adjusting Patch Volume.

If not, then see if the **Volume** of **AMP** is set to a suspiciously high or low value. In every case I've tested, this parameter is a is a "pure" volume and does not affect any power tube distortion/output transformer simulation, so can be freely adjusted. (Nevertheless, be prepared to make tweaks since merely changing the level in your monitors will alter your impression of the sound's tonal balance.)

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[4] You can see this by loading them in REAPER's Reaverb IR effect. It's why that effect has a -18dB option.

HELP! SOMETIMES MY SIGNAL BLASTS INTO THE RED WITH A DIGITAL SCREECHING SOUND, OR JUST FADES AWAY.

This seems to be a firmware bug that can happen (with some amp models[5]) if you set AMP Volume or Gain to 100. Try backing the knob off to 95.

HELP! I LOADED UP A NAM PROFILE AND IT SOUNDS VERY DIFFERENT FROM HOW IT SOUNDED IN THE OFFICIAL PLUGIN.

NAMs are not supported natively, but are converted. Read all about it in the Appendix.

HELP! WEIRD THINGS HAPPEN WHEN I'M IMPORTING OR COPY-PASTING PATCHES IN THE EDITOR.

The editor normally tracks what you're doing on the device automatically: if you change patches or adjust a knob, the editor view updates to match. However this isn't perfect and in particular if you move to a given patch using the footswitches or PARA knob, pasting or importing into that patch slot can fail.

It's safest to either do everything on device (edit and save a patch) or everything via the editor (selecting a patch, editing it, saving it).

If you are copying and pasting or importing patches, it's safest to avoid the slot with the currently loaded patch on the device - or at least to double-check that slot after the paste/import.

HELP! I LOADED AN IR AND IT SOUNDS EXACTLY THE SAME AS THE PREVIOUS IR I LOADED.

Some users have reported this rare issue, and it's not clear what the exact cause is. Try restarting the editor software after importing each IR. It only appears to have happened on Windows PCs, so try with a Mac if you have access to one.

Also try exiting the editor software, deleting the entire folder `C:\ProgramData\HTCache`, and then restarting. (This is safe as that folder is only used by the software - HT is short for Hotone)

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[5] The JC120 and Dark Twin models have been mentioned; it seems to be amps that have a bright cap as part of the model.

HELP! THE EDITOR DOESN'T CONNECT TO MY GP200.

On MacOS you may run into an issue where the editor doesn't detect the GP200. You can avoid this by starting the editor first, with the GP200 disconnected, and only plugging in the USB cable once the editor has started up and is sitting on its initial screen.

If you have other connectivity issues, especially inconsistent ones, then ensure you are using a high quality data-rated USB-C cable and avoid using USB hubs or docks (Standard USB connectivity advice!)

THE OTHER FAQ (OR "WHY DOES IT SUCK?")

Warning: some of these answers are tongue in cheek. But even they have a kernel of truth.

WHY DO THE PITCH SHIFTER EFFECTS SOUND LIKE GARBAGE?

1. The GP200 does everything on a single ARM chip with no dedicated DSP. Great pitch shifting is a complex algorithm and there isn't enough DSP to do it. Sorry. The **Hammy** effect can give decent results if you are just looking to drop tune a semitone or two.

2. Some of the pitch shifter effects are models of old BOSS pedals, and *they* sounded like garbage. So the emulation is accurate.

WHY ARE THE HIGH-GAIN AMPS SO NOISY?

They aren't.

No, really.

The amp models are all digital and have no self-noise. If you have a dead silent signal going into the 5150 model, and turn *every knob to 100*, it is still silent.

What the high gain amps are, is very high gain. Which means that the self-noise in the analogue input stage, and any noise *you* inject, gets amplified like crazy.

Make sure your signal going into the GP200 is as loud and clean as you can get it. Use a high quality transparent boost/buffer pedal. Make sure you have eliminated all ground hum, radio pickup, power supply buzz from the input signal.

WHY DOES THE NOISE GATE SUCK?

1. Because you are using GATE 1 or GATE 2 instead of the newest GATE 3. Just use GATE 3, seriously.

2. Because you have not read the previous FAQ about noise and you're asking it to do the impossible.

3. Because no-one told you that the side-chain is always fed with the raw guitar signal.

4. Because no-one told you that the side-chain is also fed from the FX return signal, regardless of where the NR is in the chain.

WHY DO ALL THE BUILT IN CABINETS SUCK?

1. Because you have left CAB Match on in Global Settings and you are being subjected to the original developers guess as to which cab sounds best with each model.

2. Because you are scrolling through a collection of wildly different IRs, captured with different microphones and different post production, and not re-adjusting your EQ each time.

3. Because a unit like this has to offer a very wide range of cabinets to cover all possible tastes and so it's inevitable that most of them will sound terrible *to you*.

4. Whether you use a built-in cab or your own IR, consider following the advice of Leon Todd and Mike Scuffham: find a cab you like and use it for all presets, regardless of amp.

WHY DOES THE TUNER SUCK?

1. The tuner is unbiased in the sense that when you set it to A=440Hz, the centre of the green zone really is 440Hz. However, it is far too easy to change this setting from touching the quick adjust knob or tapping on the wrong footswitch by mistake. So always check the display.

2. Unfortunately, the tuner has a very wide "landing zone". You will get a perfect green with the note as much as 5 cents off either way, which could be very obvious to someone with sensitive pitch. So the tuner doesn't enable you to tune accurately.

3. See this tip to get accuracy within 1.5 cents.

4. If you are feeling brave, check out the https://gp200-reversing.hashnode.dev/ blog for a firmware patch to fix the tuner landing zone.

WHY DOESN'T IT SOUND AS GREAT AS ALL THE DEMOS I SEE ON YOUTUBE?

1. You are not using the same guitar.

2. You are not the same person playing that guitar. :)

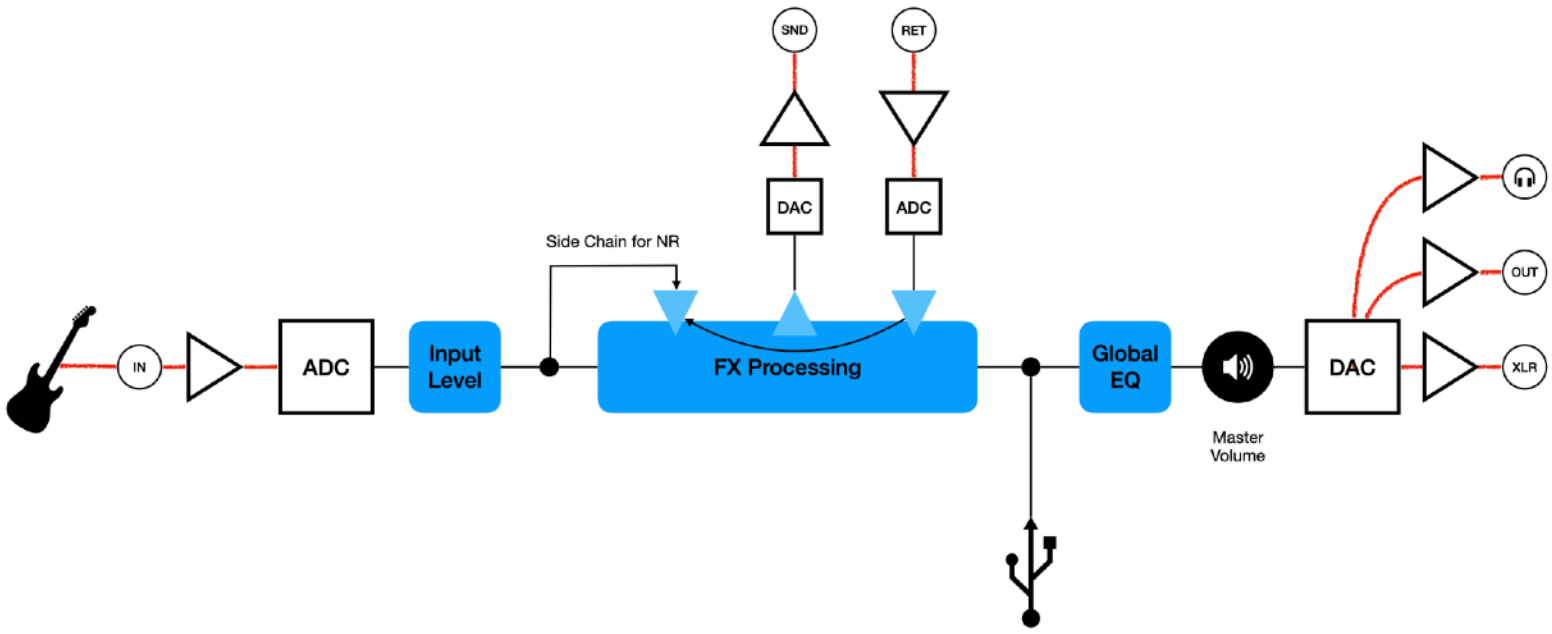
3. You are not using the same speaker/power amp.

4. You are not using the same post-production (did you check if the video explicitly said you were hearing direct sound with no post production?)

5. If the demo was from someone selling you presets: maybe they were actually playing an AxeFX.

SIGNAL PATH

While the official Valeton manual has a number of handy diagrams in the “Application Scenarios” section to explain how to connect the unit to other devices, it doesn’t include a block diagram showing what’s happening inside the device. So here you are:



Signal path block diagram of the GP200

SETTING INPUT MODE

The analogue signal at the INPUT jack first hits an input loading stage with switchable impedance: this is the **Input Mode** in Global Settings Input/Output. The Input Mode does *not* change nominal input level, so the purpose of this setting is purely to match the input impedance to the instrument. As a rule of thumb, you’d use **A.GT** for passive piezo pickups, **E.GT** for traditional passive electric pickups, and **Line** for active electronics.

What if you get it “wrong”?

If you set the input impedance too high (e.g. set to “**A. GT**” when connecting a keyboard) then you may lose some input level and therefore experience more background noise. In all honesty, it’s unlikely to ruin your day.

Setting the input impedance too low, on the other hand, definitely can, because if a guitar pickup doesn’t see enough of a load, it messes up the frequency response. You can get a dull, lifeless sound.

From there, the analogue signal goes through a non-adjustable buffer and into a 24-bit ADC. The buffer’s gain is fixed at a level that makes it very hard to clip the ADC input - it exploits

the fact that 24 bits gives a lot of dynamic range. The downside of this is that with any normal guitar, you have so much headroom that you don't optimise your signal-to-noise level.

We are now in the digital domain, and the first thing that happens is that the **Input Level** set in Global Settings is applied, -20dB to +20dB. This is nothing more than level adjustment applied to a digital signal: *turning it up or down does not alter the S/N ratio*. It's simply there to allow you to compensate for level differences between multiple instruments as you swap guitars/basses/keytars/whatever. For what it's worth, the calibration here is accurate: what you set is what you get.

It is this signal, as tweaked by Input Level, which is fed to the side chain of the NR block, regardless of the position of the block in the digital signal chain. This design decision allows the noise gates in the NR to work better with highly compressed signals, but has the downside of giving unintuitive results when the gates are placed after time-based effects.

The signal now moves through the various active effect blocks in the order in which you set them. When we reach the patch's Send point, the signal is branched off and sent to one of the DACs and to the SEND jack. Similarly, the RETURN jack feeds a buffer and another ADC, and its output is routed(or merged) into the digital signal path. (See THE FX LOOP for details.) The return signal (if there is one) is also sent to the side chain of the NR block, again regardless of the position of the NR block. (This can cause some strange side effects if you have the NR block before the return, since then it is listening to its own output.)

At some point the digital signal typically gets stereo content — a reverb, a spacious delay — and so at the end of the digital FX chain, we have a stereo output signal. This is initially routed to the USB audio interface (which *does* apply dither, in case you wondered); then the signal is further routed optionally through the Global EQ and from there to another DAC which feeds the headphone, unbalanced and balanced outputs. The master volume knob controls the level for all three analogue output pairs (headphones, unbalanced jacks, balanced XLRs), which can cause headaches as sometimes the right level for the main outputs results in deafening headphone levels.

INPUT NOISE

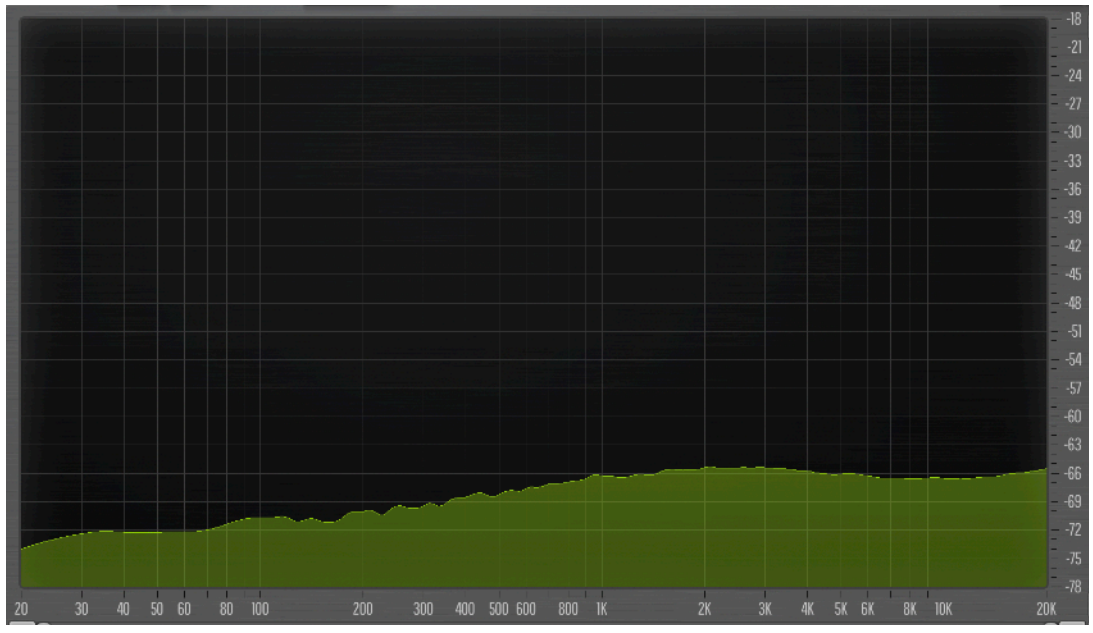
Now we understand the signal path inside the unit, a few things become clearer about input noise, or "hiss".

Turning down Input Level in Globals will reduce noise, but only because turning down the level of a signal turns down the noise within that signal. The relative level of signal to noise stays the same.

So what happens? Your guitar becomes too quiet (clean amp) or the sound gets too clean (dirty amp) and you turn up the gain on the amp… which brings back the noise.

If you keep wanting to turn your input level down, then you may just have the gain on the amp model too high.

The other thing to observe is that the only source of hiss is the residual noise in the input stage before the ADC, which has a fixed gain. In other words, there is a fixed amount of inevitable noise at the start of the digital signal path. It's possible to see this by recording the USB out (which, by definition, has no added noise from any analogue outputs) with nothing plugged into the unit:



Residual Noise, with Input Level at +20dB (boosted in DAW by 48dB)

This inherent noise, when the unit has a good PSU and is kept away from strong RF/EMI sources, doesn't contain any strong individual frequencies. It's just hiss. Any hum, buzz, clicks or pops in your signal are therefore *problems you can fix*.

SETTING GLOBAL INPUT LEVEL

Given that the Input Level in global settings doesn't alter the GP200's signal to noise ratio, there's a good technical reason for leaving global input level at 0dB, which is related to the FX loop: the main input and the FX return will then have the exact same nominal level, which makes it easier to move your other pedals around relative to the GP200.

(In other words, if you have a favourite dirt pedal with its knobs set just how you like them, then leaving global input at 0dB means you can place the pedal either in front of the GP200 or in the FX loop and get the same amount of dirt.)

You may want to use a different Input Level if:

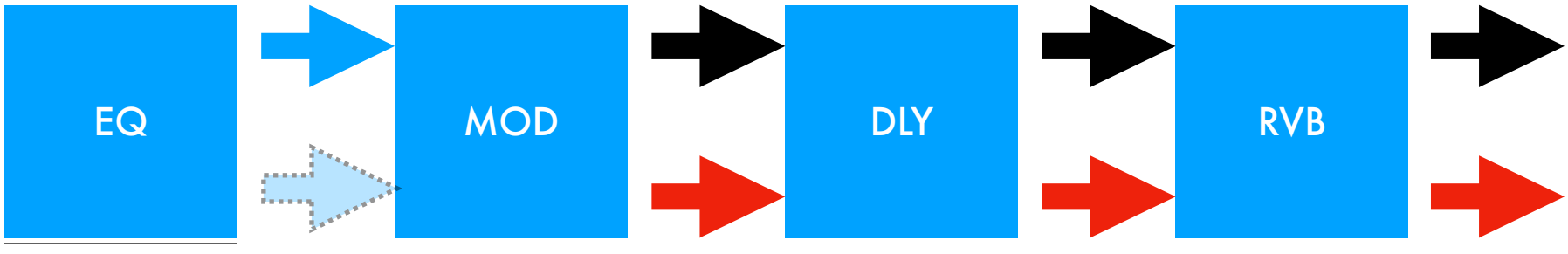
- Your guitar is unusually high or low output;
- You make heavy use of a set of NAMs from one particular source, and a particular Input Level makes them react the most accurately;
- You have a *very* high quality buffer between your guitar and the GP200 which boosts the guitar's level without adding noise[6]. (If this is the case, then lowering Input Level can improve overall noise performance)

MONO AND STEREO

The GP 200 has stereo outputs and several effects generate stereo output: many (but not all) in the MOD and DLY blocks and all in the RVB block.

However, the digital signal path is not fixed in the GP200: the blocks can be placed in any order. Given that all the effects are purely in software, this is achieved in practice by making all the effects have the same digital 'connections'[7]: every effect can be thought of as having a stereo input and the stereo output - that way they can placed in any order without hassle. What varies is what the effects do with their two inputs. Some will make use of both L and R channels, and some only make use of the L channel, ignoring what is on R.

It follows that, just as would be the case in a traditional analogue pedal board, the sequence of effects can have an impact on the end stereo image. Let's start by considering the default signal path, where a dry AMP+CAB signal is EQ'ed (all mono) and then passes through MOD, DLY and RVB.



[6] Be careful here. Many buffers lower output impedance (good) but add noise (bad)

[7] This is a simplification of what's actually going on, but explaining exactly how signal processing happens in software with blocks, buffers, DMAs etc. would take pages of text without making the description more useful in this context.

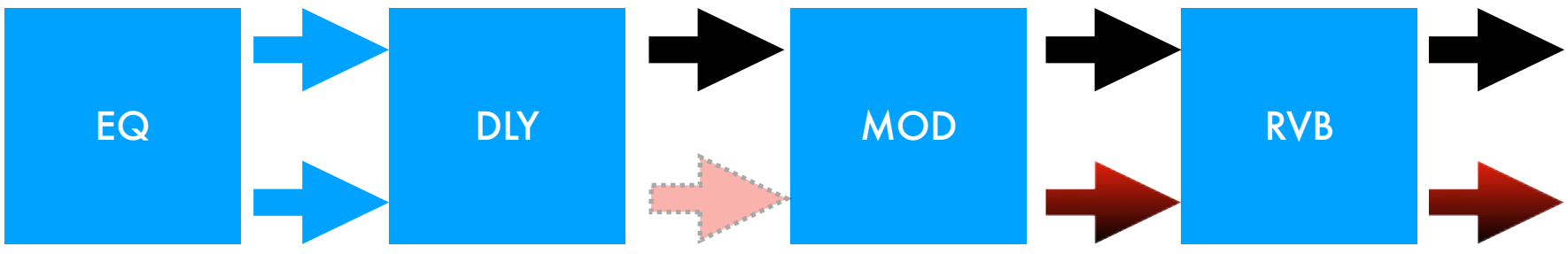
First the EQ block sends a mono signal - that is, identical L and R - into MOD. For example, the **C-Chorus** module (an emulation of the Boss Dimension-C pedal). This effect ignores what's incoming on the R channel, but this doesn't matter here because R is the same as L. It creates different output signals for left and right

This stereo signal goes into the DLY block. In *most* cases for DLY, that signal is passed direct to the DLY block's output and the delay algorithms act independently on the L and R signals. The delayed L and delayed R signals are mixed with the direct signal. So maximum 'stereoness' is preserved: everything that gives the sound its spread across the speakers is there.

The L and R outputs then go into the RVB block, where something slightly different happens. The direct signal remains stereo, but the reverb algorithm acts on the mono sum of the L+R inputs. This gives the most natural-sounding result in any case.

The combined effect of this is that you hear a nice stereo chorus, a wide stereo delay, and some stereo reverb on top, for a lush sound - assuming of course you then have stereo amps.

What if we swap the blocks round, though? Then we run into an issue, because as I mentioned, the C-Chorus doesn't run in true stereo. In fact, none of the effects in MOD actually do. (Try it for yourself by running a ping pong delay into a chorus.)



This time, the mono EQ goes into the DLY, which outputs distinct L and R content, but then the MOD block ignores the content in R (shown by pink arrow) and only uses the L content. The MOD block still outputs separate L and R signals, but the 'stereoness' of the overall effect is diluted.

With some tedious experimentation, it's possible to divide almost all the effects in the GP200 into four categories, with a handful of outliers:

The biggest category is **Mono to Mono** - only the L channel is processed as if it were a mono input, and the mono output is sent to both L and R (in case the next the effect in the chain is true stereo)

- Everything in **PRE** apart from the Harmony 1 and Harmony 2;

- Everything in **WAH**, **DST**, **AMP**, **NR**, **CAB** and **EQ** (yes, the CAB block is mono)

- In MOD: **Bit Smash, Auto Swell, Hold, Freeze**

- In DLY: **M Echo, Sweet Echo, 999 Echo**

Then we have **Mono to Stereo** - only the L channel is processed as if it were a mono input (incoming R channel is thrown away) and we get a stereo output.

- Everything in **MOD** except for Detune

Next, **Semi Stereo** - the L and R channels are summed to mono before processing, but also passed untouched as the dry signal with which the effected signal is mixed. The result doesn't seem to lose any width, because you still hear the stereo in the original signal along with the processed one.

- Everything in **RVB**

- In DLY: **Ambience 1, Ambience 2, Ice Delay**

The last big group is **True Stereo** - both the L and R channels entering are processed separately, preserving all stereo content. All of these are in the DLY block.

- In DLY: **BBD Delay S, Digital Delay S, Tape Delay S, Pure, Analog, Tape, Slapback, Sweep, Ring, Tube, Vintage Rack, Lofi Echo, Broken Delay** and (of course) **Dual Echo**.

Now for the oddballs:

In PRE: **Harmony 1 and Harmony 2**: These pass both L and R channels, but only apply the harmony effect to the L channel. R remains dry.

In MOD: **Detune**: This applies detune independently to both L and R channels, but the dry signal is taken from only L, which is made mono.

In DLY: **Rev Echo**: The dry signal preserves both L and R, but the mono delayed signal is based only on L.

Also in DLY: **Ping Pong**: The dry signal is *swapped*[8] - L channel is sent to R and R to L - and all the delayed signals are taken from L channel only.

STEREO AND THE FX LOOP

Where does the FX loop come into this, since it's mono? You'll note that the default position is just after AMP and before CAB, where a mono loop wouldn't cause an issue. If you move the send point after a stereo effect, *only the L channel is routed to SEND*. The RETURN channel, although mono, is of course routed to both L and R. In other words, the FX loop behaves like a "mono to mono" effect in our list earlier.

"NO CAB" CONFUSION

If you have read this far, you may be starting to wonder how "No CAB" mode works. Suppose you have "No CAB" turned on for L channel. One might initially imagine this is done by having the CAB block output differently on L and R outputs. But this couldn't work unless all the effect blocks after it were true stereo… and they aren't. And we've already seen the CAB block is mono. So how does the GP200 handle this?

By lying to you.

When you enable No CAB mode for either L or R channel (but not both), the GP200 secretly moves the CAB block to the **end** of the chain! It can get away with this because the CAB is just a fancy EQ, and with a typical signal chain where there are only time-based effects downstream of the CAB, the difference between EQ before or after processing is either zero or not easily noticed.

Here's the simplest experiment[9] you can do to prove it to yourself:

- Set the FX send point to just after the CAB block.
- Run the SEND into a monitor speaker. Enable an "acoustic sim" IR so that the difference between cab and no cab is extremely obvious. Play a bit: you can hear the IR.

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[8] There's actually a reason why this was probably done intentionally by a lazy coder. See if you can figure it out.

[9] There are others, and I did several, but this is the easiest.

- Now, remembering that the SEND is fed from the L channel, enable “no CAB” for R. The IR sound disappears from the SEND, but is still audible from L channel on main outputs. The send routing hasn’t changed - but the CAB processing is now happening later.

In summary:

When you turn **No CAB** on for L, the cab block is moved to the end of the chain and this happens:

L signal is passed direct to the L output;

L signal is *also* passed through CAB and from there to the R output.

Yes, you read that right. The R channel is thrown away. Why? Because when you are using this mode, you are probably running one channel to an onstage monitor and the other to the PA. You want the same signal going to both, don’t you?

If you really, really want to have different signals, then you turn **No CAB** on for R instead. Then this happens:

L signal is passed through CAB and to the L output.

R signal is passed direct to the R output.

CONCLUSIONS AND SUGGESTIONS

The GP200’s signal chain has been engineered to meet the needs of a typical rock/metal/pop/jazz musician within the cost constraints of the unit. The individual effects ‘expect’ to be placed in traditional positions, with all the “in front of amp” effects mono (for all intents and purposes) and the “post production” effects (stereo modulation, delay, reverb) various flavours of stereo. Therefore, if you want to run a stereo rig, you will be constrained in what you can do without messing up the stereo image.

If you use the No CAB mode, be aware that it can cause subtle shifts in the sound, depending on effect sequencing. You may want to move the CAB block explicitly to the end of the chain in your patches so that the sound remains consistent whether you are using headphones or connecting to your onstage setup.

BLOCK AND PATCH OUTPUT LEVELS

In most places where you see it, ‘Volume’ acts like a linear taper pot, with 50 being unity gain

If you need more make-up gain after a quiet block, you can (mis)use the "S comp" or the "Sine Trem" modules to get +9db or +6dB respectively

The Patch Level Meter is not a reliable indicator of clipping: don’t go into the amber

When programming patches, it usually pays to manage gain staging throughout; setting things up, where possible, so that the apparent volume of the patch remains constant (unless you specifically want it to change for a solo boost) as modules are turned on and off. This gives you maximum flexibility to map blocks to footswitches in stomp mode, and makes it more likely that balancing levels between patches will be easy.

To this end, both the patch as a whole and many effect modules have a Volume parameter, which is presented as a 0-100 scale. (There are a few exceptions, which are noted in the detailed effects table.)

The **Volume** parameters act like a linear pot. 50 represents unity gain, and on one side 100 gives +6dB of level; but on the other side, -6dB is at 25 and -12dB at 12. 0, of course, mutes - something you’re unlikely to need.

UNDERSTANDING THE PATCH VOLUME METER

Along the top of the screen is a digital VU meter that fills left to right. The manual on page 6 calls it a “Patch volume monitor”:



This shows the final output of the digital signal chain (after Patch Volume). There isn’t an explicit scale, but by passing a sine wave through the unit via the USB inputs and outputs we can obtain the following approximate reference points:

- Line between “EXP1” and “EXP2” sections = -18dBFS
- Line between “EXP2” and “BPM” sections = -9 dBFS

• Line between “BPM” and “Patch VOL” sections = -5 dBFS

• Clipping indicator comes on if signal exceeds -1.7 dBFS

However, this meter is actually based on a compensated RMS signal[10], not actual peaks. This means that the digital output can be clipping even when the meter is apparently bouncing safely around the -5 marker.

Given the internal processing is 24-bit, you can afford to leave considerable headroom. I would suggest that having the output meter around the -18dBFS line, and not crossing the -9dBFS marker, is a good rule of thumb to avoid any risk of clipping; that still gives you at least 21 bits of resolution. Or, if you are comfortable running the GP200 with a DAW, you can use the USB interface to get a true indication of the output peaks.

It’s also important to note that whether you use the built-in meter or you monitor the USB signal in a DAW, you are watching the output level only: the GP200 does not have an indicator for any clipping *within* the signal chain.

RECOMMENDED GAIN STAGING FOR YOUR OWN PATCHES

Unless you have a specific reason or need, I would recommend that you begin by programming up a basic clean tone: a clean AMP into your favoured CAB and maybe a touch of room reverb (especially if you are monitoring on headphones).

If you are using a built-in cab, set its **Volume** to the recommended level in Appendix 2. If you are using your own IR, first adjust the High and Low Cut filters, and then set the volume so that it sounds about the same level as “TWD SUP” with Volume on 51. Leave **Patch Volume** on 50.

Set the AMP Gain and tone stack how you like, and then adjust the Volume of the AMP block until you are metering around the -18dB mark.

Save the patch as “Clean Level Reference”.

Then whenever you create a new patch, leave Patch Volume on 50, CAB Volume on its recommended setting, and adjust AMP Volume until the patch sounds similar to the

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[10] By this I mean the displayed level is an RMS (averaged) level but adjusted upwards by 3dB so that it matches the peak level when the input signal is a sine wave. If this confuses you, don’t worry, just keep the meter in the green and be happy.

reference clean patch. (Usually this will involve turning the volume down below 50 for overdriven and high gain sounds).

Why do it this way?

1. Sticking to the recommended CAB levels means that you can quickly switch CABs and dial in the appropriate adjustment to keep levels the same.

2. Leaving Patch Volume at 50 gives you the most wiggle room for fine adjustments later using the quick access knobs.

3. If you use the FX loop, and put it in its most common position (after AMP+CAB), this ensures that you get a consistent level signal going into your pedals, reducing the risk of overloading them and saving you having to juggle Send Level from patch to patch.

Once you have a library of your own patches, you can do a second fine-tuning pass where you step through them and adjust the Patch Volume so that you get the precise level balance you want. (For example, you might intentionally make your 'solo' patches a few dB louder than your backing and rhythm patches.)

ADDING A LEVEL BOOST ON A FOOTSWITCH

You may have a patch where you'd like to have two output volume levels without a change of tone. Ideally, this would be doable by having a footswitch toggle between two Volume settings for the AMP or CAB modules; but this isn't an option in the current firmware. If you have either the PRE or MOD blocks unused in your patch, you can make use of those to create a pure level boost block which can then be assigned to a footswitch:

- For the PRE module: Select the S Comp, set threshold and ratio to 0 (effectively turning off the compressor), and set blend to 50 (this is loudest). Volume at 100 gets you 9dB of clean gain.

- For the MOD block: Set it to "Sine Trem" and set depth to 0. Now volume 50 = unity gain and volume 100= +6dB.

EQ BLOCK

All of the EQs are transparent (do nothing when set flat)

The plain pedal EQs are all -12dB to +12dB per band. Knob at +50 is +12dB, knob at +20 is +6dB (linear pot style)

The bands on Guitar EQ1 overlap, making this perhaps the most "musical" EQ

The 8k band on Bass EQ1 is NOT 8k! It is more like 4.5kHz.

The MESA-style EQ is... weird, perhaps because it is trying to accurately emulate the original.

The Hyper EQ's bands are correctly calibrated for dB, but sadly the first and last bands are not shelving.

Unlike its more expensive cousin, the Ampero, all the EQs in the EQ block are mono, so you can't EQ the output of RVB without losing stereo. Therefore you'll find yourself leaving it just after CAB 99% of the time.

Contrary to casual expectations, none of the EQs are really models[11]. Even the ones which are clearly inspired by pedal EQs are perfectly flat when set flat, and don't give any rolloff at the extreme top end in the way a model of a BOSS EQ might. They are all pure digital EQs with frequency bands that match familiar use cases.

PEDAL STYLE EQS

For **Guitar EQ1**, **Guitar EQ2**, **Bass EQ1** and **Bass EQ2**, the frequency band adjustments are all roughly the same: the knob goes from -50 to +50 and acts like a linear taper in each direction. So, with 0 (centre) being flat, -50 is -12dB, -20 is -6dB, +20 is +6dB and +50 is +12dB. This is perhaps annoying as you would normally want to have more resolution in the smaller adjustments, but that's life.

In almost all cases the bands are peaking filters with quite a wide Q (bandwidth). As a result, on Guitar EQ1, where the bands are close together, pushing adjacent bands has an overlapping effect. For example, if you boost both 400Hz and 800Hz to (nominally) +12dB, the actual result is that everything from 400 to 800 is boosted by about 18dB. This might be

[11] Reinforcing this is that the Ampero 2 has EQ modules which *are* pedal models.

useful if you are using the EQ as a boost in front of an amp. In any case, it does make the overall result a smooth EQ curve, so EQ1 is quite a "musical" tone shaping tool.

Guitar EQ2 has more widely spaced bands and so they operate more independently. For example, if you push both 1kHz and 3kHz, you don't really increase 2kHz by that much. It's also unique in that the top and bottom bands are shelving rather than peaking, so this is a good one for general high-end and low-end trimming.

The Bass EQs operate in the same way, with different frequency bands and (at least to my eyes) broader curves.

There two important errors to bear in mind: Bass EQ 1's so-called '8k' band is actually around 4.5kHz. I don't know if this is a mistake in the documentation (although the onscreen display and the manual match) or the implementation. And similarly, Guitar EQ 2's "3k" is really 4kHz.

THE "MESS" EQ

Moving onto the "Mess" EQ, this is based on the Mesa/Boogie's famous EQ (the one everyone sets to a V shape). This module appears to be trying to model it, warts and all, but it has some odd behaviour: it neither does what it says, nor does what a real Boogie does!

The "240Hz" band is actually centred on around 350Hz. Funnily enough, on a real Mesa Mark the 240Hz slider actually does 370Hz, so the GP200 is trying to be true to life here. Maximum boost is about +10dB, but maximum cut is only -6dB.

At the other end (6.6k), there is +10dB boost yet -12dB cut. The mysterious thing here is that this band on the real amp adjusts 4kHz, so the GP *doesn't* emulate the Mark's 'wrong' behaviour.

HYPER EQ

Unlike the other EQs, the knobs here are actually marked in dB, and they are accurate.

You might have wondered, looking at the 16kHz band on Hyper EQ, if it was really a shelf that adjusts everything 16kHz and up. Nope. If you dial in -12dB at 16kHz, 20kHz is completely unscathed. Granted, this doesn't really matter a great deal when these frequencies are typically removed in the CAB module, but for non-guitar processing a shelf would have made much more sense (and that's what most DAWs do).

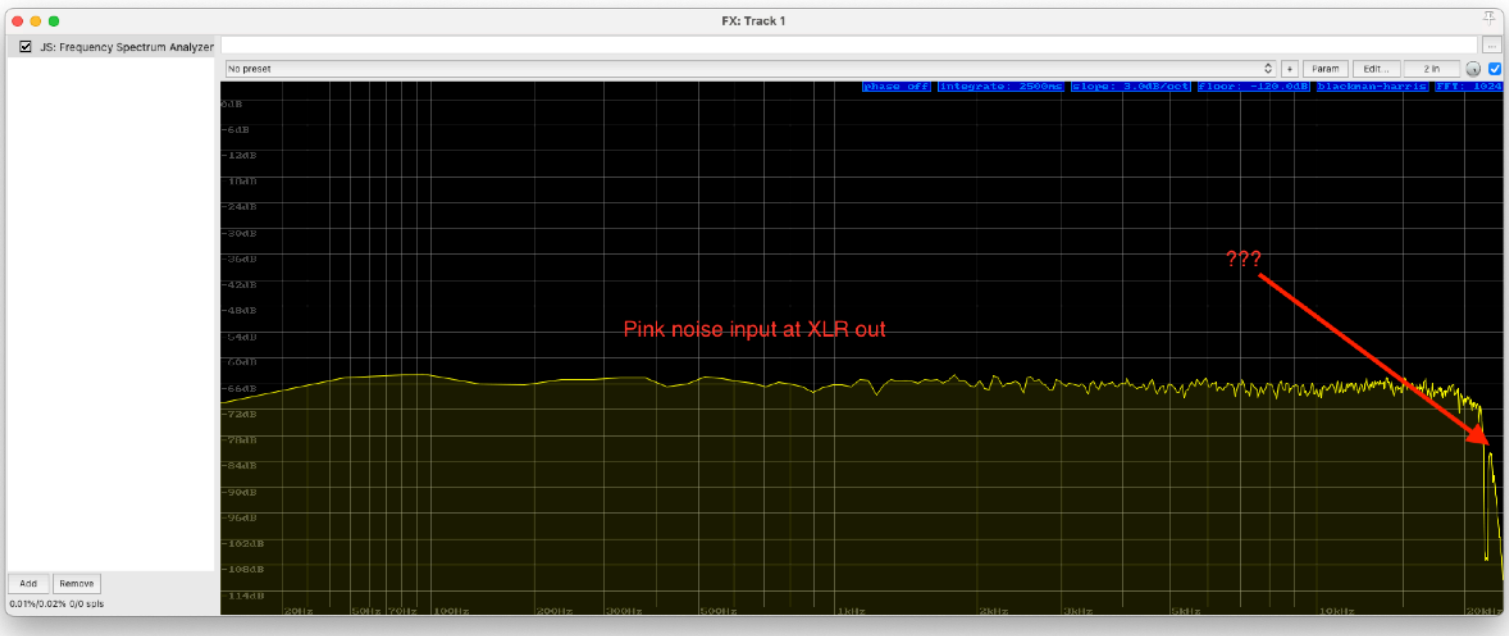
GLOBAL EQ

1. The Global EQ works as advertised, but be careful how you use it.

2. The High and Low filters are exactly the same as the CAB block

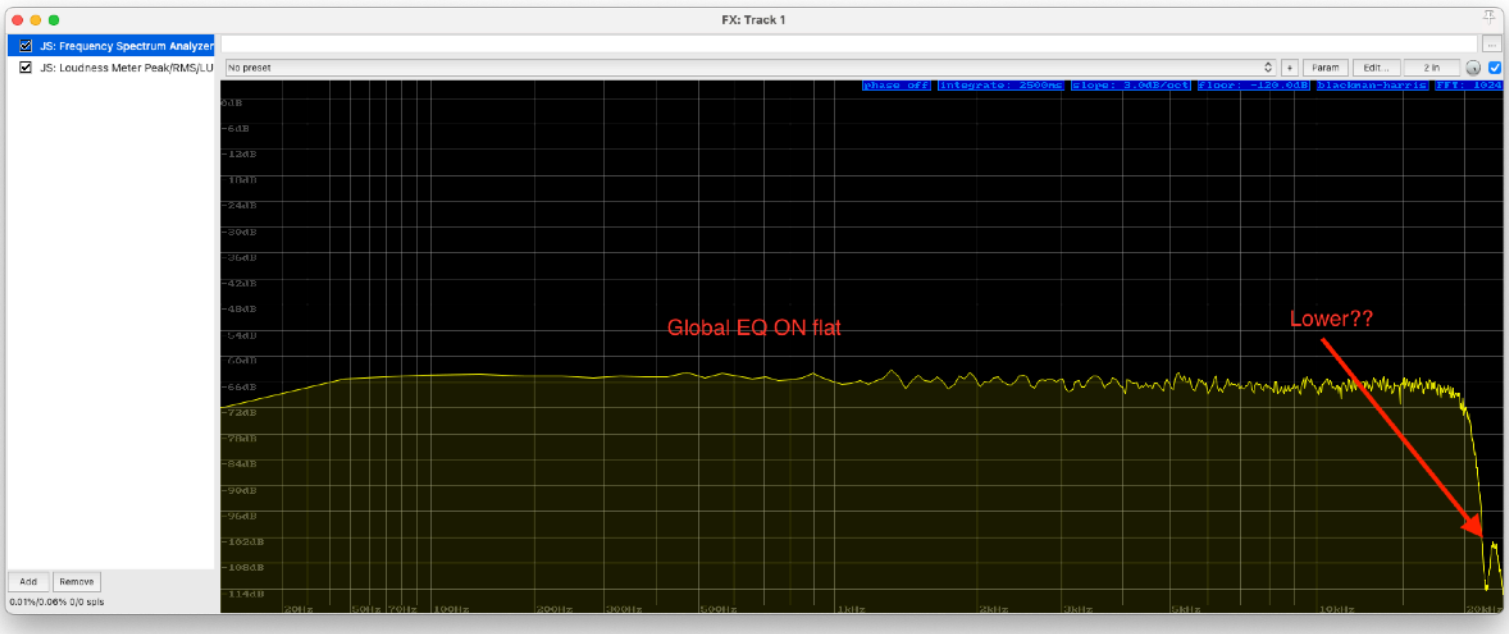
Unlike all the other modules, Global EQ is *not* applied to the USB output. (See the chapter on Signal Path). This is intentional since it is meant to be used as a way to correct for issues with the PA/FRFR/Amp you are plugged into, rather than to tweak patches when recording. (See the FAQ section on Global EQ for more.)

It also means that in order to check it, we have to plug the unit into a high-quality audio interface and measure what actually comes out of the XLRs.



Injecting pink noise into the input and measuring what is passed through with Global EQ off gives a satisfyingly flat response... *except* that there is some obvious aliasing/reflection going on. You can see a sharp dip at 22kHz (= 1/2 of 44.1) and then reflections above that. Hardly a surprise at this price point, but aliasing is one of the things that gives a "graininess" to digital modelling.

Turning Global EQ on with everything set flat does *almost* nothing. There is no change in overall level (as long as you set the volume knob to 50, which is unity gain), but running the digital signal through the Global EQ algorithm does put in place a LPF at 20kHz - the nasty peak beyond 22k is greatly reduced. I don't have good enough hearing to check this, but if you're 18 years old you may be able to hear the difference between Global EQ off and on even when set flat.



The **Volume** knob in Global EQ works like most others: 50 is 0dB (unity gain), 100 is +6dB, and 25 is -6dB.

The **Low Cut** and **High Cut** filters are first order, just like the ones in CAB: 3dB down at the corner frequency and 6dB an octave thereafter. Therefore, there's no point using them unless you find the cut filters in CAB insufficient to get the result you want.

Testing the peaking filters, these all work correctly, and the little picture you get in the UI on the unit is correct.

ANNOYANCE VALETON COULD FIX

Given that Global EQ's primary purpose is to allow the unit to be quickly adjusted to compensate for the frequency response of your monitoring solution, it would make sense for it to have not simply an on/off switch, but a handful of profiles that could be saved and recalled. One for using headphones, another for your personal FRFR cab, another for your band's PA, etc.

CAB BLOCK

They are real IRs, not just EQ filters, but have been sourced from all over the place.

The “Hi Cut” and “Low Cut” filters on CAB are first order: -3dB at the frequency set and 6dB/octave.

See the detailed table in Appendix 2 for a breakdown of all the built in cabs. If you would like to see large and detailed versions of the frequency response plots, these can be downloaded from the Files section of the **Valeton GP200** Facebook users’ group.

While the built in cabs are all actual IRs, not just crude EQ curves, it looks very likely that the dev team sourced them from various online free archives rather than licensing some from a single quality provider. The evidence for this is that the levels of the cabs vary a lot, and also some cabs clearly have high-pass and low pass filtering built in whereas others are raw.

A particularly obvious example is the "**Freed**" cab, which is allegedly a Fryette Deliverance cab. It has an unusually low level and is aggressively high-passed, making it sound small, thin and puny[12] when you are scrolling through comparing cabs.

Because the built-in cabs have very different levels of inherent filtering, it isn’t possible to give “one size fits all” recommendations for the **Low Cut** and **High Cut** parameters. Depending on your monitoring setup, personal taste, or cab and amp used, you may not use them at all, find them helpful, or even find them inadequate. Both Low and High Cut are gentle 6dB/octave filters which drop 3dB at the dialled in frequency, so you can err on the side of zealotry: 80Hz and 5kHz are a reasonable starting point and from there, use your ears.

CUSTOM IR HANDLING

In addition to the built in cabs, the GP200 supports up to 20 user-loadable IRs. The GP200 editor will happily accept mono impulse files of any length in both 44.1kHz and 48kHz formats (and possibly others), but it does not send the actual file to the unit. Instead it loads the IR into an internal plugin, sends a spike impulse through it and recaptures the response

[12] I think we all know that "small, thin and weedy" is not an accurate description of a Deliverance…

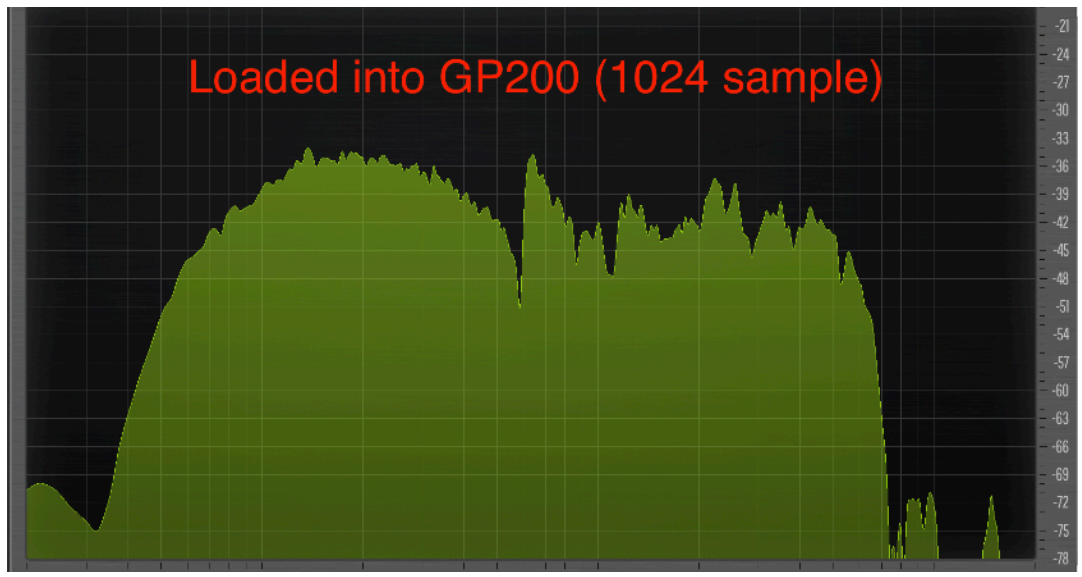
at 1024 samples, 44.1kHz mono[13]. 1024 is significantly fewer samples than pricier units like the Helix or Axe-FX. But what difference does it make?

What happens is exactly what mathematics predicts: fewer samples means lower resolution, with the most perceptible effect in the bass frequencies.

We can compare a high-quality (500ms) impulse taken from a Friedman cab, measured in an IR plug-in in a DAW, with the 1024-sample version running in the GP200. Notice that the little resonance at about 72Hz gets completely smoothed over, while the EQ signature in the mids and up is more or less preserved.



The other thing that is lost is any room "air" content in the impulse, which may or may not be desirable depending on your application. In a live context, it's unlikely to be noticeable.



[13] We know this because the temporary file created by this process is left behind on the disk.

PRE BLOCK - BOOSTS

All Three COMP modules can be used as boosts, however only "S COMP" gives you something novel (completely clean and flat)

SECRET: The S COMP Tone control is a tilt EQ, and thus useful everywhere in the chain.

MicroBoost, FAT BB, P Boost and Boost are all similar in basic sound. FAT BB and P Boost have near identical treble/bass EQ, so you may as well use FAT BB since it also has the low cut.

Unlike the other flat boosts, P Boost does saturate and compress at higher gains.

The 14Boost isn't a particularly accurate model of the Fortin Grind.

The AC Boost and B Boost are the strong mid-hump boosts.

The PRE block is something of a miscellany, but a good proportion of the modules in here are boost pedals of one sort or another. It's also worth noting that the three compressor models can all also be used as a clean boost, so I will be including them in this chapter, used as boosts only (no compression).

There are two primary uses of boosts[14] in a multi-fx:

- Used after the AMP block as a way to have a footswitchable volume jump, e.g. for a solo when the vocal drops out. This typically wants a completely flat boost, or at least one where EQ is optional.
- Used before a DST or AMP block as a way to push a gain stage harder or to alter the characteristic of gain (e.g. tighter low end). This often involves a very non-flat EQ response.

For this reason it makes sense to divide the models into two categories: Flat and coloured. Flat boosts are *mostly* flat response when set to defaults. Coloured boosts can't really be set flat, regardless of the knob settings.

[14] Yes, I know there are others, especially in traditional pedal boards where a boost can be an "always on" sweetener. The **Saturate** module might also be worth a try here.

THE 'FLAT' BOOSTS

We can start by working out the "redundant" boosts, by which I mean models that aren't necessary because another model can replicate them so closely they may as well not be there. This *does not* mean they are "bad", just that you don't need to audition them when you are crafting a patch from scratch.

COMP (with Threshold=0) and **COMP4** (with Threshold=0) can both act as clean boosts, but their frequency curve is basically the same as the Microboost, and the latter has much more gain range.

However, the **Microboost** is also almost redundant since it has no EQ, whereas the Boost model has the Bright switch. The only reason to use Microboost over Boost is if you need an insane amount of boost (> 18dB)

P Boost and **FAT BB** are extremely similar in baseline frequency response, and even their Bass and Treble controls act very very similarly. FAT BB also has a low cut switch. The big difference, however, is in their saturation behaviour. FAT BB is almost perfectly clean regardless of gain settings or how hard is the signal driving it, so can go anywhere in the signal chain. P Boost begins to compress when the input signal reaches -26dBFS, and saturates at about -15dBFS, so if placed later in the chain, engaging it may make your signal *quieter*! It also generates odd-order harmonics from that saturation.

So from a eq shaping point of view, we have three options:

FAT BB (subtle mid lift, softens highs, very low distortion)

Boost (mostly flat, softens highs, subtle saturation with higher input signal)

S COMP with Threshold=0 (completely flat, zero distortion) and Blend = 50 (maximises output level)

The base response of FAT BB is not as flat as Boost. Boost is flat from 60Hz to 6kHz, whereas over the same range, FAT BB has a gentle 3dB hump. Both roll off hard after 10k; in contrast S COMP (when used as a boost) is flat across the spectrum.

Hidden Tilt EQ

The **Tone** control in S COMP is actually a tilt EQ centred on 1kHz, so you can set the block to unity gain and use it just for that purpose. This could be useful to switch in and out in sync with a MOD pedal, since those often make the tone subjectively darker/lighter.

THE ‘COLOURED' BOOSTS

There are three boosts that are not flat at all. I can’t imagine them being used after AMP, with one exception…

14Boost is a model of the Fortin Grind, a pedal designed originally for Djent as it is intended to be placed in front of a high-gain amp. It cuts lower frequencies while boosting the upper mids and treble, tightening the sound while increasing high-end “grind”, hence the name.



Thanks to an intrepid YouTuber[15] we have actual response curve measurements of the real pedal, and in comparison the 14Boost doesn’t have quite as much level difference between low and high, plus some other differences in the exact curve. However, it’s probably close enough to do the job it’s meant for.

Since the 14Boost, if you turn it up enough, doesn’t attenuate lows relative to the dry signal, you can also use it like a treble booster, so worth trying with vintage style amps (anything where you might think of using a treble booster, really) and not just modern metal beasts.

The **B Boost** and the **AC Boost** are both extreme mid hump pedals with their tone controls set to noon. So you would use them to push a low-mid gain amp into thick overdriven lead tones. If you crank the Bass and Treble at the same time, both get flatter, but still not flat. They are similar, but different enough that in the context of pushing



AC Boost at defaults

[15] https://www.youtube.com/watch?v=8KsiHmyCsvc

mids before a gain stage, it's worth auditioning both to see which gives you the texture you like the most. They are also saturating boosts, generating additional harmonic content, which is helpful for stacking with overdrives.



B Boost at defaults

Telephone Effect

It's worth noting that the standard EQ curve of the B Boost is a strong hump centred on 1kHz combined with some saturation, so if you adjust the Gain such that it doesn't produce an actual volume increase, it can be handy as a switchable "telephone" effect. Think the start of "Wish you were here"… you could run this after an acoustic sim patch to let you begin in "radio" mode and then turn it off..

CONCLUSIONS AND SUGGESTIONS

First, the obligatory disclaimer: if it sounds good it is good. If you love the way a particular model sounds/works for you, you are not wrong! Nothing here is intended to replace your ultimate arbiter, namely your ears. Rather, to help you make choices as to which models might make sense to try first in particular contexts, and speed up your patch creation; and to make you aware that comparing boosts is tricky because you need to equalise volumes by hand.

That said, I think that you can't go far wrong with these:

1. For a post-AMP level boost, **S COMP with Threshold=0 and blend=50** is a great choice. You can get up to 9dB of pure level boost for a solo with no change of tone, and you also have a handy tilt EQ on top for overall top/bottom balance. If you want more radical EQ, FAT BB is worth trying since it adds negligible distortion.

2. For stacking into a drive pedal or amp to push that little bit more, consider **FAT BB**. Set flat, it just adds a touch of warmth together with the gain, but the Bass, Treble and Low Cut controls give you a ton of shaping options.If you can't find the sound you want, it's

probably an issue downstream. P Boost will add some more harmonics, but can't actually drive the next pedal as hard as FAT BB.

3. For the particular job you'd use a Grind, the **14Boost** is 'good enough'. It's a one-trick pony, but if you need that trick, it's your pony.

4. If you want the mid pushed sound, because you're trying to set up a Fender model with scooped rhythm (boost off) and liquid leads (boost on), I would start with the **B Boost**. It has more actual gain on tap than the AC Boost. But begin with the Bass and Treble cranked to about 75 and tweak from there, so it doesn't sound too honky. However, don't discount the AC boost if the B Boost doesn't float your boat.

5. Lastly.. if you just want to go nuts and drive the internals into clipping for some "destroyed unit" gargle-fuzz, the **Microboost** will give you ludicrous gain at 100. Just remember to turn things back down somewhere at the end of the chain…

PRE BLOCK - COMPRESSION

0 is fast and 100 slow for Attack and Decay.

S COMP is more of a "late in the chain" compressor than the pedal models.

There are three compressors in the PRE block, COMP (modelling the ROSS pedal), COMP4 (the Keeley) and the enigmatic S COMP. The latter I believe to be S for Studio, since as well as having the most control it appears to be a zero-colouration compressor.

COMP AND COMP4

These two are, let's be honest, "turn the knobs till you like it" modules, but it's still useful to understand their behaviour.

COMP (Ross) has a fixed attack and release time, attack being around 25ms and release 250ms.

COMP4 (Keeley C4) has a control "Attack" which actually adjusts both attack and release[16]. The fastest setting is 0, with a very fast attack (10ms?) and a release of 100ms. At the highest setting of 100, the attack stretches out to 50ms or so, and the release beyond 500ms. The "Clipping" parameter doesn't change these (nor should it).

Both have some form of auto make-up gain, and indeed actually add a few dB over unity even when the "Sustain" is turned down all the way (no compression)

S COMP

S COMP is the most interesting of the three since it has separate attack and release and also a blend control, plus no auto-makeup gain. The latter means you have to be careful and not set it for too much compression or you'll lose level.

The **Blend** control has some eccentricities: in the centre position you have a blend of the dry and wet signal at full volume, and since this is a "perfect" compressor with no phasing issues, that means a 3dB bump on the bypassed signal if there is no compression. As you turn the blend down towards 0 the wet signal is removed, which you'd expect, but the dry

[16] The official docs for the Keeley C4 say that the release time is set by the "sustain" control, not "attack". I don't know whether the docs are incomplete or the GP200's model is wrong. Maybe someone with a real C4 can comment?

signal also gets quieter: at 0 blend you have all dry signal, but it is 6dB down on bypass. Peculiar!

When you turn the blend control above 50, the dry signal is progressively removed, until at 100 you only have compressed signal. This time there is no attenuation, so blend=100 with no compression is the *same* level as bypass.

In practice this means the simplest way to handle things is to start with blend=100, set up the compression 'feel' the way you want, and then ease Blend back if desired.

The **Threshold** knob is awkwardly named since it isn't marked in dB and works the opposite way from a traditional threshold control. In a real studio compressor, a lower threshold means more compression, but here threshold=0 means *no* compression, and as you turn it up the actual dB threshold goes down. But without a gain reduction meter you have to go by ear anyway… Similarly, the **Ratio** control is just 0 to 100 with 100 being limiting and lower values progressively gentler slopes until 0 is no compression at all.

Attack and **Release** are separately adjustable: Attack varies from near instant to a very generous 250+ ms, and release also goes from zero to over 1000ms, so you have a lot of room for manoeuvre. As so often on the GP200, the knobs don't have a great response curve, with a lot of the action in the area between 50 and 100.

S Comp Tips

- Because the S COMP is transparent (with sensible settings), you can use it *after* the AMP module as a way of simulating power amp sag and bloom. Try setting a lower gain amp to edge of breakup, put S COMP between the AMP and CAB modules, and tinker. It's subtle, but can make amp modules (including NAM head captures) 'breathe' a little more.

- S COMP's potential for more subtle compression, including parallel compression, also makes it the more logical choice for electroacoustic patches.

NR BLOCK

There's no reason to use GATE 1 or 2, just use GATE 3

The gate position matters less than you think, but it *must* go before any delay.

If you use the FX Loop, you *must* put the gate after the loop return.

There's not a lot one can write about the gates in the GP200, since there are only three of them and only one is any good. All attempt to use some form of downward expansion, in that once the signal falls below the gate threshold, the level is turned down (at a rate determined by 'Release') to a level dependent on how much the input signal is below the threshold. This ought to give a more natural decay to a note than a naive mute. Unfortunately, two of the implementations in the GP200 have no hysteresis, and so if you try to finesse the threshold you get the dreaded "popcorn" effect. This is one reason, I suspect, why so many hated the gates until firmware 1.8 - but there is another hidden reason… of which more later.

As is often the case with the GP, the manual does not explain what the 0-100 values actually mean for any parameters - you are expected to tune by ear. It doesn't even say whether 0 attack is fast or slow.

GATE 1 has non-adjustable attack and decay. The attack is about 5ms, release 125ms. No wonder a lot of people hated it, especially high-gain users. The attack is too slow to preserve pick attack, and the release too slow to prevent annoying "pssht" right after a palm muted chug.

GATE 2 is adjustable. **Attack** goes from around 2ms (at 0) up to 20ms (at 100), the latter useful more as a special effect than anything else. **Release** goes from about 12ms (at 0) to a luxuriating 250ms (at 100).

GATE 3, which was introduced in the 1.8.0 firmware, is even more configurable and benefits from having **Attack**, **Release**, **Hold** wide-ranging and directly settable in milliseconds instead of vague 0-100 values. It even includes a **Hysteresis** parameter (even though the latter doesn't really work well)

As far as I can tell, there's no reason to use GATE 1 or GATE 2, since GATE 3 can do everything they can and sounds significantly better since it uses a proper downward expander. Dialling it in, though, needs a different approach from the traditional gates.

HOW TO SET UP GATE 3

For a typical use case (a high gain amp model) start by setting **Threshold** to 20, **Attack** to 1ms, **Release** to 100ms, **Hold** to 0ms and leave **Hysteresis** at -3dB.

Then, keeping the strings muted, turn **Ratio** initially to 0 and slowly increase it until the level of background hiss is low enough that it doesn't bother you. You don't have to get to complete silence if you don't want to, and the lower Ratio is set the better for sustained notes.

Now you can fine-tune Threshold and Release to taste. You'll find adjusting Release much easier via the PC/Mac editor software.

WHERE SHOULD THE NR BLOCK GO?

The default position of the gate is between AMP and CAB, which reflects the common wisdom of putting a gate in an amp's effects loop. For a real high-gain amp, this makes a lot of sense: the pre-amp itself has a significant amount of self-noise and so even if the input is muted, you hear hiss unless you put the gate between preamp and power amp.

However, in a digital modeller it's less clear-cut, because the modelled pre-map has NO self-noise, as I established in my earlier post about GP200 input noise and levels. If the input signal to the "EV 51" model is muted, you can set all the knobs to 100 and get not a whisper of noise.

Ok, so this is good right? You can put the gate earlier in the chain, depending on which signal makes most sense to be the trigger. (Dan Worrall voice) Right?

Wrong. Recall the details from Signal Path:

The gate's side chain (the signal that actually triggers the gate) is *always* a mix of the raw guitar signal immediately after the input level and the FX return, regardless of where the gate is in the chain[17]. All you are changing is where the on/off tap is, not what turns it.

Therefore:

- If you are using the recommended Gate 3 without the FX loop, then place it after the last noisy block (AMP if you're using a high-gain amp, DST if you're running a distortion into

[17] The Ampero 2 gives you a lot more control over NR side chain routing.

a clean amp) for the subjectively least noise, or at the start of the chain for a more 'organic' sound where you hear a a fading note decay into a little bit of noise.

- If you are using Gate 1 or Gate 2 without the FX loop, you may as well put them right at the start of the chain, so that what they are doing matches what your eyes see: actual sound varies far less.

- Whichever gate you use, if you put a delay early in the chain, e.g. **Analog** into **AMP**, EVH style, you *must* put NR before DLY, or it will cut off the echo repeats.

- If you are using the FX loop, you *must* put NR after the loop, otherwise the gate will be listening to its own output and this will mess up the release time badly.

- Putting this together, it means that you can't have DLY before the loop if you are using NR and get great results. Even more annoyingly, you can't put a delay pedal *in* the loop and use NR.

This mess explains why a lot of GP200 users put a noise gate pedal in the loop - because that way, they can control where the side chain is. Remember though, putting a 3rd party gate in the loop adds 3.5ms of latency, so it's worth at least trying to make the built in gate work for you.

THE FX LOOP

Set send to 95 and Return to 100 and use pedals.

The loop in Series mode adds 3.5ms latency.

Don't use Parallel unless your external unit can run fully wet.

The GP200 has a pretty flexible FX Loop, but (as with some other features) the documentation misses out on some information to help you make the best use of it.

Page 11 of the manual explains how to set the send and return points. The Blue arrow (send point) shows where the signal is tapped off to be sent out of SEND, and the purple arrow shows where what is arriving at RETURN is injected back into the signal path. The defaults for both are just after the AMP module, which gives a roughly equivalent behaviour to the FX loop in a real amp (after the preamp and before the cabinet).

Every patch has its own settings for the FX Loop, which gives you a lot of flexibility but can be annoying if you actually want to have a common setup through all patches.

The first thing to know is that unlike an amplifier's FX loop, merely plugging into the RETURN does not break the signal chain. By default, the FX Loop is set to parallel and the levels are set to zero, which effectively disables it completely.

So, to put an EQ in the loop you would set the mode to Serial (since you want all the signal to pass through the EQ), but to put a delay or reverb in the loop you should prefer Parallel (since you want to leave the dry signal passing through and then blend in the delayed signal) *if* the pedal can be set to fully 'wet'.

That much is reasonably clear, even from Valeton's own documentation, but how should you set the "Send Level" and "Return Level"?

SEND AND RETURN LEVELS

The Send and Return level knobs are actually both attenuation pots with a linear taper. That is, 100 is unity gain, 50 is -6dB, 25 is -12dB, and so on, with 0 = muted.

If your Input Level (in Global Settings) is left at the default of 0dB, then with **Send Level** set to 100, the signal that emerges from the SEND socket is about +0.5dB hotter than the

signal that entered the main INPUT socket. In other words, it's instrument-level, and suitable for guitar pedals.

If you set Send Level to 95dB, then you eliminate that 0.5dB difference and the sent signal matches your input signal in level (allowing for the influence of whatever effect blocks you have used in between)

The level that the RETURN jack "expects", when Return level is at 100, is *exactly the same* as the level the INPUT jack expects with Input Level=0dB. In other words, it is also instrument level.

What does this mean overall?

First, it means that with Send Level at 95 and Return Level at 100, if you directly connect the SEND and RETURN jacks with a patch cable, the FX loop will have no effect on overall patch level. So that's a sensible starting point. If, for example, you use the loop to connect an analogue drive pedal with the loop early in the signal chain, you'll be able to leave the drive pedal's knobs set the way you have them when used with your guitar direct. (There may be some subtle difference in tone because of the lower output impedance of the SEND jack)

It also means that the FX loop's nominal level is optimised for use with pedals, as opposed to rack gear or other pro units which expect +4dB nominal levels. This is probably the best option if they had to pick one, but it is also the noisiest option.

When would you *not* use those Send and Return Levels?

In theory, you would lower the Send level if the signal coming out of the SEND jack is too hot, but in practice that ought not to happen if you are sensibly adjusting the output Volume of blocks so that your patch doesn't jump about alarmingly when turning modules on and off.

If at all possible, you want to keep the Send Level high (95 or 100) because it is just a digital attenuation, and the lower it is, the worse the signal-to-noise ratio. (This is another reason why I often see users complain the FX loop is noisy)

The Return level you would only reduce if the external effect you are using isn't unity gain and makes the signal hotter relative to dry, or when you are in Parallel mode and you are using it to control how much wet signal you want to blend in.

Summary

When in Series Mode: Send at 100 (or 95 if you want to be super accurate), Return at 100.

When in Parallel Mode: Send at 100 or 95, Return as required, external effect set fully wet (reason why follows..)

LATENCY

Using the FX Loop involves the signal travelling through another DAC/ADC pair and so inevitably adds latency. Perhaps unsurprisingly, this extra latency is about 3.5ms[18], the same as the end-to-end latency of the unit when the FX loop is not used.

So if you use the FX Loop in Series mode, you are adding 3.5ms of latency to the standard 3.5ms, making a total of 7ms or so. And of course, if your external pedal is also digital (like a GP5!) then that adds even more latency on top. So if you are sensitive to latency, be cautious about using the loop in series.

If you use the loop in Parallel mode, then while the wet signal will have that extra 3.5ms, of course your dry signal won't. But, if your external unit can't be set fully wet, you'll have a problem in that the dry signal from the unit will comb filter with the dry signal inside the unit, creating a hole in the sound at around 150Hz and nasty phasing throughout the midrange.

IMPEDANCE CONSIDERATIONS

Unlike the main INPUT jack, the SEND and return jacks do not have switchable impedance. The RETURN jack has a 'compromise' impedance of 100kΩ, which covers both line and instrument signals, and the SEND jack has a low impedance. The latter is more important as it means it will interact very differently from a guitar when used with an impedance-sensitive effect like fuzz or wah.

LOOP QUALITY

The loop uses a DAC/ADC pair that is identical to the main input and output: 24-bit, 44.1kHz. By connecting the SEND and RETURN jacks with a quality patch code and running test signals through the unit, we can verify that the frequency response is ruler-flat

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[18] Using firmware 1.8.0 or later. Earlier firmwares had 4.6ms FX loop latency.

and there is no significant distortion. So the loop is transparent and in that regard, not a "tone suck".

However, because of the low nominal signal level and the fixed impedance, it can't be ruled out that the loop might interact with external effects in a way that subtly alters the overall tone. Also the added latency may change how the sound "feels" under your fingers.

FX LOOP TRICKS AND HACKS

Don't forget ultimately the FX loop is just a way of routing audio, and it can be useful in other ways even if you don't want to put external effects in it.

1. **Wet/Dry/Wet** - if you place the SEND point after the AMP (or after the CAB, depending on your amplification setup) you can take a dry mono feed out of the unit prior to all your stereo effects and put that through your onstage 'dry' cab, then make the downstream stereo effects mostly or entirely 'wet' and run them into the PA (or another set of stereo speakers). Can give you a huge sound while remaining focused; great for trio/duo gigs.

2. **SEND as main OUT** - the SEND output's signal level is nominally the same as the main unbalanced jack outputs, but it is always at the level the mains would be if the volume knob was cranked. You can use the Send Level to attenuate it. So, if you only need one output and don't use Global EQ, you can place the Send point at the end of your signal chain and use that as the main output, using Send Level instead of Patch Volume to equalise between patches. Why do this? Because then your output is completely fixed and not vulnerable to knocking the master volume knob (which is very close to the expression pedal).

3. **SEND as stage output** - this is a variant of the previous trick. If you have an onstage monitor as well as running into the PA, you can use the FX out to drive that monitor while still running stereo from the main XLRs into the PA. Place the SEND point either before CAB (if you're going into a regular amp) or after CAB (if you're going into a FRFR monitor). Not only does this give you separate volume control for the outputs, but also allows for some interesting volume pedal tricks.

4. **RETURN as second input** - as long as it's low impedance (e.g. an active bass, or a guitar via a buffer pedal) you can plug an instrument into the RETURN jack, route the return point to the *start* of the chain, and then you can swap instruments by changing

patches rather than unplugging or buying another A/B switch. Just set the loop to Series to disable the main input. For a more detailed version of this, look at Appendix 5.

5. **Patch cable hack** - do you wish the **C Chorus** had a mix control? Put the loop into parallel mode, set the send point to just before MOD, the return point to just after, connect SEND and RETURN with a patch cable, and use return level to blend dry signal back into the chorused signal. Watch out for comb filter effects though.

6. **Split signal bass trick** - this needs an extra pedal but may be worth it for bassists. Set up a standard chains of PRE (compressor) -> AMP (some kind of bass preamp) -> CAB, but set the FX SEND point to after AMP and the return point at the end of the chain. Put a bass EQ pedal of some kind in the loop (set to **parallel**) and roll off everything above 800Hz or so. Back in the main GP200 chain, put EQ and DST after cab, roll off everything below 800Hz and add dirt. Now balance the full undistorted low end with the gritty, overdriven high end. **Important** - you need to use EQ to keep the two paths in separate frequency areas so that you don't have comb filtering ruining the sound.

7. **Reverb Sculptor** - if you have EQ and other blocks spare, you can gain additional creative reverb effects by placing the blocks together and bypassing them with the FX loop. For example: have DST-EQ-RVB at the end of your chain. Set the FX send point to before DST and the FX return point to after RVB. Connect the FX loop with a patch cable, set the loop to parallel, FX send and return levels to 100. Set the reverb to 100% wet and use the EQ Volume to control the blend of reverb against the direct signal. As long as you can stomach the additional latency on the direct signal, you can now have sculpted fuzz in the reverb only. Experiment!

ANNOYANCES VALETON COULD FIX

Obviously there are hardware changes that would improve the FX loop, like balanced connections, switchable impedance/nominal level, stereo operation… but here are three things that could be entirely fixed in software:

Send and Return make-up gain

The send and return levels are just digital trims, and instead of having 100 as unity gain, 50 should be unity gain just like it is everywhere else. That would make 100 give +6dB makeup gain. Invaluable for pedals like the Yamaha Magicstomp where the wet signal is always a bit weaker than the bypass.

Inability to control the FX loop

There are so many parameters that are controllable on the GP 200, it boggles the mind why you can't do the following:

a. Make a CTL footswitch turn the FX loop on and off

b. Assign the expression pedal to the return level (to fade a parallel effect in and out)

c. Assign a Quick Knob to the return level (to make easy adjustments live)

Noise Gate side chain control

When the loop is placed after a high-gain amp, the noise coming back into the RETURN is fed to the sidechain of the NR block. This forces you to place the NR block later in the chain, and even if you do can give poor results as you have to set the Threshold unnecessarily high. It would make sense if, as is the case on the Ampero 2, the user could choose the sidechain source for the NR block: either raw input or FX return.

THE EXPRESSION PEDAL

Recalibrate the expression pedal periodically

You can set up a parameter to go from high to low

The A/B switching can be used as an extra volume switch

The built-in expression pedal on the GP200 has some limitations, most specifically that it is not based on a continuous analogue value; there is a rotary encoder inside which after calibration generates discrete steps 0-100. This means that for some applications you get audible “zippering” as you change a parameter. In addition, the calibration does drift - I have calibrated the pedal, only to find a few months later that it only sweeps between 0 and 93. So make sure to recalibrate it every couple of months to be on the safe side.

Despite these annoyances, for some applications the pedal is very useful, since it can be programmed on a per patch basis to control up to three parameters in each of two modes A and B. By default, A is set up to control volume and B to control wah. Pressing hard on the toe of the pedal toggles between them, although you can also map a footswitch to perform the same function if you don’t want to be forced to move to the end of the travel to make the change.

The range of a parameter can be entered “backwards”, i.e. have the heel be the higher value and the toe the lower value. So you can make it fade between a delay mix and a reverb mix, or simultaneously move two/three bands on an eq to morph between a scooped rhythm and a mid-rich lead sound.

VOLUME PEDAL TRICKS

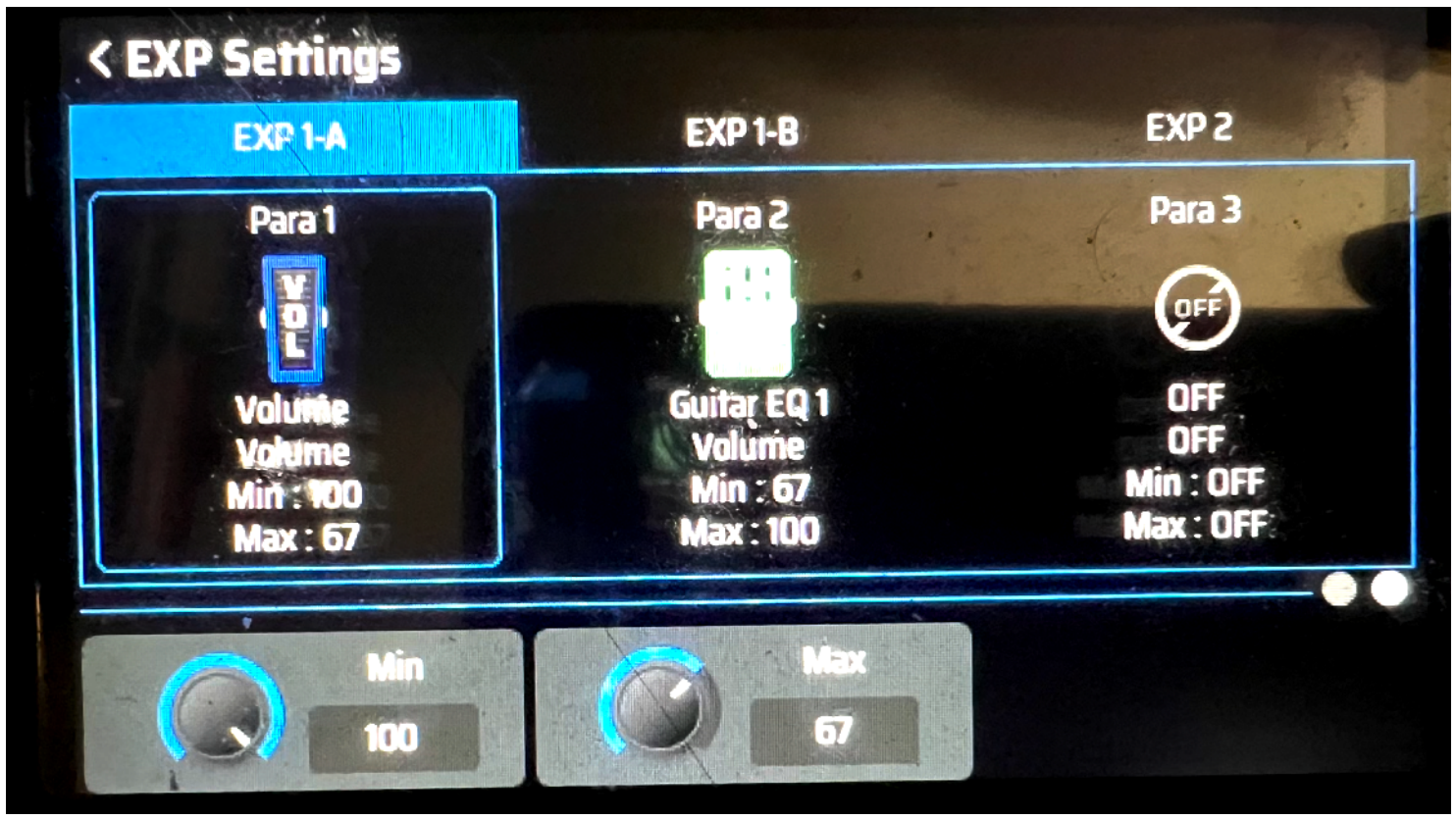
Not everyone uses the built-in expression pedal on every patch, or even much at all. So here are three tips in ascending order of complexity that you may find useful or spark your own ideas:

Range adjustment: Never forget that you don’t have to have the volume pedal sweep from 0 to 100. If you set the extremes to 50 and 100, you can keep the pedal back normally and then rock it all the way forward for a 6dB soloing boost. For a less intrusive 3dB boost, use values of 67 and 100.

Fade plus boost: The default patch sets up the expression pedal A (green) mode to be Volume and B (red) mode to be wah. If you remove wah from the B mode, then it just turns the volume pedal off. Now set your volume pedal range at 0 to 67. When in green mode, it operates as normal: but press down hard to flip into B mode and you get a 3dB level boost for solo. If you don't like using the "toe squidge" technique, you can set a foot switch to toggle between Expression A and B.

Sneaky boost: In the FX Loop chapter I wrote about the hack to use the FX send as an output to an onstage powered cab, giving you monitoring with independent volume control and leaving both L&R main outputs to go to FoH.

If you use this setup, then place the FX send point just before Volume, and make the pedal modify two parameters: its own Volume (range 100 to 67, going backwards) and the Volume of EQ, (range 67 to 100):



Adjust patch level to compensate if needed.

Now when you rock the pedal from all the way back to all the way forward, your level to FoH stays the same (so your sound guy doesn't hate you) but the level to your onstage monitor goes up by 3dB. Great for a subtle reinforcement that's kind to everyone's in-ears. (Inspired by an FM9 tip from John Cordey) Just keep the pedal at one extreme or the other, because in the middle the level will be slightly louder.

You can of course use other ranges, like 50 to 100 for a 6dB boost, as long as you have the same range inverted on the two blocks.

ANNOYANCES VALETON COULD FIX

The biggest annoyance here is that almost everything regarding the expression pedals is per-patch configurable (good) except the response curve, which is bad. Certain effects make most sense with linear, some with log, and a few with exponential (wah comes to mind); but you are forced to pick a single setting globally, which is fine if you use the expression pedal for the same purpose on every patch but is a frustration if you want to use it more creatively.

THE LOOPER

The looper works as designed - but unfortunately that design is pretty terrible, making it not really usable for anything other than looping a riff while you dial in a patch. If you use loops for performance, you'll be better off getting a good looper pedal and putting it in the FX loop.

More than a few people have torn their hair out over the looper, especially the undo feature. Just search the GP200 Facebook or Reddit groups to see their frustration...

The looper is a handy feature when working on a patch (play a little riff and leave it looping while twiddling the knobs) But if you are trying to use it for live performance then there are a number of annoyances.

SYNCHRONISING WITH THE DRUMS

You can get a loop to synch reliably with the drums, but it requires very precise timing. In a good looper with rhythm patterns, when you hit record the recording actually starts on the next beat or bar. The GP200 just starts instantly, though, so you have to hit REC *exactly* on the downbeat and then again (to stop recording) *almost exactly* on the downbeat after you're down.

Once you have that initial loop locked in, overdubs become straightforward.

However, it's an annoying "once". You have to be very exact for this to work, and if you aren't then the loop won't be a whole number of bars and it will get out of sync immediately.

Doing this in an actual performance context is a drag, because unlike a quality looper where the tempo automatically sets the loop length, you have to get the loop length right yourself, and that means remembering to tap your foot on the footswitch on a beat rather than just before or after.

UNDO/REDO

I am baffled why Valeton did not add ONE MORE PAGE to the manual to actually explain how to use the looper, and especially how to use Undo (to the extent it exists). What follows is mostly thanks to posters on Reddit and in the Facebook group; I have just checked it and added a clarification.

First a 'gotcha', although it is briefly documented in the manual: you have to set the short recording time for undo to work at all. That is, 90s in "pre" mode and 45s in "post" mode.

Now the actual operation of the undo. To try this yourself I recommend you start with a long base loop, 20s or so.

Assuming you have the usual footswitch arrangement (FS5 is undo, FS6, is record/play, FS7 is stop/clear):

Wipe any existing loop by holding FS7.

Record a new loop by tapping FS6, play for a while, then tapping FS6 again. The loop begins (bar is green).

While the loop is close to the start, tap FS6. bar turns blue while you overdub. Tap FS6 again and you go back to green.

Here is the key concept: undo *only works within the current overdub pass*. You *must* exit overdub *and* tap undo before the end of the current pass.

Example of undo when looping

Say you have a 20s loop running. While it's running, at 5 seconds in you tap FS6 (overdub/blue), play a bit. But you mess up.

If you just tap "Undo", nothing happens (ok the pedal flashes white, but nothing actually happens)

If you allow the playback to get to the 20s mark and the loop restarts, it's too late: that mistake gets locked in.

So, if you flub a note 15s into the loop, you have five seconds to tap FS6 (to go back to green) ***and*** FS7 (undo). If you manage that before the 20s mark, the overdub is removed.

As you can see, this is of very limited use. If you make a mistake towards the end of the loop, you have no chance of hitting those two buttons one after another fast enough.

CONCLUSION AND ANNOYANCES VALETON COULD FIX

The implementation of the looper seems to have been done by people who never actually use loopers for performance. It does work as 'intended', but the intent is poorly thought through.

It would have made much more sense for FS7 to both exit overdub mode and undo all in one go, although it's still far from perfect.

Even better (are you listening Valeton??) would have been to make Undo mode cut the record time to one-third (this would still allow a full minute in Pre mode) as then Undo could always discard everything overdubbed since overdub was entered. It would make things work far better.

The drum synchronisation is poor, requiring very precise tap-dancing. Instead, when drum sync is on the ‘record’ switch should work like most other good loopers: start recording but align the actual loop point to the next downbeat, merging the pick-up notes as needed.

THE USB AUDIO INTERFACE

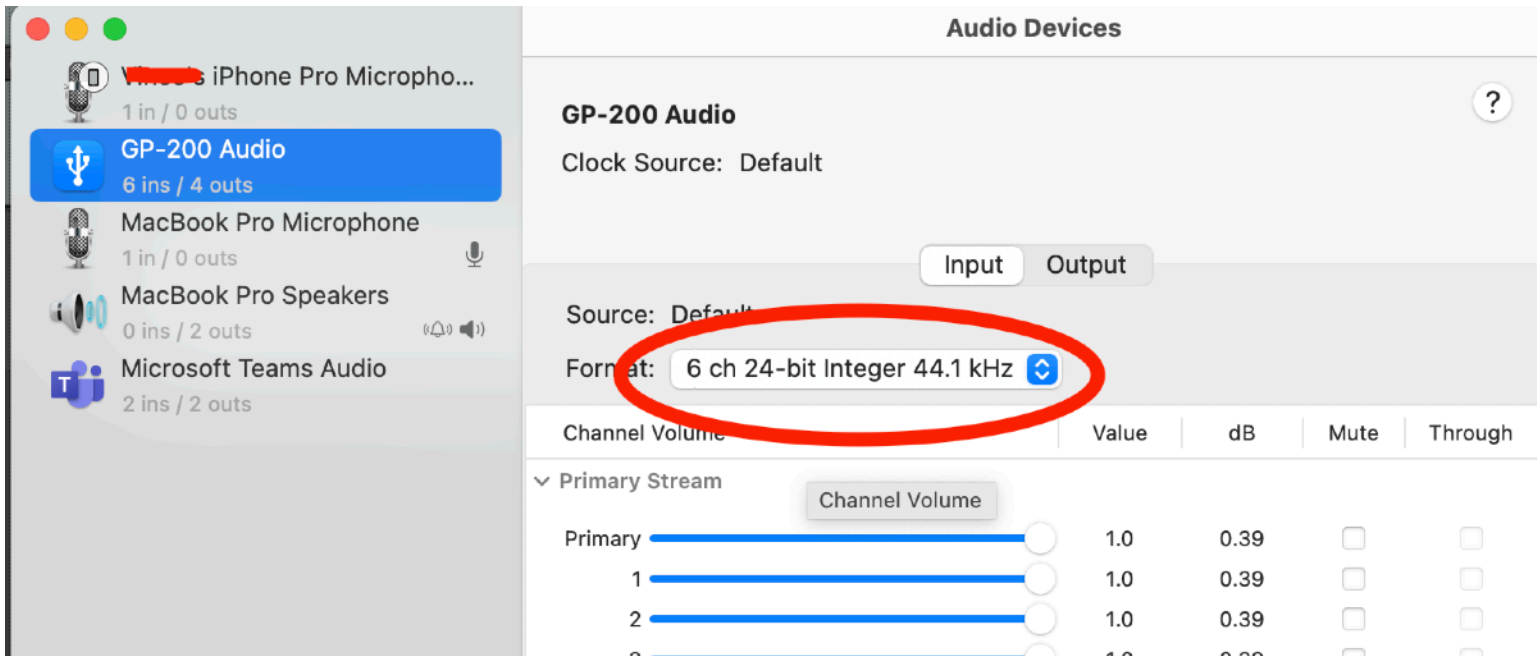
When using the GP200 with a DAW, use the Dry option in Global Settings and turn off Input Monitoring

Use the Loopback Input to check if you have the right latency compensation in your DAW

Make sure you are using all 24 bits on Mac

When plugged into a Mac (no driver required) or PC (ASIO driver from Valeton needs to be installed) the GP200 acts as a 24-bit, 44.1kHz audio interface. Unlike the Ampero, only 44.1kHz is supported[19].

There is a gotcha on Mac: the device out of the box can show up as only 16 bits.



Go into Audio/MIDI Settings and make sure the 24-bit option is selected.

The official manual has a very brief section on page 14 which describes two use cases but doesn't really explain what the USB inputs and outputs do.

The USB interface is described as 6 in and 4 out: in practice, you should think of it exclusively as 3 stereo inputs and 2 stereo outputs. Also remember that the naming is from the point of view of your computer, so the "USB inputs" relate to *outputs* of the GP200, and the "USB outputs" relate to *inputs* of the GP200.

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[19] The Ampero always runs internally at 44.1kHz but has additional hardware to resample on the fly so that you can easily use it with a DAW session that was recorded at 48, 96 or 192.

USB channel pair	Function
Output 1&2	Audio sent here from the computer is heard at the analogue outputs (XLR, unbalanced, headphones) Global EQ is NOT applied, but USB Audio "Monitor Level" is.
Output 3&4	Audio sent here from the computer is routed to the start of the processing chain, along with any guitar signal from the IN jack. Global input level is not applied
Input 1&2	This sends audio to the computer and is either the output of the processing chain OR the dry guitar signal from the IN jack, depending on Global Settings
Input 3&4	This sends audio to the computer and is always the output of the processing chain regardless of Global Settings
Input 5&6	This sends audio to the computer and is a unity gain mix of the output of the processing chain and the audio received on the "Output 1&2" channel

The channel pairs and defaults have been arranged so that the simplest use cases "just work". If you plug the GP200 into your computer and select it as your main audio device, then any music you play from e.g. Spotify will come into your headphones, mixed with your guitar, allowing you to play along. Similarly, if you hit record on a voice note app and just play, the default Rec Mode setting of Wet means that you will record your guitar with all effects active (but not Global EQ, which is never applied to the USB audio)

The default setting for **Rec Mode** is however not the most flexible, since it makes Inputs 1&2 exactly the same as Inputs 3&4. If you want to make full use of the GP200 in combination with your DAW, you will want to set Rec Mode to *Dry*. This will allow you to record Inputs 1&2 onto one track, and 3&4 simultaneously onto another track, giving you the GP200 output in full stereo and also a completely dry track you can use for re-amping, either with the GP200 or with other plugins.

MONITORING IN THE DAW

When working with a DAW, it is important to set up track monitoring correctly. Many DAWs default to enabling record monitoring by default, so that you can hear what you are recording back through the monitors: this is of course essential when using software plugins. But when you are using the GP200 to process the guitar signal, you will already be hearing your guitar directly, so you must turn input monitoring off completely or you will hear two copies of the audio, which aren't sample aligned. This will result in strange phasing sounds, excessive volume, or both.

If you are using the GP200 purely as an interface - in other words, all processing of the guitar signal is being done within the DAW via plugins - then use the "Dry" inputs 1&2 in

the DAW and set the Patch Volume to 0 on the GP200, so you don't hear your direct guitar signal.

THE LOOPBACK INPUT

Inputs 5&6 are described as the "LOOPBACK function", and carry a simple mix of the audio received on Outputs 1&2 and the processed guitar signal on Inputs 3&4.

This is primarily there for live-streaming, allowing you to playback a backing track from the computer, play along and have that fed into your streaming software like OBS.

However we can also use it to verify how accurate the reported roundtrip latency is:

- Play back a test track to Outputs 1&2 via the DAW, recording Inputs 1&2 to a new track.
- Play both the original track and the recorded track at the same time with the recorded track's phase flipped
- Nudge the recorded track until the two tracks null perfectly.

If the round trip latency report is perfect, then the tracks should null immediately without adjustment: but this is unlikely. In my test with REAPER, I had to nudge the recorded track back an additional 85 samples to align it. Your results may vary depending on your platform, DAW and block size.

Whatever result you get, the same will apply to recordings made from Inputs 1&2 or 3&4 - all three input pairs are sample-aligned.

USB AUDIO LEVELS

There are two level controls in Global Settings for USB Audio: **Rec Level** and **Monitor Level**. These are both digital trims and if you are working with a DAW I would recommend you leave them both at 0dB, since this passes the digital audio "as-is" and you will be able to adjust levels more conveniently in your software. In particular, resist any temptation to boost Rec Level, since this could result in you hearing good audio in your headphones only to find out your recorded file has clipping.

If on the other hand you are not using a DAW but just want to plug into your computer so you can play along to Spotify/YouTube/Apple Music etc., then you will probably find it helpful to lower Monitor Level. This is because the nominal level of the processed guitar signal is well below 0dBFS (to have plenty of headroom) but incoming professionally

mastered music will be close to that and thus incredibly loud in comparison. Not to mention the level of any system notification sounds, which might also sound in your headphones depending on your system setup. Start at -20dB and then turn it up to get a good balance.

THE SNAPTONE LOADER (NAM SUPPORT)

You can load amp NAMs into the DIST block and pedal NAMs into the AMP block.

The EQ in the Snaptone loader is post-model

The Gain and Volume knobs are very wide ranging

The GP200 can import NAM files into both the AMP and DST blocks, with room for 5 in each, and does so by converting them to Hotone's Snaptone capture format[20]. That much is common knowledge. See Appendix 4 for an in-depth analysis.

Despite the slight UI differences, the Snaptone module in DST is exactly the same as in AMP. They both have a pure digital gain/trim that adjusts the level going into the capture (this is very important if you want your capture to react 'realistically' to your guitar compared with the real thing), a three-band EQ which is post processing, and another gain/trim pot to adjust for differences in NAM calibration.

SNAPTONE EQ

The tone stack on a Snaptone amp is fixed regardless of loaded profile, and is applied to the output of the module, so it can’t be used to tighten up a high-gain model.

Bass: +/- 6dB peaking at 120Hz

Middle: +/-6dB peaking at 750Hz

Treble: +/- 6dB shelving at 3kHz

SNAPTONE INPUT LEVELS

When you load a NAM it is actually converted to Hotone's own SnapTone format, so you can't directly translate the original NAM input signal specification; at least, Hotone hasn't given any information on that. So you will need to go by ear.

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[20] Bizarrely, even though NAMs are converted to Snaptone before loading, there is no facility to directly load a Snaptone created by an Ampero unit and shared on Hotone’s website. I can only assume this is deliberate to encourage users to buy the more expensive Ampero. 3rd party solutions are currently under development.

It’s unwise to mess with the input level control in Global, unless you are always using NAMs with the same input calibration. Otherwise, whatever level you set will be wrong for other NAMs or the built in models. If you can get the levels right just within the preset it will be easier to manage down the road

The **Gain** knob on the Snaptone module is actually a clean gain before the capture - it's *not* attempting to model the behaviour of the gain control on the real amp - so the first thing to try is turning that up or down as needed

For example, suppose you have loaded a NAM that says "captured with edge of breakup tone”; then you adjust the Gain until you get edge of breakup.

The **Gain** and **Volume** controls are distinctive amongst knobs on the GP200 in that they both range from a -24dB cut (at 0) through unity (50) to a +24dB boost (at 100), and have natural stepping: in other words, +12dB is at 75 - halfway between 0 and +24dB. You wonder why they didn’t just make it display the actual decibel value, but at least this gives you a lot of headroom to compensate for any level matching issues.

SNAPTONE TRICKS

Ok, so you can load an amp NAM into the DST block. So what?

So, two things:

1. You can load the NAM for the clean channel of an amp and the NAM for the dirty channel into the same preset, then by setting a control footswitch to switch both, you can have a footswitch to toggle between them. You now have a preset which is a channel-switching amp.

2. You can move the AMP block *before* the DIST block and load an amp NAM into the DIST block. Why on earth do that? Because you can then set the AMP block to be a clean Fender-style amp (try the JC120 model), and thus use the amp tone stack as if it were a boost pedal. Which is basically how a Dumble ODS works.... and anyway, gives you a whole universe of new boost topologies to play with.

MISCELLANEOUS TIPS

HEADPHONE EQ

When you're using your GP200 with headphones, you may hit the issue that most budget or even mid-price headphones are not linear. They often have a V-shaped/smiley-face curve or something else unnatural. For example, my Sony MDR7506s have a lot of extra high mids (2k-5k) which sounds very harsh.

"Sounds like a job for global EQ", you say, except that global EQ uses up some DSP and may result in the dreaded overload message, and the four parametric bands (no shelving) may not be enough to accurately match the calibration curve needed.

But there is a solution!

- Find the right correction curve for your headphones from the AutoEq website or some other database (or indeed, your own experiments)
- Start a new empty project in your DAW at 44.1kHz sample rate, and drag your favourite speaker IR into a track.
- Apply the EQ correction you determined to the track, and bounce it to a new mono 44.1kHz WAV.
- Trim the WAV down to a sensible length if needed.
- Use that IR in some special "headphone practice" patches.

..and you have perfectly tuned sound just for your headphones, with the benefit that if you create a nice preset, just save a copy and swap the IR out for the original speaker IR and it's good to go.

TROUBLESHOOTING NOISE

High gain amp models will, necessarily, amplify any noise in the incoming signal as well as the actual guitar sound; so even minor noise sources can become very irritating, especially when using headphones. Tracking down sources of noise isn’t always easy: if you use your GP200 at several locations, comparing noise levels purely from memory is subject to human error and the influence of background sounds, not to mention your own stress levels.

Here's a systematic approach that also gives you a numerical measurement you can use to compare setups and scenarios:

Set up a special preset

Create a preset with only two components in it: **Gate 1** in NR, and **Eagle 120+** in **AMP**. Turn the Gain in the amp to 100. Save this as "noise check" for re-use later.

Residual noise check

Now set the Input Level to 0dB in Global Settings and unplug from the input of the GP200; yes, not even a guitar *lead* plugged into INPUT.

Monitoring through headphones, you shouldn't hear anything because of the noise gate. Turn it off temporarily, and you should hear just a soft hiss of white noise, like an untuned radio, not too loud. That's normal, it's called 'residual noise' or 'self noise'.

> If you hear any other sort of noise that changes as you turn the GP200 around or move it to another room, then you have an issue with interference. Turn other devices off (or move to another location) to isolate the culprit.
>
> If you hear a constant 100Hz buzz, then your PSU is the problem.
>
> If you hear non-constant noises that *don't* change as you turn/relocate the GP200, especially pops and clicks that can be triggered by tapping or vibrating it, then you may unfortunately have a manufacturing fault (cold or cracked soldering on the PCB). Get a warranty return if possible.

Activate the gate again and then slowly lower Threshold until you start to hear some sputtering noise breakthrough, just turn it back up so you find the gate stays closed.

You should end up with Threshold at 9 or 10. **If it's higher, then you have noise breakthrough from your PSU or an earth loop.** Stop and try to fix this before you go further.

Quick guitar lead test

Next step: disable the gate and plug in the guitar lead; but just the lead, not the guitar! The level of hiss shouldn't change, but you will hear some hum. hold the barrel of the jack at the other end, not touching the tip but shielding it with the rest of your hand. The hum should almost go away entirely.

If the level of hiss changes, the hum remains loud, or you hear any buzz, clicks, pops that weren't there before... your lead has worn out and the shielding isn't working, so swap it out.



Shielding cable tip with your hand

Testing with a guitar

Once you're happy with the lead, plug it into your guitar and select a humbucker pickup if possible. With the guitar volume rolled all the way down, you should hear no hum, just the same soft hiss as when nothing was plugged in at all. If you hear any buzz, hum, etc then you have dodgy guitar wiring.

Roll the guitar volume up to max, mute the strings with a sock or towel, and turn on the noise gate again. It will very likely stay open. Now go into Global Settings and slowly turn the Input Level down until the gate closes again. This will give you a reading of how much noise your guitar and cable are picking up: the more you have to turn down from 0dB, the more noise pickup you have. With a good cable and a well-wired guitar with hum bucking pickups, you shouldn't need to turn down more than 4dB.

You can now experiment with different cables, PSUs, locations, or turning other appliances off and on, and get a repeatable measurement to home in on what are the big sources of noise and interference in your setup.

The "headphone hum" gotcha

One special case you need to be aware of is if you get a lot of buzzing when you are using *only* headphones, which goes away if you connect the unit to a traditional guitar amp. This is caused by what's known as *floating ground*. Without going into technical details, the simple solution (there are others) is to leave the unit plugged into the amp (switched off) even if you're just practicing through headphones.

MORE PRECISE TUNING

As mentioned earlier, the tuner in the GP200 is very forgiving: a string will be shown as "in tune" (central green indicator lit only) when it is within 5 cents of the target. This is too loose for several applications, where precise tuning is important, and you may find that running a dedicated tuner pedal is necessary.

If you find yourself stuck without an alternative, there is however a trick to get a little more resolution out of the tuner:

Tune the string until it goes green, then use the 7 & 8 footswitches to flick the calibration first to A=441 and then to A=439. If the note stays green on those two settings, then it's very close (within 1.5 cents). If it slides off green at A=441, the note is flat: if it slides off green at A=439, it's sharp.

Don’t forget to put things back to A=440 before you finish, though!

For a more elegant solution, if you are comfortable with installing 3rd party firmware patches, check out https://github.com/tntexplosivesltd/gp200-patcher.

MAXIMUM CTL FOOTSWITCH FLEXIBILITY

Since the GP200 doesn’t have spillover between patches, one workaround is to program footswitches as CTL1 up to CTL8 and use them to toggle effects within a patch. Keeping track of which footswitch controls which effect in each patch is not easy, since the footswitch LEDs cannot be custom coloured.

If you can get access to a dual momentary footswitch to connect to the EXP/FS jack, then you can maximise your CTLs as follows:

1. Configure the two extra footswitches pedals as **BANK-** and **BANK+**

2. Set the **Bank Sel** mode to **Wait**.

3. Now all eight footswitches on the GP200 are free to allocate. I'd suggest keeping the TAP switch as tap tempo/hold for tuner, and set the other seven to CTL 1-7 for use in controlling individual effects in patches. Having seven footswitches means you can give each one a fixed role, e.g.: D is always DLY, C is always MOD, B is always PRE (boost), and so on,

4. In **Bank Sel** mode, when you tap your the BANK+/BANK- switches the screen changes to show the four patches in each bank, and the A-D switches *temporarily* change role to let you pick the patch. So you can switch patches between songs very efficiently.

5. If you're someone who needs to change patches quickly *mid song* then tweak this by making footswitch D **PATCH+**, and arrange patches on the device so that the patches for a given song are in sequence (duplicating if needed). That way you can just go through the song by tapping on D, and you still have 6 effect control footswitches.

6. Note that either way, you are able to put the footswitches that would be most disruptive if accidentally pressed mid-song physically separate from the ones you would operate mid-song, reducing the scope for human error - especially if you have large feet.

APPENDIX 1:
DETAILED EFFECT LIST

The Valeton manual lists all available effects, models and IRs, but there are some typos, errors, and missing information that is useful for a reference.

This appendix attempts to give a comprehensive guide including both information I have discovered through research or experiment and contributions from other group members (for which I am eternally grateful). Marketing fluff from the Valeton manual has been removed (especially since it can be inaccurate when the model does not track the hardware it emulates accurately)

Effects with noticeably high DSP use have been marked with a 🔥 symbol. Too many can cause the “Overload” screen to appear.

Some effects continue to use DSP even when turned off. These have been marked with a 🐷 symbol. Do not leave these effects selected in a block that you aren’t using in a patch, as they waste resources.

A few amp models are suspected to be unstable when Gain is set to 100. These have been marked with 🧨

For **Description**, where the effect is based on a real device, the device name is given in red. This is largely taken from Valeton’s own documentation; but in cases where they were unable (BOSS pedals) or forgot to give trademarked names, and it has been possible to deduce, information has been added. Apparent errors have also been corrected. (For example, ‘Foxy 15TB’ both by name and by the available controls is obviously the AC15 top boost channel and not the AC100.)

For **Routing**, see the section on Stereo.

In **Parameters**, the parameters are listed together with any useful calibration information such as the actual range in milliseconds or dB. Unless otherwise stated, **Volume** is a pure output level adjustment where 25 = -6dB, 50 = unity gain, and 100 = +6dB.

- **Sync** is a common setting for time-based effects, allowing the Time or Rate parameter to be a subdivision of the tapped or configured tempo.

- **Trail** is another common setting for delay and reverb effects: when engaged, the effect will naturally die away if the block is turned off *within* a patch, not when changing patches[21].

- For amp models, '**Standard GBMTPV**' means that the amp offers a standard set of gain, bass, mids, treble, presence, volume which map directly onto the small convenience buttons on the unit. '**Standard GBMTV**' is similar but without presence control.

[21] This is a limitation because of lack of DSP power in the GP200.

FX Name	Type	Routing	Description	Parameters	Notes
	PRE BLOCK				
COMP	Compressor	Mono	Ross Compressor	Sustain Volume	Fixed attack of 25ms and release 250ms
COMP4	Compressor	Mono	Keeley C4 compressor	Sustain Attack (0 = 10ms, 100 = 50ms) Volume Clipping (= input gain)	'Attack' also affects release time. 0 = 100ms release, 100 = 500ms release
S-Comp	Compressor	Mono	Studio-style compressor	Threshold Ratio Volume Attack (0 = 2ms, 100 = 250ms) Release (0 = 2ms, 100 = 1000ms) Blend (0 = all direct signal, 100 = all compressed signal) Tone (tilt EQ centred on 1kHz)	Tone works on blended signal so is active even when Blend is set to 0, enabling use as just an EQ
Micro Boost	Flat Boost	Mono	MXR M133 Micro Amp 2	Gain (20 = unity gain, 100 = +42dB)	Adds both even and odd order harmonics, even at unity gain
AC Boost	Mid Boost with saturation	Mono	Xotic AC Booster	Gain (29 = +18dB @ 800Hz, 100 = +21db) Volume Bass Treble	Boost with 800Hz centre, even and odd order harmonics
B-Boost	Mid Boost with saturation	Mono	Xotic BB Preamp	Gain (20 = +14dB @ 1kHz, 100 = +27dB) Volume Bass Treble	Boost with 1kHz centre, odd order harmonics
P-Boost	Flat Boost with saturation	Mono	Xotic RC Booster	Gain (0 = unity, 20 = +6dB, 100 = +15dB) Volume Bass (0 = -9dB, 100 = +9dB) Treble (= -12dB, 100 = +12dB)	Very similar to FAT BB in frequency response, but with odd-order harmonics from saturation
14 Boost	Tight Boost	Mono	Fortin Grind	Gain (20 = tone shape only, 100 = +33dB)	
FAT BB	Flat Boost	Mono	BOSS FA-1 FET Amplifier	Bass (0 = -12dB, 100 = +6dB) Treble (0 = -12dB, 100 = +12dB) Volume (15 = unity, 20 = +3dB) Low Cut on/off (-6dB/oct at 200Hz)	Volume' here is more like 'Gain'. Almost no distortion/saturation
Boost	Flat Boost	Mono	Xotic EP Booster	Gain (0 = unity, 100 = +15dB) +3dB on/off Bright on/off (off rolls off above 2kHz)	Subtle saturation effect with odd-order and some even-order harmonics
OD 9	Overdrive	Mono	Ibanez TS9 Tube Screamer	*See entry in DST*	*See entry in DST*
Yellow OD	Overdrive	Mono	BOSS OD-1	*See entry in DST*	*See entry in DST*
Penesas 🔥	Overdrive	Mono	Klon Centaur	*See entry in DST*	*See entry in DST*
Super OD	Overdrive	Mono	BOSS SD-1	*See entry in DST*	*See entry in DST*
Blues OD	Overdrive	Mono	BOSS BD-2	*See entry in DST*	*See entry in DST*
AC Refiner	Acoustic	Mono	Similar to the TC Electronic BodyRez	Shape	

FX Name	Type	Routing	Description	Parameters	Notes
AC Sim	Acoustic Simulator	Mono	BOSS AC-2 Acoustic Sim	Body Top Volume Mode (Standard, Jumbo, Enhanced, Piezo)	
T-Wah	Auto Wah (dynamic)	Mono	Auto wah: envelope-following filter	Sens Range Q Mix Mode (Guitar, Bass)	
A-WAH	Auto Wah (periodic)	Mono	Auto Wah - filter frequency driven by LFO	Depth Rate Volume Low High Q Sync on/off	With adjustment can yield a sound reminiscent of harmonic trem - but isn't the same thing.
Step Filter	Filter Effect	Mono	Filter driven by 4-step sequencer	Step 1 (centre frequency) Step 2 (centre frequency) Step 3 (centre frequency) Step 4 (centre frequency) Rate Sync on/off	
OCTA	Octaver	Mono	Simple octave effect	Oct 1 Oct 2 Dry	
Pitch	Pitch Shift	Mono	Dual fixed pitch shifter	Hi Pitch (0 to +24 semitones) Low Pitch (-24 to 0 semitones) Dry (dry signal level) Low Volume High Volume	Sounds terrible Wet signal has 17ms latency relative to dry (Thanks to Tony Loyo for measurements)
P-Bend	Pitch Shift	Mono	Pitch shifter suitable for expression pedal control	Hi Pitch (0 to +12 semitones) Low Pitch (-12 to 0 semitones) Wet (wet signal level) Dry (dry signal level) Range (0 = shifted to low pitch, 100 = shifted to high pitch)	Map expression pedal to 'Range'. Sounds terrible Wet signal has 24ms latency relative to dry (Thanks to Tony Loyo for measurements)
Hammy	Pitch Shift	Mono	Digitech Whammy	Range Harmony on/off (on = also output dry signal) Volume Position	Map expression pedal to 'Position. Wet signal has 4ms latency relative to dry (Thanks to Tony Loyo for measurements)
Harmony1	Pitch Shift	Special	Diatonic pitch shift, one harmony voice	Mix Key Mode (scale type) Interval Smooth Mode on/off	Passes L and R direct signal, applies harmony to L only
Harmony2	Pitch Shift	Special	Diatonic pitch shift, two harmony voices	Mix Key Mode (scale type) Interval 1 Interval 2 Smooth Mode on/off	Passes L and R direct signal, applies harmony to L only
Ring Mod	Special	Mono	Ring modulator	Mix Freq Fine (frequency fine tune) Tone	

FX Name	Type	Routing	Description	Parameters	Notes
Saturate	Special	Mono	Tape Saturation	Saturation Mix Volume High Cut	Adds odd-order harmonics even with Saturation at 0
Auto Swell	Special	Mono	Similar to Boss Slow Gear, but not a clone	Attack (80 - 4000ms) Curve (Line, Exp, Log)	
Hold	Special	Mono	Freezes the sound at the moment of activation as a short loop	Volume Activate	Percussive elements are preserved
Freeze 🔥	Special	Mono	Freezes the sound at activation as a tone	Volume Attack Release Activate	Tone and pitch are preserved, transients smoothed. High DSP usage
	WAH BLOCK				
V-Wah	Wah	Mono	VOX V846	Range Q Volume	
C-Wah	Wah	Mono	Dunlop CryBaby	Range Q Volume	
P-Wah	Wah	Mono	J Petrucci's custom CryBaby	Range Q Volume EQ on/off	
S-Wah	Wah	Mono	Hotone Soul Press (wah mode)	Range Q Volume	
B-Wah	Wah	Mono	Hotone Bass Press (wah mode)	Range Q Volume	
Hammy	Pitch Shift	Mono	Digitech Whammy	*See entry in PRE*	*See entry in PRE*
	DST BLOCK				
Green OD	Overdrive	Mono	Ibanez TS808	Gain Tone Volume	
OD 9	Overdrive	Mono	Ibanez TS9 Tube Screamer	Gain Tone Volume	
Yellow OD	Overdrive	Mono	BOSS OD-1	Gain Volume	
Penesas 🔥	Overdrive	Mono	Klon Centaur	Gain Tone Volume	
Swarm	Overdrive	Mono	Providence SOV-2 Stampede	Gain Tone Volume	
Super OD	Overdrive	Mono	BOSS SD-1 (MIJ)	Gain Tone Volume	
Scream OD	Overdrive	Mono	Hotone original Tube Screamer style drive	Gain Tone Volume Fat on/off Air on/off	Fat and Air represent popular capacitor mods on the TS

FX Name	Type	Routing	Description	Parameters	Notes
Blues OD	Overdrive	Mono	BOSS BD-2	Gain Tone Volume	
Force	Overdrive	Mono	Fulltone OCD	Gain Tone Volume Mode (HP, LP)	
Tube	Overdrive	Mono	B. K. Butler Tube Driver	Gain Volume Bass Treble	*New in firmware 1.8*
Blues Master	Overdrive	Mono	Marshall Bluesbreaker	Gain Volume Bass Middle Treble	
Master OD	Overdrive	Mono	Based on Marshall Drivemaster	Gain Tone Volume	
TaiChi OD	Overdrive	Mono	Hermida Zendrive	Gain Tone Volume Voice (tunes upper frequencies)	
Timmy OD	Overdrive	Mono	Paul Cochrane Timmy	Gain Volume Bass Middle Treble Mode (I, II, III)	
Precise OD	Overdrive	Mono	Horizon Devices Precision Drive	Gain Tone Volume Attack (1 - 6) Gate	
Empire OD 🐷	Overdrive	Mono	AnalogMan Prince of Tone	Gain Tone Volume Mode	Why didn't they put this in the PRE block too so we could make a KoT?
Lazaro	Fuzz	Mono	ElectroHarmonix Big Muff Pi	Sustain (fuzz) Tone Volume	
Red Haze	Fuzz	Mono	Dallas Arbiter Fuzz Face	Fuzz Volume	
Sora Fuzz	Fuzz	Mono	Sola Sound Tone Bender	Fuzz Volume	
Plustortion	Distortion	Mono	MXR M104 Distortion+	Gain Volume	
SM Dist	Distortion	Mono	BOSS DS-1 (MIJ)	Gain Tone Volume	
Darktale	Distortion	Mono	ProCo The Rat, LM308 version	Gain Filter (turn up for less highs) Volume	
Chief	Distortion	Mono	Marshall Guv'nor	Gain Volume Bass Middle Treble	

FX Name	Type	Routing	Description	Parameters	Notes
Master Dist	Distortion	Mono	Marshall Shredmaster	Gain Volume Bass Contour Treble	
La Charger	Distortion	Mono	MI Audio Crunch Box	Gain Tone Volume	
Revolt 🐷	Distortion	Mono	Suhr Riot	Gain Tone Volume Mode (I, II, III)	
Flagman Dist 🐷	Distortion	Mono	Friedman BE-OD	Gain Volume Bass Treble Presence Tight	
Flex OD	Bass Overdrive	Mono	Hotone-designed original bass overdrive	Gain Tone Volume Mode (Normal, Scoop, Edge) Blend	Appears on Ampero as “Solid Steel”
Bass OD	Bass Overdrive	Mono	BOSS ODB-3 Bass Overdrive	Gain Blend Volume Bass Treble	
Black Bass	Bass Preamp	Mono	Darkglass Microtubes B7K	Gain Blend (0 = dry, 100 = overdrive) Volume Low (-10dB to +10dB @100Hz) Lo-mid (-6dB to +6db @ 1.5kHz) Hi-Mid (-12dB to +12dB @3.3kHz) Treble (-20dB to +20dB @10kHz) Attack (Cut, Boost, Flat)	Low shelf starts @ 800Hz High shelf starts @ 400Hz “Attack” only affects overdrive sound, not dry EQ affects blended sound
Bass Hammer 🐷	Bass Preamp	Mono	Aguilar Tone Hammer Front	Gain Master (= Volume) Bass (-18dB to +18dB @ 40Hz) Middle (-18dB to +18dB) Mid Freq (0 = 150Hz, 100 = 900Hz) Treble (-24dB to +24dB @ 4kHz) Drive on/off	Sharp rolloff above 10k
Micro Boost	Flat Boost	Mono	MXR M133 Micro Amp 2	*See entry in PRE*	*See entry in PRE*
AC Boost	Mid Boost	Mono	Xotic AC Booster	*See entry in PRE*	*See entry in PRE*
B-Boost	Mid Boost	Mono	Xotic BB Preamp	*See entry in PRE*	*See entry in PRE*
P-Boost	Flat Boost	Mono	Xotic RC Booster	*See entry in PRE*	*See entry in PRE*
14 Boost	Tight Boost	Mono	Fortin Grind	*See entry in PRE*	*See entry in PRE*
FAT BB	Flat Boost	Mono	BOSS FA-1 FET Amplifier	*See entry in PRE*	*See entry in PRE*
Boost	Flat Boost	Mono	Xotic EP Booster	*See entry in PRE*	*See entry in PRE*
	AMP BLOCK				

FX Name	Type	Routing	Description	Parameters	Notes
Tweedy	Clean Amp	Mono	Fender Tweed Deluxe 5E3	Gain Tone Volume	
Bellman 59N	Clean Amp	Mono	Fender 59 Bassman (5F6A), Normal input	Standard GBMTPV	
Bellman 59B 🧨	Clean Amp	Mono	Fender 59 Bassman (5F6A), Bright input	Standard GBMTPV	Bright effect lessens with increased Gain (correctly)
Dark Twin 🧨	Clean Amp	Mono	Fender 65 (Blackface) Twin Reverb	Standard GBMTV Bright on/off	
Dark DLX	Clean Amp	Mono	Fender 65 (Blackface) Deluxe Reverb	Gain Volume Bass Treble	
Dark Vibra 🧨	Clean Amp	Mono	Fender 63 (Brownface) Vibroverb 6G16	Gain Volume Bass Treble Bright on/off	Bright control switches between normal and bright inputs
Silver Twin 🧨	Clean Amp	Mono	Fender 70s (Silverface) Twin Reverb	Standard GBMTV (Valeton's manual is wrong)	Bright cap is permanently on: effect reduces at higher Gain
SUPDual CL	Clean Amp	Mono	Supro Dual-Tone 1624T Channel 1 (clean)	Gain (see note) Volume Tone	There is a strange glitch: Gain actually drops between 12 and 13
SUPDual OD	Driven Amp	Mono	Supro Dual-Tone 1624T Channels 1&2 jumpered	Gain 1 Gain 2 Tone 1 Tone 2 Volume	
Foxy 15TB	Clean Amp	Mono	VOX AC15 Top Boost channel	Gain Tone Cut Volume Bass Treble	
Foxy 30N	Clean Amp	Mono	VOX AC30 Normal channel	Gain Volume Tone Cut Bright on/off	
Foxy 30TB	Driven Amp	Mono	VOX AC30 Top Boost channel	Gain Tone Cut Volume Bass Treble Char (Cool, Hot)	
J-120 CL 🧨	Clean Amp	Mono	Roland JC120 Jazz Chorus	Gain Bright on/off Bass Middle Treble	No Volume control as this is a perfectly clean preamp: set Gain to desired level. Idealised Fender style passive tone stack: • Bass 0 Mid 100 Treble 0 is flat. • Mid scoop @ 600Hz

FX Name	Type	Routing	Description	Parameters	Notes
Match CL	Clean Amp	Mono	Matchless Chieftain 212, lower gain input	Standard GBMTPV	Presence here is 'Brilliance' on the real amp
Match OD	Driven Amp	Mono	Matchless Chieftain 212, higher gain input	Standard GBMTPV	Presence here is 'Brilliance' on the real amp
L-Star CL	Clean Amp	Mono	Mesa/Boogie Lone Star, channel 1	Standard GBMTPV	
L-Star OD	Driven Amp	Mono	Mesa/Boogie Lone Star, channel 2	Input (additional input gain) Standard GBMTPV	
BogSV CL	Clean Amp	Mono	Bognor Shiva 20th Anniversary, Channel 1	Gain Presence Volume Bass Treble Bright on/off	
Bog SV OD	Driven Amp	Mono	Bognor Shiva 20th Anniversary, Channel 2	Standard GBMTPV	
Bog BlueV	Driven Amp	Mono	Bognor XTC, blue channel, Structure switch on Vintage	Standard GBMTPV	
Bog BlueM	Driven Amp	Mono	Bognor XTC, blue channel, Structure switch on Modern	Standard GBMTPV	
Bog RedV	High Gain Amp	Mono	Bognor XTC, red channel, Structure switch on Vintage	Standard GBMTPV	
Bog RedM	High Gain Amp	Mono	Bognor XTC, red channel, Structure switch on Modern	Standard GBMTPV	
Z38 CL	Clean Amp	Mono	Dr. Z Maz 38 Sr, lower gain input	Standard GBMTV Tone Cut	
Z38 OD	Driven Amp	Mono	Dr. Z Maz 38 Sr, higher gain input	Standard GBMTV Tone Cut	
Knights CL	Clean Amp	Mono	Grindrod Pendragon PG20C, Normal Channel, Bright Off	Standard GBMTV	
Knights CL+	Clean Amp	Mono	Grindrod Pendragon PG20C, Normal Channel, Bright On	Standard GBMTV	
Knights OD	Driven Amp	Mono	Grindrod Pendragon PG20C, Drive Channel	Standard GBMTV	
Bad-KT CL	Clean Amp	Mono	Bad Cat Hot Cat 30, Clean channel	Gain Presence Volume	
Bad-KT OD	Driven Amp	Mono	Bad Cat Hot Cat 30, Overdrive channel	Gain Presence Volume Edge Bass Treble	
Solo100 CL	Clean Amp	Mono	Soldano SLO100, clean channel	Standard GBMTPV	
Solo100 OD	Driven Amp	Mono	Soldano SLO100, crunch channel	Standard GBMTPV	

FX Name	Type	Routing	Description	Parameters	Notes
Solo100 LD	High Gain Amp	Mono	Soldano SLO100, overdrive channel	Standard GBMTPV	
UK 45	Driven Amp	Mono	Marshall JTM45, normal input	Standard GBMTPV	
UK 45+ 🔥	Driven Amp	Mono	Marshall JTM45, bright input	Standard GBMTPV	
UK 45JP 🔥	Driven Amp	Mono	Marshall JTM45, jumpered inputs	Gain 1 (Normal input) Presence Volume Bass Middle Treble Gain 2 (Bright input)	
UK 50	Driven Amp	Mono	Marshall JMP50, normal input	Standard GBMTPV	
UK 50+ 🔥	Driven Amp	Mono	Marshall JMP50, bright input	Standard GBMTPV	
UK 50JP 🔥	Driven Amp	Mono	Marshall JMP50, jumpered inputs	Gain 1 (Normal input) Presence Volume Bass Middle Treble Gain 2 (Bright input)	
UK SLP	Driven Amp	Mono	Marshall 1959HW Super Lead, normal input	Standard GBMTPV	
UK 800	Driven Amp	Mono	Marshall JCM800	Standard GBMTPV	
UK 900	High Gain Amp	Mono	Marshall JCM900	Standard GBMTPV	
Flagman 1	Driven Amp	Mono	Friedman BE-100, BE channel, C45 On	Standard GBMTPV	Thanks to input from Jon Marsh
Flagman 2	Driven Amp	Mono	Friedman BE-100, BE channel, C45 Off	Standard GBMTPV	Thanks to input from Jon Marsh
Flagman+ 1	High Gain Amp	Mono	Friedman BE-100, HBE channel, C45 Off	Standard GBMTPV	Thanks to input from Jon Marsh This model has low output relative to the other amp models, raise Volume to compensate
Flagman+ 2	High Gain Amp	Mono	Friedman BE-100, HBE channel, C45 On	Standard GBMTPV	Thanks to input from Jon Marsh
Mess2C+ 1	Driven Amp	Mono	Mesa/Boogie Mark II C+	Standard GBMTPV	Pull switch settings subject to confirmation
Mess2C+ 2	Driven Amp	Mono	Mesa/Boogie Mark II C+, Deep on	Standard GBMTPV	Pull switch settings subject to confirmation
Mess2C+ 3	Driven Amp	Mono	Mesa/Boogie Mark II C+, Treble shift on	Standard GBMTPV	Pull switch settings subject to confirmation
Mess4 LD	High Gain Amp	Mono	Mesa/Boogie Mark IV, Lead channel (Fat on, Harmonics Voicing)	Standard GBMTPV	Pull switch settings subject to confirmation
Mess4 LD 2	High Gain Amp	Mono	Mesa/Boogie Mark IV, Lead channel (Fat On, Mid Gain Voicing)	Standard GBMTPV	Pull switch settings subject to confirmation

FX Name	Type	Routing	Description	Parameters	Notes
Mess4 LD 3	High Gain Amp	Mono	Mesa/Boogie Mark IV, Lead channel (Fat Off, Mid Gain Voicing)	Standard GBMTPV	Pull switch settings subject to confirmation
Mess DualV	High Gain Amp	Mono	Mesa/Boogie Dual Rectifier, channel 3, Vintage Mode	Standard GBMTPV	details based on Ampero II
Mess DualM	High Gain Amp	Mono	Mesa/Boogie Dual Rectifier, channel 3, Modern Mode	Standard GBMTPV	details based on Ampero II
Juice30 OD	Driven Amp	Mono	Orange AD30, dirty channel	Standard GBMTV	
Juice R100	High Gain Amp	Mono	Orange Rockerverb 100, dirty channel	Standard GBMTV Bright on/off	
EV 51	High Gain Amp	Mono	Peavey 5150, Lead channel	Standard GBMTPV	
Eagle 120	High Gain Amp	Mono	ENGL Savage 120, channel 4, contour off	Standard GBMTPV	details based on Ampero II
Eagle 120+	High Gain Amp	Mono	ENGL Savage 120, channel 4, contour on	Standard GBMTPV	details based on Ampero II
Power LD	High Gain Amp	Mono	ENGL Powerball II E645/2, channel 4	Standard GBMTPV	
Dizz VH	High Gain Amp	Mono	Diezel VH4 Blue Panel, channel 3	Standard GBMTPV	details based on Ampero II
Dizz VH S	High Gain Amp	Mono	Diezel VH4 Silver Panel, channel 3	Standard GBMTPV	details based on Ampero II
Dizz VH+	High Gain Amp	Mono	Diezel VH4 Blue Panel, channel 4	Standard GBMTPV	details based on Ampero II
Dizz VH+ S	High Gain Amp	Mono	Diezel VH4 Silver Panel, channel 4	Standard GBMTPV	details based on Ampero II
Classic Bass	Bass Amp	Mono	Ampeg SVT	Gain Midrange (220, 450, 800, 1.6k, 3k) Volume Bass Midrange Treble	Not really an amp model - has no valve colouration
Foxy Bass	Bass Amp	Mono	VOX AC100	Volume Bass Treble	Not really an amp model - has no valve colouration
Mess Bass	Bass Amp	Mono	Mesa/Boogie Bass 400	Standard GBMTV	
Mini Bass	Bass Amp	Mono	Ampeg B-15	Volume Bass Treble	
Bass Pre	Bass Amp	Mono	Alembic F-2B	Volume Bright on/off (off rolls off above 4kHz, on is flat) Bass Middle (centre 400Hz) Treble	Traditional Fender style passive tone stack: • Bass 0 Mid 100 Treble 0 is flat. • Bass 50 Mid 50 Treble 50 gives a -12dB cut at 400Hz

FX Name	Type	Routing	Description	Parameters	Notes
AC Pre	Acoustic Preamp	Mono	AER Colourizer 2 acoustic preamp	Volume Tone (0 = flat, 100 = +10dB @ 40Hz and 10kHz) Balance (0 = Bass only, 100 = treble only) EQ Freq (0 = 90Hz to 100 = 1.6kHz, 50 = 190Hz) EQ Q (0 = narrow, 100 = wide) EQ Gain (-12dB to +12dB: 50 = flat)	Turning Tone up from 0 adds a "smiley face" EQ, when Balance is 50. Adjusting Balance then dampens down either the bass or the treble boost
AC Pre 2	Acoustic Preamp	Mono	AER Colourizer 2 acoustic preamp with alternate EQ range	Volume Tone (0 = flat, 100 = +10dB @ 40Hz and 10kHz) Balance (0 = Bass only, 100 = treble only) EQ Freq (0 = 850Hz to 100 = 13kHz, 50 = 1.6kHz) EQ Q (0 = narrow, 100 = wide) EQ Gain (-15dB to +15dB: 50 = flat)	*See entry for AC Pre above*
	NR BLOCK				
Gate 1	Gate	Mono	ISP Decimator	Threshold	Has no advantages over Gate 2
Gate 2	Gate	Mono	Tunable noise gate	Threshold Attack (0 = 2ms , 100 = 20ms) Release (0 = 12ms, 100 = 250ms)	Has no advantages over Gate 3
Gate 3	Gate	Mono	Tunable noise gate based on downward Expansion	Threshold Ratio Attack (1 - 500 ms) Release (10 - 10,000ms) Hold (0 - 1000ms) Hysteresis (-10dB to 0dB)	Attack appears to do nothing, set to 1ms
Auto Swell	Special	Mono		*See entry in PRE*	*See entry in PRE*
	See separate appendix 2 for CAB details				
	EQ BLOCK				
Guitar EQ 1	Guitar EQ	Mono	Graphic EQ with guitar-focused bands (narrow spread)	125Hz 400Hz 800Hz 1.6kHz 4kHz Volume	All bands: -50 = -12dB -20 = -6dB 0 = flat 20 = +6dB 50 = +12dB
Guitar EQ 2	Guitar EQ	Mono	Graphic EQ with guitar-focused bands (wide spread)	100Hz (shelf) 500Hz 1kHz 3kHz (actually 4kHz) 6kHz (shelf) Volume	All bands: -50 = -12dB -20 = -6dB 0 = flat 20 = +6dB 50 = +12dB
Bass EQ 1	Bass EQ	Mono	Graphic EQ with bass-focused bands	33Hz 150Hz 600Hz 2kHz 8kHz (actually 4.5kHz) Volume	All bands: -50 = -12dB -20 = -6dB 0 = flat 20 = +6dB 50 = +12dB

FX Name	Type	Routing	Description	Parameters	Notes
Bass EQ 2	Bass EQ	Mono	Graphic EQ with bass-focused bands	50Hz 120Hz 400Hz 800Hz 4.5kHz Volume	All bands: -50 = -12dB -20 = -6dB 0 = flat 20 = +6dB 50 = +12dB
Mess EQ	Guitar EQ	Mono	Mesa/Boogie graphic EQ	80Hz 240Hz (actually 350Hz) 750Hz 2.2kHz 6.6kHz	
Hyper EQ	Generic Graphic EQ	Mono	10-band graphic with bands evenly spaced in octaves	31Hz 63Hz 125Hz 250Hz 500Hz 1kHz 2kHz 4kHz 8kHz 16kHz Volume	All bands are -12dB to +12dB
	MOD BLOCK				
A-Chorus	Chorus	Mono->Stereo	Arion SCH-1	Depth Rate (0.1 - 10Hz) Tone (affects > 1kHz) Sync	New in firmware 1.8
G-Chorus	Chorus	Mono->Stereo	BOSS CE-1 (chorus mode)	Depth Rate (0.1 - 10 Hz) Volume Sync on/off	
C-Chorus	Chorus	Mono->Stereo	BOSS DC-2 Dimension C	Mode (1, 2, 3, 4)	
B-Chorus	Chorus (Bass)	Mono->Stereo	BOSS CE-2B	Depth Rate (0.1 - 10 Hz) Volume Sync on/off	
M-Chorus	Chorus	Mono->Stereo	Very similar to Free the Tone TA-2H TRI AVATAR	Mix Rate (0.1 - 10 Hz) Filter Depth L Depth C Depth R Sync on/off	
Jet	Flanger	Mono->Stereo	Generic Flanger	Depth Rate (0.1 - 10 Hz) PreDelay (centre delay time) Feedback Sync on/off	
B-Jet	Flanger (Bass)	Mono->Stereo	Generic Flanger applied to upper frequencies only	Depth Rate (0.1 - 10 Hz) PreDelay (centre delay time) Feedback Sync on/off	
N-Jet	Flanger	Mono->Stereo	Generic Flanger with negative feedback	Depth Rate (0.1 - 10 Hz) PreDelay (centre delay time) Feedback Sync on/off	

FX Name	Type	Routing	Description	Parameters	Notes
Trem Jet	Flanger/Tremolo	Mono->Stereo	Flanger and Tremolo effect combined	Flg Depth Flg Rate (0.1 - 10 Hz) Feedback Trm Depth Trm Rate (0.1 - 10 Hz) Flg Sync on/off Trm Sync on/off	
V-Roto	Vibrato	Mono->Stereo	BOSS VB-2	Depth Rate (0.1 - 10 Hz) Sync on/off	
G-Roto	Vibrato	Mono->Stereo	BOSS CE-1 (vibrato mode)	Depth Rate (0.1 - 10 Hz) Volume Sync on/off	
Vibrato	Vibrato	Mono->Stereo	Generic vibrato	Depth Rate (0.1 - 10 Hz) Volume Sync on/off	
Vibrato T 🐷	Vibrato	Mono->Stereo	Level-sensitive vibrato: Vibrato is less for loud playing and increases as note fades	Sense Rate (0.1 - 10 Hz) Output (= Volume) Sync on/off	Increasing "Sense" means *less* vibrato
O-Phase	Phaser	Mono->Stereo	MXR M101 Phase 90	Rate (0.1 - 10 Hz) Sync on/off	
G-Phase	Phaser	Mono->Stereo	BOSS PH-1 Phaser	Depth Rate (0.1 - 10 Hz) Sync on/off	
S-Phase	Phaser	Mono->Stereo	ElectroHarmonix Small Stone	Color (Warm, Sharp) Rate (0.1 - 10 Hz) Sync on/off	
Pan Phase	Phaser	Mono->Stereo	Combined Phaser and Autopan/tremolo	Phs Depth Phs Rate (0.1 - 10 Hz) Pan Depth Pan Rate (0.1 - 10 Hz) Phs Sync on/off Pan Sync on/off	Pan will sound like Tremolo in a mono setup
M-Vibe	Phaser	Mono->Stereo	Voodoo Lab Micro Vibe (Univibe clone)	Depth Rate (0.1 - 10 Hz) Sync on/off	
Vibe	Phaser	Mono->Stereo	Shin-Ei Uni-Vibe	Depth Rate (0.1 - 10 Hz) Volume Mode (Chorus, Vibrato) Sync on/off	
O-Trem	Tremolo	Mono->Stereo	Demeter TRM-1 Tremulator optical trem	Depth Rate (0.1 - 10 Hz) Sync on/off	
Sine Trem	Tremolo	Mono->Stereo	Pure Tremolo with sine waveform	Depth Rate (0.1 - 10 Hz) Volume Sync on/off	
Triangle Trem	Tremolo	Mono->Stereo	Pure Tremolo with triangle waveform	Depth Rate (0.1 - 10 Hz) Volume Sync on/off	

FX Name	Type	Routing	Description	Parameters	Notes
Bias Trem	Tremolo	Mono->Stereo	Approximation of Princeton-style bias trem	Depth Rate (0.1 - 10 Hz) Volume Sync on/off Bias (changes waveform)	
Detune	Pitch Shift	Special	Simple detuner effect	Detune (-50 - +50 cents) Wet Dry	Passes L dry to both L & R, detunes L & R separately
Bit Smash	Special	Mono	Digital lo-fi effect	Mix Krush (sample rate reduction) Bit (bit depth reduction) Hi Cut Lo Cut	
Auto Swell	Special	Mono	*See entry in PRE*	*See entry in PRE*	*See entry in PRE*
Hold	Special	Mono	*See entry in PRE*	*See entry in PRE*	*See entry in PRE*
Freeze	Special	Mono	*See entry in PRE*	*See entry in PRE*	*See entry in PRE*
	DLY BLOCK				
BBD Delay S	Delay	True Stereo	Stereo BBD delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Time R% Spread (-100 = L/R swapped, 0 = delays panned centre, +100 = default) Level (=Volume) Sync on/off Trail on/off	Use negative 'Spread' to have longer delay in R channel
Digital Delay S	Delay	True Stereo	Stereo digital delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Time R% Spread (-100 = L/R swapped, 0 = delays panned centre, +100 = default) Level (=Volume) Sync on/off Trail on/off	Use negative 'Spread' to have longer delay in R channel
Tape Delay S	Delay	True Stereo	Stereo tape delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Time R% Spread (-100 = L/R swapped, 0 = delays panned centre, +100 = default) Wow & Flutter Age (Past, Present, Future) Drive Level (=Volume) Sync on/off Trail on/off	Use negative 'Spread' to have longer delay in R channel
Ambience 1	Delay	Semi-stereo	Multi-tap modulated delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Level (=Volume) Mod Tone Sync on/off Trail on/off	

FX Name	Type	Routing	Description	Parameters	Notes
Ambience 2	Delay	Semi-stereo	Multi-tap modulated delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Level (=Volume) Mod Tone Sync on/off Trail on/off	Tonal variation on Ambience 1
Pure	Delay	True Stereo	Pure digital delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sync on/off Trail on/off	
Analog	Delay	True Stereo	Analogue (BBD) delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sync on/off Trail on/off	
Tape	Delay	True Stereo	Tape delay (transistor)	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sync on/off Trail on/off	
Ping Pong	Delay	Special	Ping pong echo (repeats alternate L and R)	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sync on/off Trail on/off	Direct signal swaps L & R channels. All delay is originated from the L input signal
Slapback	Delay	True Stereo	Short delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 300ms) Sync on/off Trail on/off	Really only exists to make the adjustment range more comfortable for short delays
Sweep Echo	Delay	True Stereo	Filter sweep over delay repeats	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sweep Depth Sweep Rate Swp Sync on/off Time Sync on/off Trail on/off	
Ring Echo	Delay	True Stereo	Ring modulator on delay repeats	Dly Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Ring Mix (0 = delay clean, 100 = delay ring) Freq Tone Sync on/off Trail on/off	
Tube	Delay	True Stereo	Tube-driven tape delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sync on/off Trail on/off	
M-Echo	Delay	Mono	Multi-head tape delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Tone Mode (1 - 12) Sync on/off Trail on/off	'Mode' chooses a multi-tap variation, like the old Roland Space Echo

FX Name	Type	Routing	Description	Parameters	Notes
Sweet Echo	Delay	Mono	BOSS DM-2 Delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sync on/off Trail on/off	
999 Echo	Delay	Mono	Maxon AD999 Analog Delay	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Sync on/off Trail on/off	Corrected with tip from Ampero
Vintage Rack	Delay	True Stereo	1980 rack delay with reduced fidelity feedback	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Mod Tone Sync on/off Trail on/off	
Lofi Echo	Delay	True Stereo	Delay with bit crusher on repeats	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Crush (lower sample rate) Bit (lower bit depth) Sync on/off Trail on/off	
Rev Echo	Delay	Special	Delay with reversed repeats	Mix (0 = dry, 100 = wet) Feedback Time (20 - 4000ms) Volume Sync on/off Trail on/off	Passes L and R direct signal, applies reverse to L+R (mono)
Dual Echo	Delay	True Stereo	Completely independent L & R delay lines	Mix A (0 = dry, 100 = wet) FB A Time A (20 - 4000ms) Mix B (0 = dry, 100 = wet) FB B Time B (20 - 4000ms) A Sync on/off B Sync on/off Trail on/off	A is L and B is R
Ice Delay 🐷	Delay	Semi-stereo	Stereo delay with pitch shifted slices	Mix (0 = dry, 100 = wet) Time (20 - 2000ms) Feedback Mod Tone Pitch (-12 to +12 semitones, plus -50, -20, 0, +20, +50 cents) Slice (Short, Medium, Long) Direction (-100 to 100) Blend (blend between normal and pitch shifted repeats) Smooth (smooths attack of repeats) Level (=Volume) Sync on/off Trail on/off	
Broken Delay 🐷	Delay	True Stereo	Stereo Delay with glitching effect on repeats	Mix (0 = dry, 100 = wet) Time (20 - 4000ms) Feedback Latch (glitch length, 0 = no glitch) Cut (glitch speed, 0 = no glitch) Sync on/off Trail on/off	Feedback sets the level of the first repeat: Feedback = 0 results in no delay signal New in firmware 1.8

FX Name	Type	Routing	Description	Parameters	Notes
	RVB BLOCK				
Room	Reverb	Semi-stereo	Digital room	Mix (0 = dry, 100 = wet) Pre Delay (0 - 100ms) Decay Trail on/off	
Hall	Reverb	Semi-stereo	Digital hall	Mix (0 = dry, 100 = wet) Pre Delay (0 - 100ms) Decay Trail on/off	
Church	Reverb	Semi-stereo	Digital large hall	Mix (0 = dry, 100 = wet) Pre Delay (0 - 100ms) Decay Trail on/off	
Plate	Reverb	Semi-stereo	Plate reverb	Mix (0 = dry, 100 = wet) Decay High Damp Trail on/off	
Spring	Reverb	Semi-stereo	Generic spring	Mix (0 = dry, 100 = wet) Decay Tone Trail on/off	
Tube Spring 🔥🐷	Reverb	Semi-stereo	Spring with tube drive and recovery	Mix (0 = dry, 100 = wet) Pre Delay (0 - 300ms) Decay Low Damp Hi Damp Mod Trail on/off	
Amp Spring 🔥🐷	Reverb	Semi-stereo	Combo amp spring	Mix (0 = dry, 100 = wet) Pre Delay (0 - 300ms) Decay Low Damp Hi Damp Mod Trail on/off	
Studio	Reverb	Semi-stereo	Digital reverb with modulation	Mix (0 = dry, 100 = wet) Pre Delay (0 - 300ms) Decay Low Damp Hi Damp Mod Trail on/off	
Club	Reverb	Semi-stereo	Digital reverb with modulation	Mix (0 = dry, 100 = wet) Pre Delay (0 - 300ms) Decay Low Damp Hi Damp Mod Trail on/off	
Concert	Reverb	Semi-stereo	Digital reverb with modulation	Mix (0 = dry, 100 = wet) Pre Delay (0 - 300ms) Decay Low Damp Hi Damp Mod Trail on/off	

FX Name	Type	Routing	Description	Parameters	Notes
Arena	Reverb	Semi-stereo	Digital reverb with modulation	Mix (0 = dry, 100 = wet) Pre Delay (0 - 300ms) Decay Low Damp Hi Damp Mod Trail on/off	
N-Star	Reverb	Semi-stereo	Reverb with light chorus on trails	Mix (0 = dry, 100 = wet) Decay Trail on/off	
Deepsea	Reverb	Semi-stereo	Reverb with slow vibrato/ flange on trails	Mix (0 = dry, 100 = wet) Decay Trail on/off	
Sweet space	Reverb	Semi-stereo	Reverb with gentle modulation	Mix (0 = dry, 100 = wet) Pre Delay (0 - 100ms) Decay Low End (-50 to +50) Hi End (-50 to +50) Trail on/off	
Shimmer	Reverb	Semi-stereo	Reverb with trails pitch-shifted up	Mix (0 = dry, 100 = wet) Pre Delay (0 - 100ms) Decay Low End (-50 to +50) Hi End (-50 to +50) Trail on/off	

APPENDIX 2:
DETAILED CAB LIST

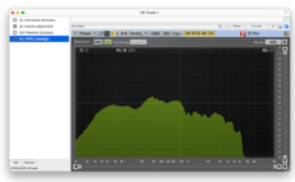
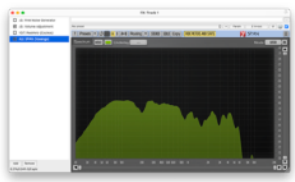
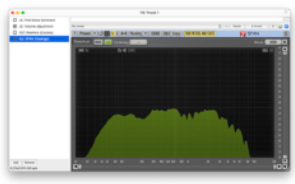
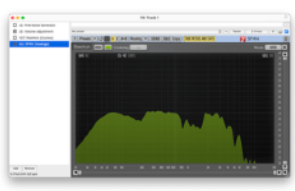
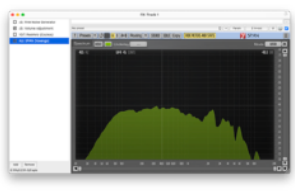
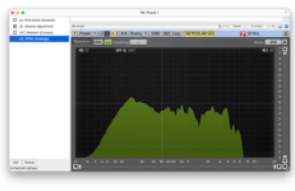
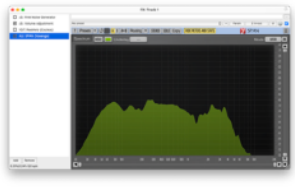
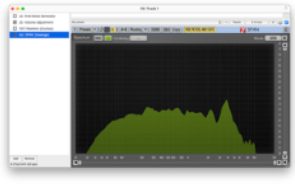
This table shows details of the built-in cabinet IRs in the GP200 firmware. The special acoustic simulation IRs are not shown. A point of note is that the GP200's IRs appear to be a completely different set from those in the Ampero 2 range, in contrast to the other effects which are a subset of the Ampero's algorithms.

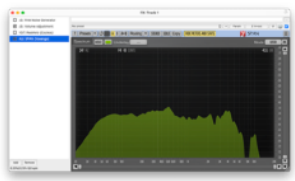
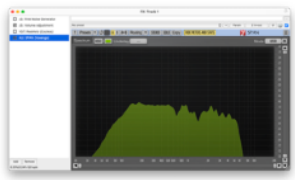
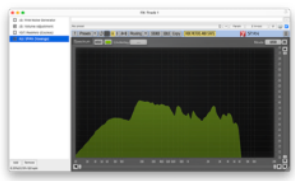
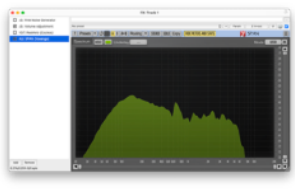
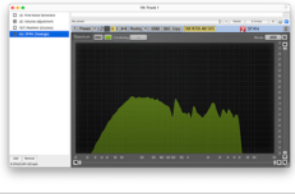
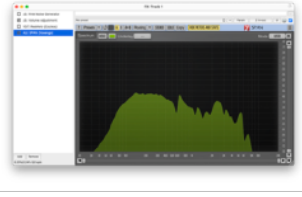
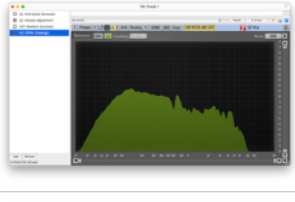
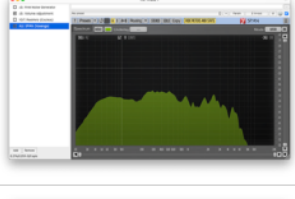
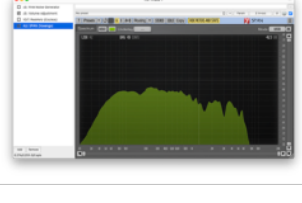
Curve is a thumbnail of the overall frequency response of the impulse, which is also summarised in **Shape**. These are provided to help the user choose which other cabinets to try, given that they like one. For example, if you find the "Dark VT" cab works well with an amp, you might find it worthwhile also trying the "TWD LEX" , since they are both flattish, but maybe not the "TWD SUP" since that has a low-end hump.

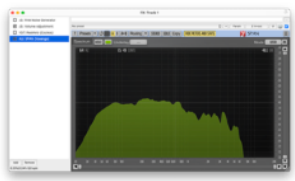
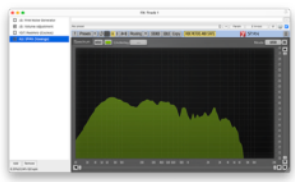
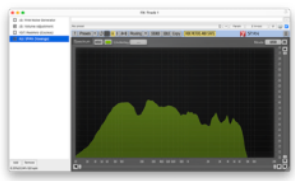
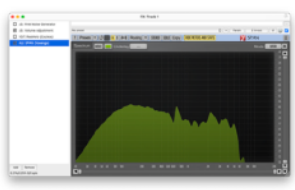
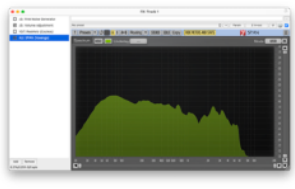
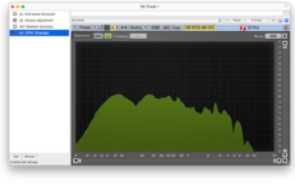
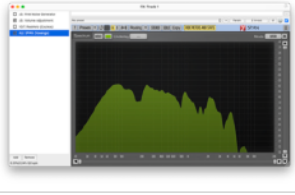
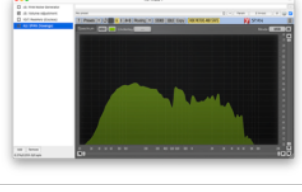
The cabs vary greatly both intrinsically and also in the amount of included filtering. **Low Reach** and **Top End** give an idea of the effective frequency range of the cab. A cab with a low frequency for Low Reach will sound fuller with clean sounds, but might sound boomy or flabby with high gain sounds. Conversely, a higher frequency here would sound tighter or thinner depending on the circumstances. Cabs with more top end will have more sparkle for cleans, but also more fizz for gain.

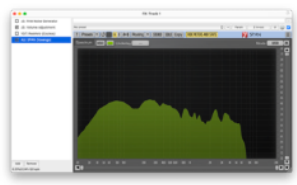
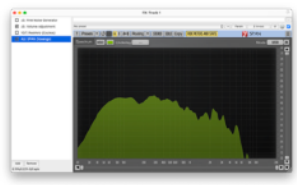
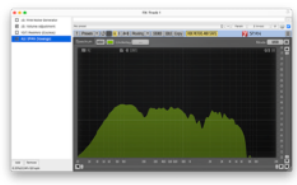
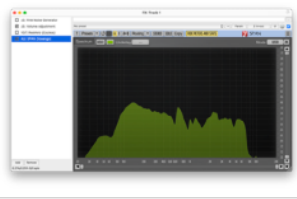
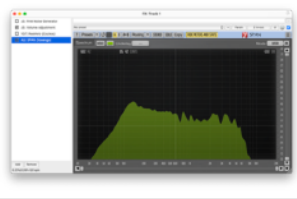
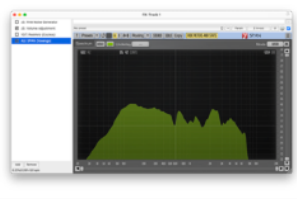
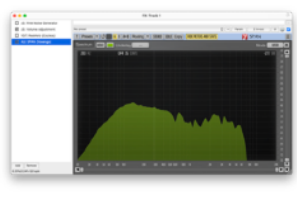
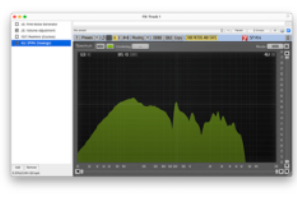
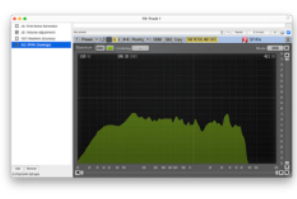
The **Level Drop** field compares the approximate subjective loudness of the IR with the bypassed signal, as measured by LUFS with pink noise. For example, the SUP ZEP cab sounds 3.9dB quieter than the bypassed signal, so its Level Drop is 3.9. The bigger the Level Drop, the bigger the perceived volume drop when you engage CAB - in other words, cabs with higher values sound 'quieter'. This can be relevant if you are using "No CAB" mode and so making use of outputs both with and without CAB simultaneously.

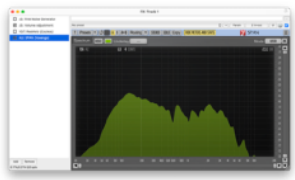
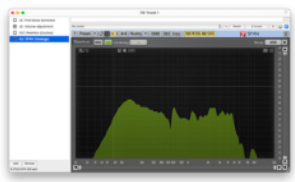
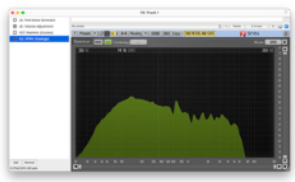
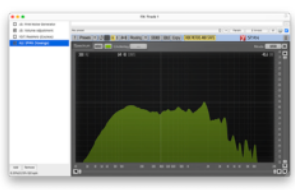
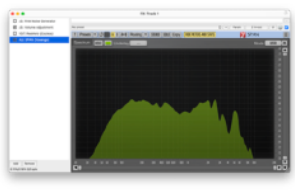
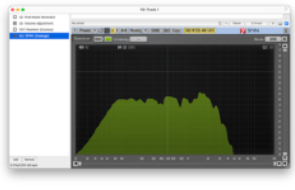
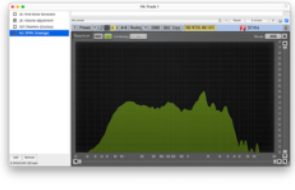
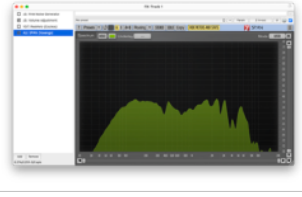
The **Vol Match** field is the suggested value you set the Volume parameter in the CAB block so that patches with different cabs have similar subjective volume. This is, of course, an inexact science, so treat these values as starting points. Perhaps the most useful role of this number is to make you aware of which cabs are much quieter than others, so that when comparing cabs you don't fall into the 'louder is better' trap.

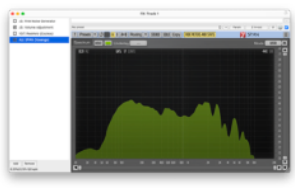
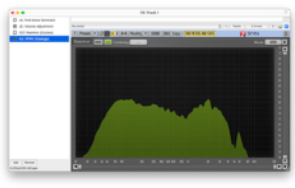
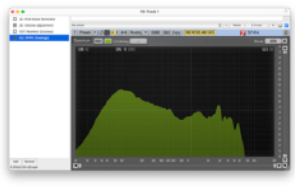
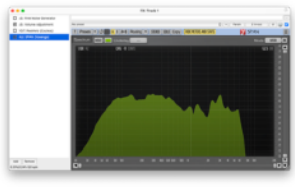
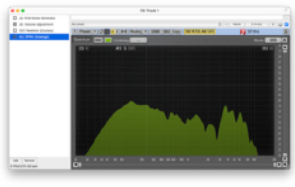
Name	#	Curve	Type	Shape	Low Reach	Top End (k)	Level Drop (dB)	Vol Match	Valeton Description	Open/Closed
SUP ZEP	1		1x6	Hump	40	9	3.9	55	Supro® 1x6" cabinet with oval speaker	Open back
TWD CP	2		1x8	Flat	80	8	3.4	52	Vintage Fender® Champ 1x8"	Open back
TWD PRC	3		1x10	Box	35	7	2.9	49	Vintage Fender® Princeton 1x10"	Open back
TWD SUP	4		2x10	Hump	30	6.5	3.2	51	A custom Fender® Tweed 2x10"	Open back
TWD LUX	5		1x12	Flat	55	11	7.4	83	Fender® Tweed Deluxe 1x12"	Open back
Dark LUX	6		1x12	Flat	55	9	6.3	73	Vintage Fender® Deluxe 1x12"	Open back
Dark VT	7		1x12	Flat	35	9	4.8	62	Vintage Fender® Vibrolux 1x12"	Open back
Dark Twin	8		2x12	Flat	60	12	5	63	Vintage Fender® ’65 Twin Reverb 2x12”	Open back
Dark CS	9		2x12	Hump	65	7	3.9	55	Custom modified Fender® 2x12"	Closed Back
Bellman1	10		2x12	Flat	35	9	4.6	60	Vintage Fender® “Piggyback" Bassman® 2x12"	Closed Back
Bellman2	11		4x10	Box	60	10	6.9	78	Fender® '59 Bassman® 4x10"	Open back

Name	#	Curve	Type	Shape	Low Reach	Top End (k)	Level Drop (dB)	Vol Match	Valeton Description	Open/Closed
J-120	12		2x12	Flat	30	12	6.6	76	Legendary "Jazz Chorus" 2x12"	Open back
UK G12	13		1x12	Flat	70	6.5	4.6	60	Marshall® 1x12"	Open back
UK GRN1	14		2x12	Box	50	6	4.4	59	Marshall® 2550 2x12"	Open back
UK LD	15		4x12	Hump	60	7.5	4.4	59	Marshall® 1960AV 4x12"	Closed Back
UK TD	16		4x12	Hump	40	7.5	1.8	44	68 Marshall® Basketweave 4x12"	Closed Back
UK MD	17		4x12	Hump	50	7	3.1	51	Custom modified Marshall® 4x12"	Closed Back
UK GRN2	18		4x12	Hump	60	6.5	4.2	57	Vintage Marshall® 4x12” with Celestion® Greenback® speakers	Closed Back
UK 75	19		4x12	Hump	60	8	3.3	52	Marshall® 4x12” with Celestion® G12T-75 speakers	Closed Back
UK Dark	20		4x12	Hump	40	8	2.3	46	1968 Marshall® 4x12"	Closed Back
FOXY1	21		1x12	Flat	35	8	3.7	54	Vintage VOX® AC15 1x12"	Open back
FOXY2	22		2x12	Hump	40	7	3.7	54	Vintage VOX® AC30 2x12"	Open back

Name	#	Curve	Type	Shape	Low Reach	Top End (k)	Level Drop (dB)	Vol Match	Valeton Description	Open/Closed
ROUT	23		1x12	Flat	40	7	4.8	62	Carr® Rambler 1x12"	Open back
BogSV	24		1x12	Hump	30	8	0.5	37	Bogner® Shiva 1x12"	Open back
Bad-KT	25		1x12	Hump	35	7	2.8	49	Bad Cat® Hot Cat 1x12"	Open back
Match	26		2x12	Hump	30	7	2.5	47	Matchless® Chieftain 2x12"	Open back
TOM OPEN	27		1x12	Hump	45	8	4.6	60	Swart® Atomic Space 1x12"	Open back
ACE	28		1x12	Hump	40	7	4	56	Morgan® AC-20 Deluxe 1x12"	Open back
Mess	29		4x12	Hump	40	6	3.9	55	Mesa/Boogie® Rectifier® 4x12"	Closed Back
D STAR	30		1x12	Hump	25	8	2.9	49	Mesa/Boogie® Lonestar 1x12"	Open back
SUP Star	31		2x12	Hump	35	10	3.6	54	Mesa/Boogie® Lonestar 2x12"	Open back
US STO	32		1x12	Hump	35	8	-0.4	34	1980's Mesa/Boogie® 1x12"	Open back
BOUTI	33		2x12	Hump	45	9	2.6	48	*A unique custom 2x12" cabinet*	

Name	#	Curve	Type	Shape	Low Reach	Top End (k)	Level Drop (dB)	Vol Match	Valeton Description	Open/Closed
SUP	34		2x12	Hump	60	7	2.8	49	Supro® 1624T 2x12" (Note... the 1624T is a 1x12)	Open back
MATT TWD	35		2x12	Hump	25	7	3.1	51	Matchless® 2x12"	Open back
Freed	36		2x12	Scoop	60	10	6.6	76	Fryette® Deliverance 2x12"	Closed Back
DB Rock	37		2x12	Hump	35	7	1.4	42	Two-Rock® 2x12"	Open back
Blue SK	38		2x12	Flat	45	7	5.4	66	A custom 2x12" with Celestion® Alnico Blue speakers	
EV	39		4x12	Scoop	45	8	6.4	74	Peavey® 6505 4x12"	Closed Back
Bog	40		4x12	Scoop	70	8	5.5	67	Bogner® 4x12"	Closed Back
Eagle	41		4x12	Hump	50	8	5.1	64	ENGL® 4x12"	Closed Back
Uban	42		4x12	Scoop	40	7	4.4	59	Bogner® Uberkab 4x12"	Closed Back
Solo	43		4x12	Hump	35	7	2.1	45	Soldano® 4x12"	Closed Back
Juice	44		4x12	Flat	35	7.5	6.7	77	Orange® PPC412 4x12"	Closed Back

Name	#	Curve	Type	Shape	Low Reach	Top End (k)	Level Drop (dB)	Vol Match	Valeton Description	Open/Closed
H-WAY	45		4x12	Scoop	50	8	2.6	48	Vintage Hiwatt® SE4123 4x12"	Closed Back
Way	46		4x12	Flat	80	10	5.1	64	Vintage WEM® 4x12"	Closed Back
Dumb	47		4x12	Hump	25	6	1.9	44	Dumble® 4x12"	Closed Back
Dizz	48		4x12	Scoop	55	10	4.5	59	Diezel® 4x12"	Closed Back
TRP	49		4x12	Scoop	65	8	4.9	62	Hughes & Kettner® Triamp 4x12"	Closed Back
King	50		4x12	Hump	40	7	3.2	51	Mesa/Boogie® Road King® 4x12"	Closed Back
ADM1	51		1x15 B				2.7	48	David Eden® 1x15"	
ADM2	52		4x10 B				1.7	43	David Eden® 4x10"	
Workman1	53		1x15 B				1.3	41	SWR® 1x15"	
Workman2	54		4x10 B				4.3	58	SWR® Workingman's 4x10"	
US BASS	55		2x10 B				4.8	62	Mesa/Boogie® 2x10"	

Name	#	Curve	Type	Shape	Low Reach	Top End (k)	Level Drop (dB)	Vol Match	Valeton Description	Open/Closed
MATT	56		2x10 B				3.2	51	Mark Bass® 4x10"	
F-TOP	57		1x15 B				2.8	49	Ampeg® PF-115HE 1x15"	
AMPG1	58		4x10 B				1.8	44	Ampeg® SVT-410HE 4x10"	
AMPG2	59		8x10 B				0.3	37	Ampeg SVT-810E 8x10"	
HACK	60		4x12 B				6.2	72	Hartke® 4x12"	

APPENDIX 3: AMP MODEL HEADROOM

Some amp models are intrinsically louder than others, in terms of the maximum digital signal level they pass. We can test this by passing a pure sine wave at -6dBFS into the signal chain (using the USB interface) and running it through each amp model in turn, with **Volume** at 50 (no post module gain adjustment) and turning the **Gain** up until the signal meters stop rising. This can be recorded as the **Saturation Level**. A higher level means the amp model has more headroom to pass on high signal levels coming into the block; a lower level means that the amp will saturate/compress a high input signal more readily. In a couple of amps, continuing to increase Gain past the saturation point actually reduces the output level - in these cases the table also shows the output level at max gain, labelled **Sag**. A model with a value in the Sag column will, in some circumstances, sound more "squishy" than the others.

Among the high-gain models, which are running in saturation all the time, the significance of Saturation Level is that a model with a lower level will sound quieter when A/B'ed with one with a higher level. This can trick your ear into which is "better". Take care to raise the Volume of quieter models so you make a fair comparison.

The AMP headroom is of relevance if you are placing a boost module of some kind (boost pedal, EQ, drive pedal) before the amp: if the amp is already running near to saturation level with the boost off, then turning it on will not give you an increase in volume, just saturation. This is no different from a real amp, of course: but on a real amp saturation level is usually *really freaking loud*, whereas on the GP200 a saturated high gain amp is just normal volume, so you can blunder into this by accident.

Headroom is also an important consideration if you are running the GP200 as a preamp (with the output connected to the FX return or power amp in of a tube amp) and run analogue pedals in front. In this instance, and especially if you obtain your drive sounds from pedals and are looking for a pedal platform, you will want to pick an amp model that has the maximum headroom. Good choices here are the **Match CL**, **Z38 CL** and **Mini Bass**: all will output right up to 0dBFS but also have some colouration or 'character'. Note that unlike their real-world counterparts, the Fender clean amp models are *not* the go-to picks.

The **J-120 CL** model is a special case, as it isn't really an amp model at all: it is an idealised clean amp with 0% distortion at all Gain settings, and a Fender-style tone stack. If you want no pre-amp colouration, this could be the one to go for - but also check out **Classic Bass** and **Foxy Bass**.

Name	Saturation Level (dBFS)	Sag (dBFS)
Tweedy	-5.6	-9.3
Bellman 59N	-10.5	
Bellman 59B	-14.4	
Dark Twin	-12.7	
Dark DLX	-6.3	
Dark Vibra	-10.7	
Silver Twin	-12.7	
SUPDual CL	-4.7	
SUPDual OD	-7.7	
Foxy 15TB	-15.3	
Foxy 30N	-8.5	
Foxy 30TB	-17.4	
J-120 CL	No THD up to 0dBFS	
Match CL	0	
Match OD	-14.4	
L-Star CL	-9	-12
L-Star OD	-14.9	
BogSV CL	-10.5	
Bog SV OD	-10.6	
Bog BlueV	-9	
Bog BlueM	-11.8	
Bog RedV	-14.6	
Bog RedM	-16.0	
Z38 CL	0	
Z38 OD	-11.6	
Knights CL	-2.1	
Knights CL+	-5.5	
Knights OD	-15.7	
Bad-KT CL	-7	
Bad-KT OD	-10.7	
Solo100 CL	0	
Solo100 OD	-9.7	
Solo100 LD	-12.9	
UK 45	-8	
UK 45+	-18.7	
UK 45JP	-17.8	

Name	Saturation Level (dBFS)	Sag (dBFS)
UK 50	-16	
UK 50+	-17.7	
UK 50JP	-17.7	
UK SLP	-8.7	
UK 800	-14.5	
UK 900	-12.3	
Flagman 1	-20.2	
Flagman 2	-18.2	
Flagman+ 1	-24	
Flagman+ 2	-20.1	
Mess2C+ 1	-12.1	
Mess2C+ 2	-11.9	
Mess2C+ 3	-12.4	
Mess4 LD	-15.5	
Mess4 LD 2	-15.0	
Mess4 LD 3	-15.0	
Mess DualV	-17.8	
Mess DualM	-15.9	
Juice30 OD	-12.8	
Juice R100	-13.6	
EV 51	-12.3	
Eagle 120	-12.7	
Eagle 120+	-10.7	
Power LD	-15.1	
Dizz VH	-11.8	
Dizz VH S	-13.7	
Dizz VH+	-13.5	
Dizz VH+ S	-13.2	
Classic Bass	No THD up to 0dBFS	
Foxy Bass	No THD up to 0dBFS	
Mess Bass	-22.5	
Mini Bass	0	
Bass Pre	No THD up to 0dBFS	
AC Pre	No THD up to 0dBFS	
AC Pre 2	No THD up to 0dBFS	

APPENDIX 4: NAM TO SNAPTONE CONVERSION

The conversion process actually involves loading the NAM file into a hidden NAM player and using that to train a Snaptone.

As of editor version 1.8.1 or later, both A1 and A2 format NAMs can be imported.

If you are careful, you can edit the input training file to include your own playing and raw guitar sound to (in theory) get more optimised results.

There is also a generic slightly improved file available which yields marginally more accuracy in the high end response

IMPORTANT! This information is supplied without warranty. If you don't know what you're doing, you can easily break your GP200 editor app, and I can't accept responsibility.

Although the GP200 allows you to load a NAM profile into the DST and AMP blocks (providing 5 slots in each block), the fact that the slots are labelled "Snaptone" and not "NAM" hints at the underlying truth - the GP200 doesn't have enough DSP power to run NAM files natively (even 'feather' captures) and so some form of conversion is taking place.

To understand what's going on, it's worth first looking at the standalone Soundclone program that Hotone provides[22] to allow you to do your own pedal/amp captures for the Ampero 2. This walks you through a process as follows:

- Checking the level of your audio interface;
- Inviting you to (optionally) record 20 seconds of your own guitar strumming;
- Connecting the equipment to be captured to the audio interface;
- Choosing whether the equipment is "clean" or "distorted";
- Running the capture process.

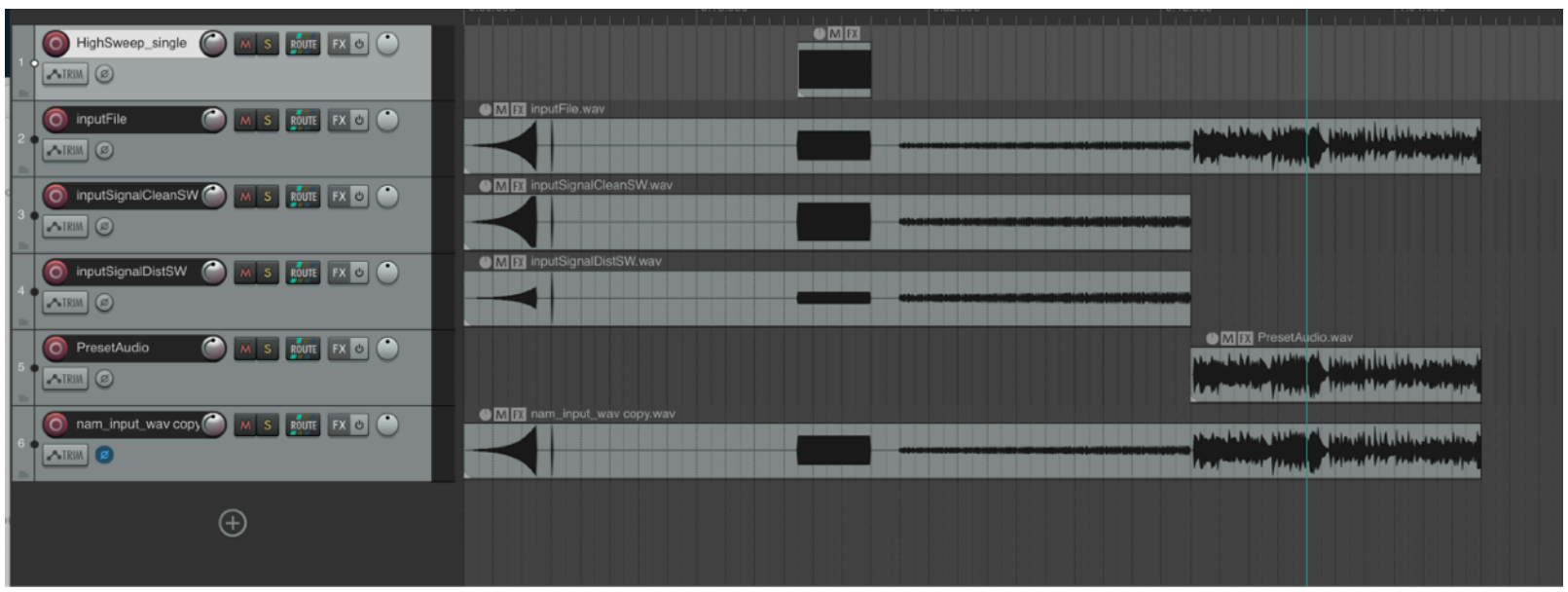
During the capture, if you are able to hear what's going on, you will hear the following:

1. A tone swept upward in volume from silent to loud;

[22] https://soundclone.net

2. A click;
3. A period of soft, undulating hiss;
4. A loud frequency sweep from 20Hz to 10kHz;
5. Several sets of rising synthetic tones (the sets are not identical);
6. The 20 seconds of either default or your custom strumming. This appears to be used to drive a form of EQ matching to tune the black-box model.

If we dig around in the installation folders of the Soundclone program, we find files that correspond to these stages:



If you select ‘clean’ as the device type, then what is used for the capture process is inputSignalCleanSW + PresetAudio (or your custom strumming). If you select ‘dist’, then inputSignalDistSW + PresetAudio (or your custom strumming). The difference between the two is the input signal levels, with inputSignalDistSW having less dynamic range.

There is also a mysterious file called simply inputFile, which is somewhere in-between inputSignalCleanSW and inputSignalDistSW. This doesn’t seem to be used by Soundclone itself, at least not anymore, but this exact file is also present in the installation of the GP-200 - where it is called “`nam_input_wav.wav`”. Why? Because as a halfway house, it presumably works tolerably well with both clean and dirty amps.

What, then, is happening when you import a NAM is the following:

1. The GP-200 editor loads the NAM into a hidden instance of the open-source NAM player;

2. The generic input WAV file is played through it (much faster than realtime) to create an output WAV;

3. The input WAV and the output WAV are passed to the Soundclone training algorithm to generate a Soundclone file;

4. This file is transferred to the GP200 as a Snaptone.

This capture of a capture, as one might expect, is of lower fidelity than the original NAM, although most of the time it still sounds fine, even good. But we know that it can be improved. Why? Because in the main Soundclone software, we are allowed to pick the input file from two choices, and encouraged to record our own guitar. Therefore it is reasonable to deduce that altering the input file ought to influence the import recapture process.

CREATING AN OPTIMISED TRAINING INPUT FILE

The default `nam_input_wav.wav` file is a 16-bit, stereo, 44.1kHz WAV exactly 1:10 long (3087000 samples) and any replacement must be the same to avoid the risk of crashes.

On MacOS, it is located in
`/Library/Application Support/Valeton/GP-200/File`

On Windows, it is located in
`Program Files\Valeton\GP-200\Resource\GP-200\File`

Here is a recommended procedure for creating it in REAPER:

- Create a new blank project at 44.1kHz.

- Load the desired base input signal to a new track: the safest option is to use a copy of the original default file and crop off the last 20 seconds, but you can also try the files from Soundclone - either inputSignalCleanSW or inputSignalDistSW depending on the type of amp/pedal to be captured.

- Plug your guitar into the interface and record a dry track of you strumming in a style that matches your preferred genre. You should play chords, a mixture of strums and ringing arpeggios - you want overlapping notes and harmonic complexity

- Make sure the dry track is clean, without any clipping, and is free from extraneous noise - you can edit out unnecessary gaps between playing.

- Trim and edit this track to *precisely* 20 seconds of playing without any significant gaps.

- Move the item start time so that it plays directly after the base input signal.
- Render out the project as 16-bit 44.1kHz stereo WAV.
- Double-check the resulting file with sox --info. It should come back with:

```
Channels       : 2
Sample Rate    : 44100
Precision      : 16-bit
Duration       : 00:01:10.00 = 3087000 samples = 5250 CDDA sectors
File Size      : 12.3M
Bit Rate       : 1.41M
Sample Encoding: 16-bit Signed Integer PCM
```

  It is **very important** that the file is **precisely** 3087000 samples long. Other lengths can make the editor crash when you try to import NAMs.

- Listen to the file as a final sanity check.

You can now back up the original `nam_input_wav.wav` file in your GP-200 editor installation (**important**!) and replace it with the new file. Then start the GP-200 editor and import the NAM as usual.

(Of course, doing this will not have any effect on NAMs you have already imported, unless you re-import them)

If you do things right, the effect will be subtle, but hopefully in a good way.

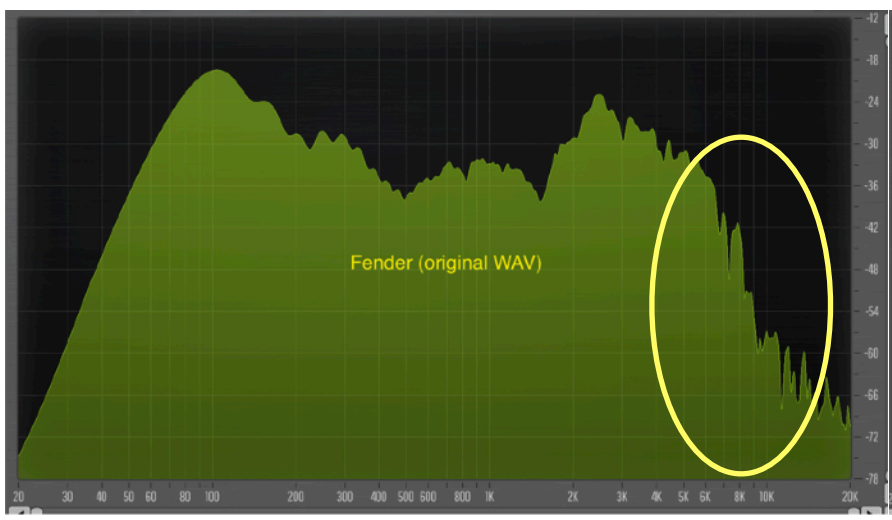
LAZY OPTION - OPTIMISED GENERIC FILE

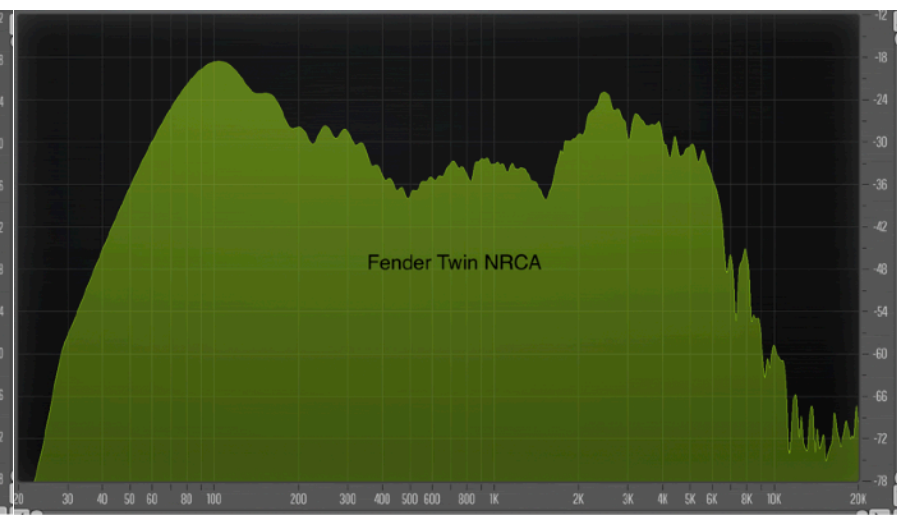
If you don't want to experiment with this, then another option is to use my alternative **`nam_input_wav.wav`** file. This is the result of experimentation which found that using 20 seconds of shaped pink noise often works even better than guitar strumming (despite Hotone's intentions) to get accuracy in the upper mids and treble.

For example, here is the frequency response of a Fender Twin NAM, and imports done with the default file and the alternative one. The Fender, being a very clean amp, is one of the easier ones for the Snaptone conversion process to handle.

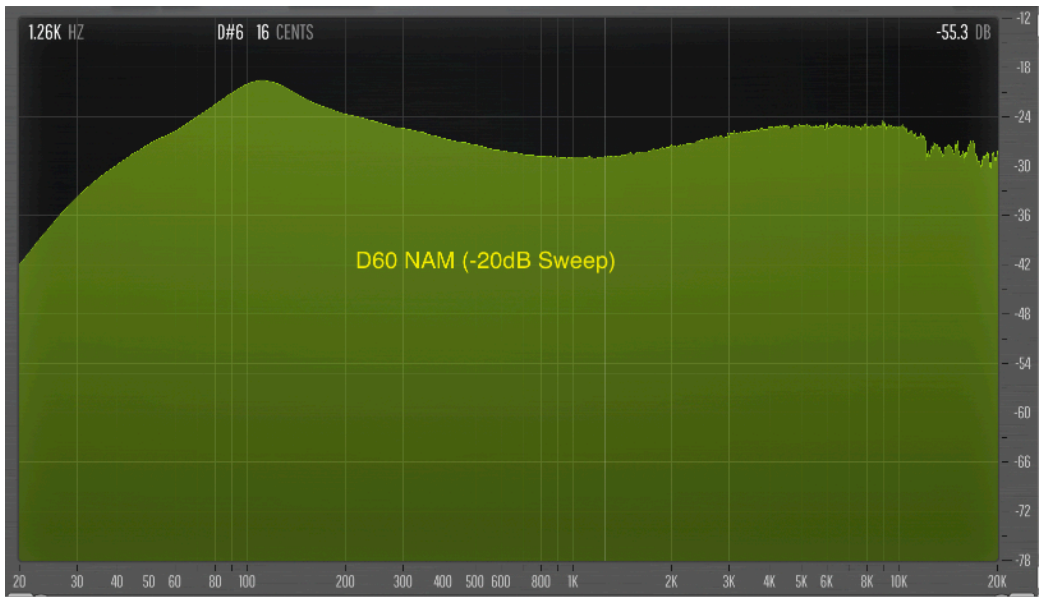
You can see that the alternative file is closer to the original in the 5kHz to 10kHz band. Note the 8kHz peak, which is at -48dBFS in the NAM, -42dB in the default import, and the correct -48dB in the import with the replacement file.

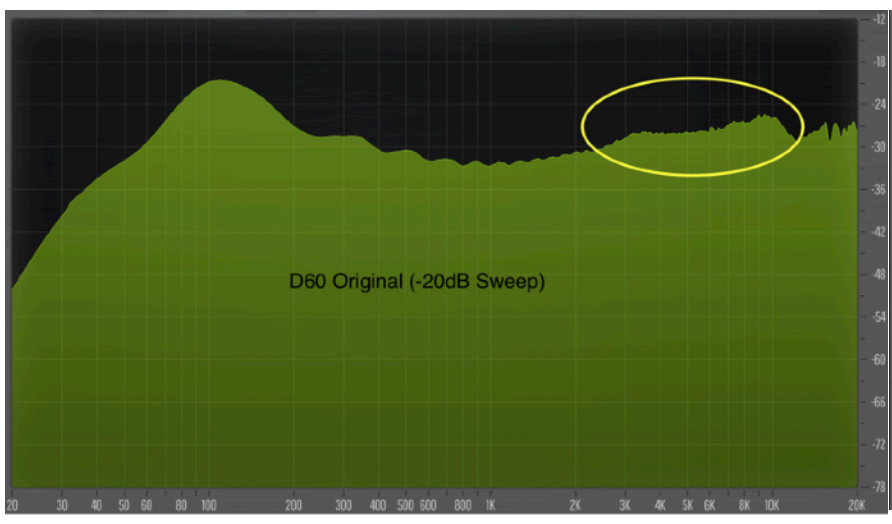


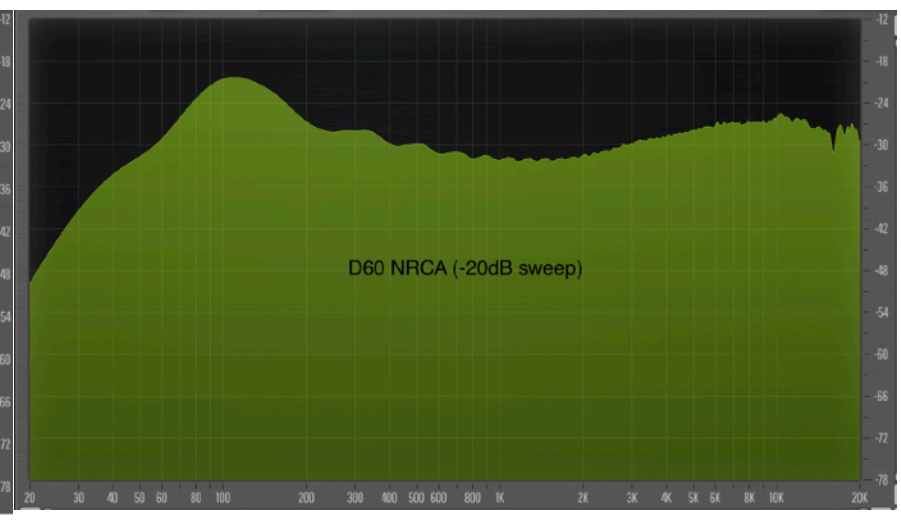




The same is true for higher gain amps. Here's the response of a head-only NAM capture of a Deliverance 60:







Now the original Snaptone import has a strange pointy peak in the upper mids and treble, while the alternative file allows for a closer match to the overall voicing.

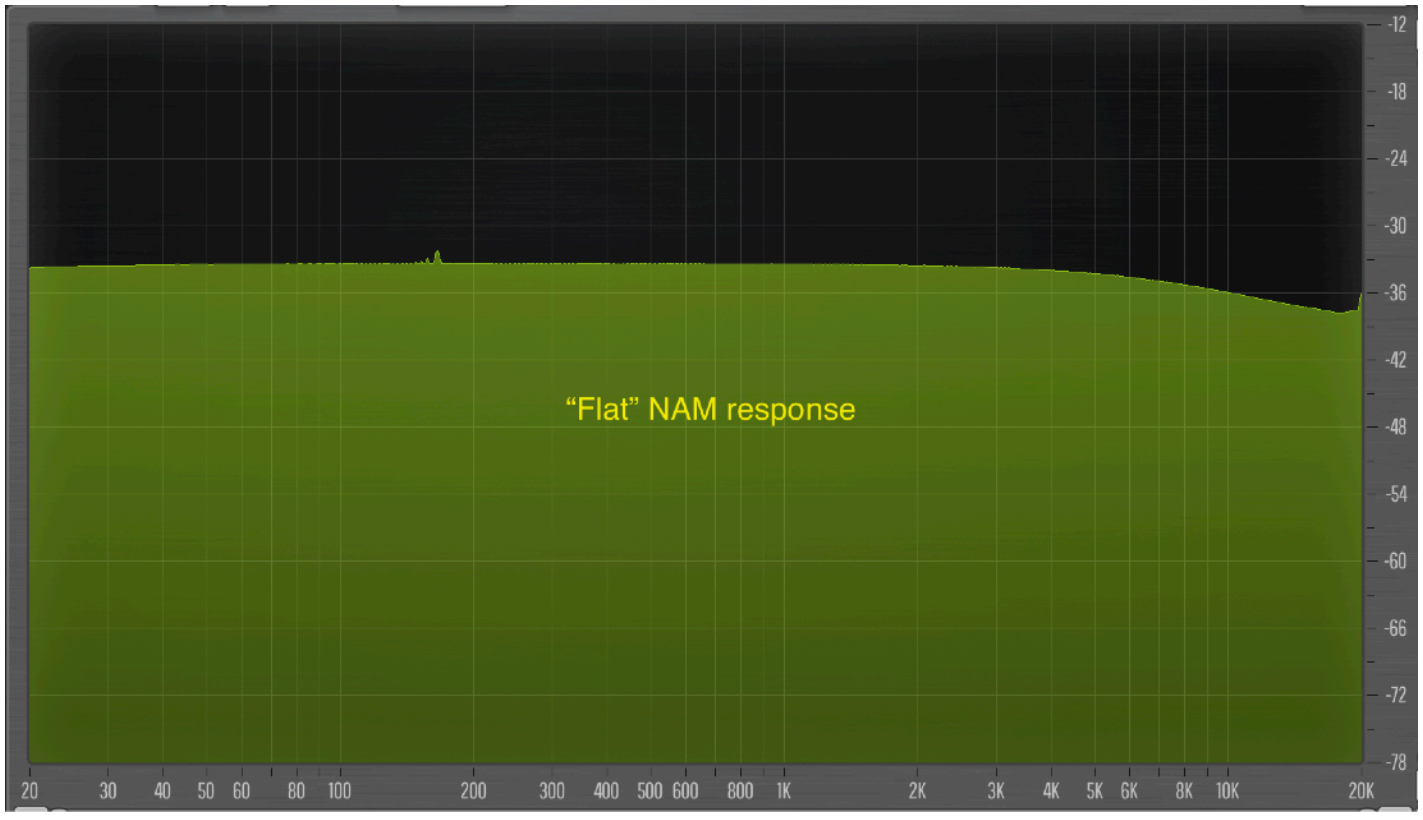
You'll also have seen a recurring theme with all Snaptone imports: an emphasis in the bass, usually centred around 100 to 120Hz.

You can find the replacement file in the Files section of the GP200 public Facebook group - the same place this ebook is distributed - or as a free download from the shop on my buymeacoffee page.

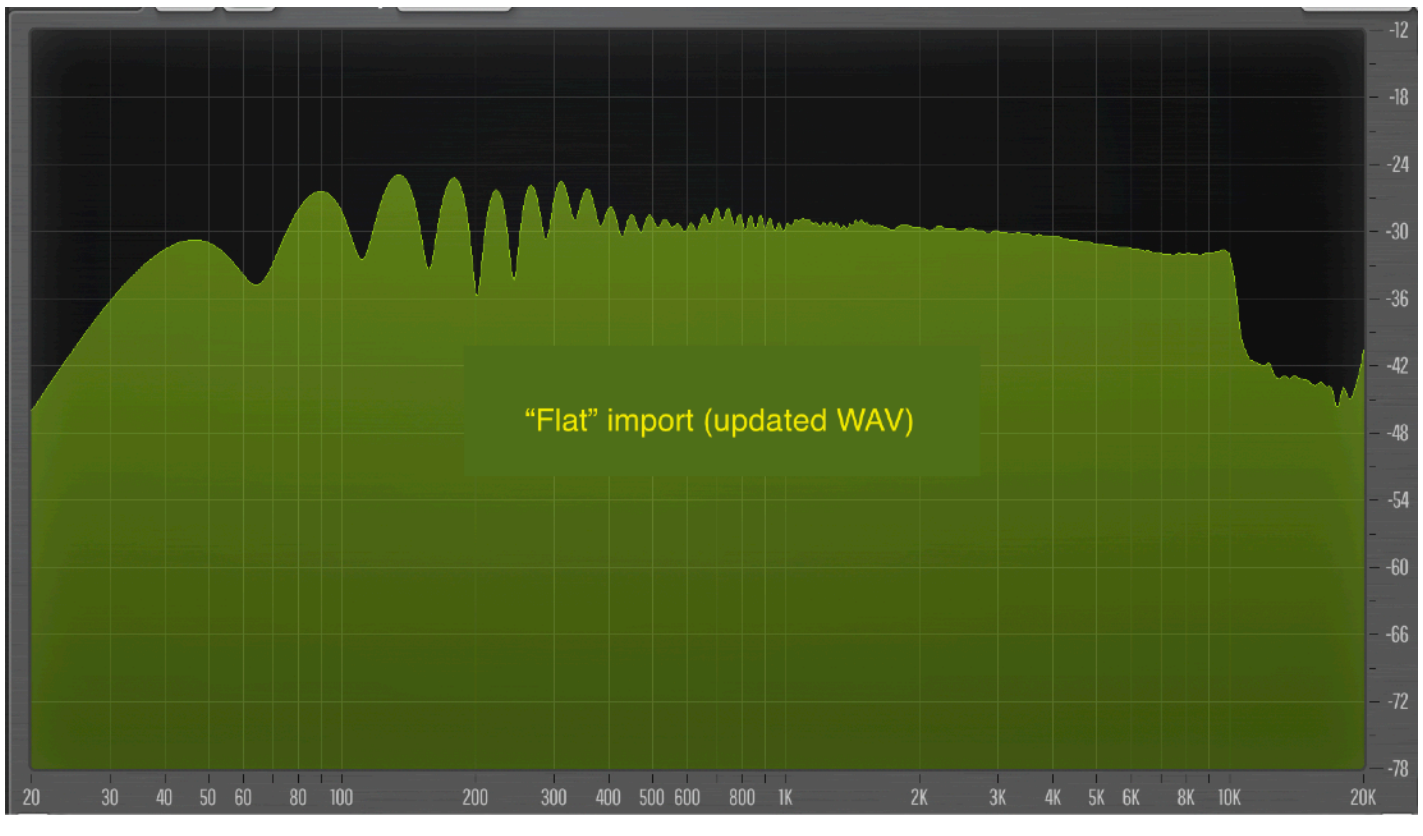
WHICH NAMS CONVERT THE BEST?

If we try converting some unusual NAMs, this gives us an insight into the foibles and limitations of the conversion process.

By creating an A1 NAM with the same output file as input file, we should get a completely flat response. As it happens, we don't - we get a mild rolloff - but it is nevertheless fairly flat and certainly very smooth.

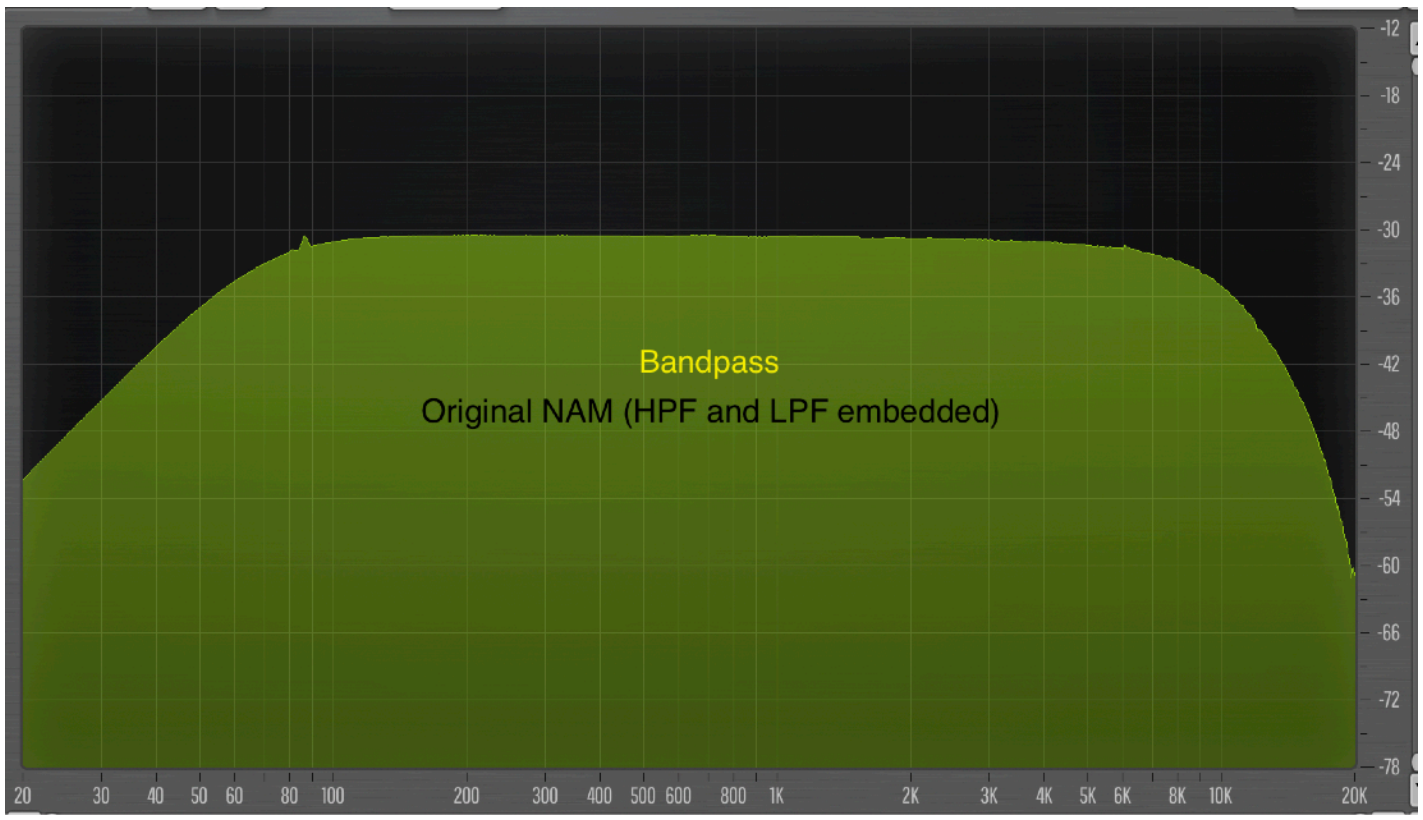


Importing this to the GP200, however, yields a response that is anything but smooth:



Two things are particularly evident: a very lumpy and exaggerated low-end response (with some of the artefacts of a poorly created 1024-sample IIR filter[23] - the peaks are at multiples of ~45Hz) and a complete breakdown of frequencies above 10kHz. The latter should not surprise us, since the frequency sweep in the conversion process only goes to 10kHz.

We can instead try a NAM where the output file is just the input file with a HPF and LPF applied: its frequency response looks like this:



When this is loaded, the result is overall better behaved:



[23] The working theory is that this filter is created in part during the EQ matching process, something backed up by analysis from Jesus Serra and confirmed by code analysis from Thomas Bronwyn Phillips.

The low end has far less ripple. (+/- 2dB instead of +/- 6dB). However, the high end is if anything *less* accurate - stripping off the high frequencies has given the EQ matching stage of the Snaptone import less to work with, and it ends up with randomness. In fact pretty much *all* imported NAMs have issues in the extreme highs (above 10kHz). The frequency response there is inaccurate; and what is worse, input frequencies close to 20kHz create noticeable aliasing artefacts.

Conclusions:

- High-gain full-rig NAMs can be problematic, since the import process doesn't control the treble well if the NAM doesn't pass it. This means that the harshness from the modelled distortion breaks through, giving a fatiguing and grainy sound. There is a possible remedy using a custom IR, for which see later in this chapter.

- Clean full-rig NAMs are more likely to work, since while the theoretical treble response is still inaccurate, there are far fewer upper harmonics to deal with in the first place.

- Head-only NAMs work best in practice, because (a) the head passes a lot of high frequencies to the EQ matcher and (b) the mess above 10kHz will be largely removed by the CAB IR you use.

- If you are creating NAMs of your own amps, or have access to the raw "output.wav" used so you can run it yourself through the trainer, then high-pass filtering the output.wav prior to training may yield a NAM that imports better; NAMs that pass a lot of low end seem to confuse the import process (I suspect the algorithm was tested only on guitar amps). In view of the bias in the conversion process, it may also be worth cutting a few dB centred on 100Hz.

- The Snaptone module has aliasing issues. These will be most dangerous if you are placing a distortion or fuzz in front of the AMP block, since they create a lot of high-frequency content. Therefore, if you are able to, filter off these > 10kHz frequencies *before* they reach the AMP block.

CORRECTIVE EQ

Even with all this optimisations in mind, a "good" import still has a tendency to emphasise lows and (to a lesser extent) de-emphasise upper mids. Therefore if you have some EQ spare, placing EQ immediately before or after the AMP block may get you even closer to the original NAM.

By way of example, here's the result of using the S COMP in PRE as a pure tilt EQ in front of the converted Fender NAM mentioned earlier:



The peaks at 100Hz and 2.5kHz are now the same height, as with the original NAM. Please note that the S COMP tilt EQ settings will be different for each NAM.

If you are using PRE but have the EQ block spare, then Guitar EQ 2 is the most useful since its 120Hz band is a shelf, making it good for taming low-end.

Finally, if you are using "full rig" NAMs and therefore have the CAB block spare, you can create a custom IR to apply a detailed EQ correction - as long as you aren't trying to also use the "no CAB" mode - as follows:

Using EQ match to fine-tune a Full Rig NAM

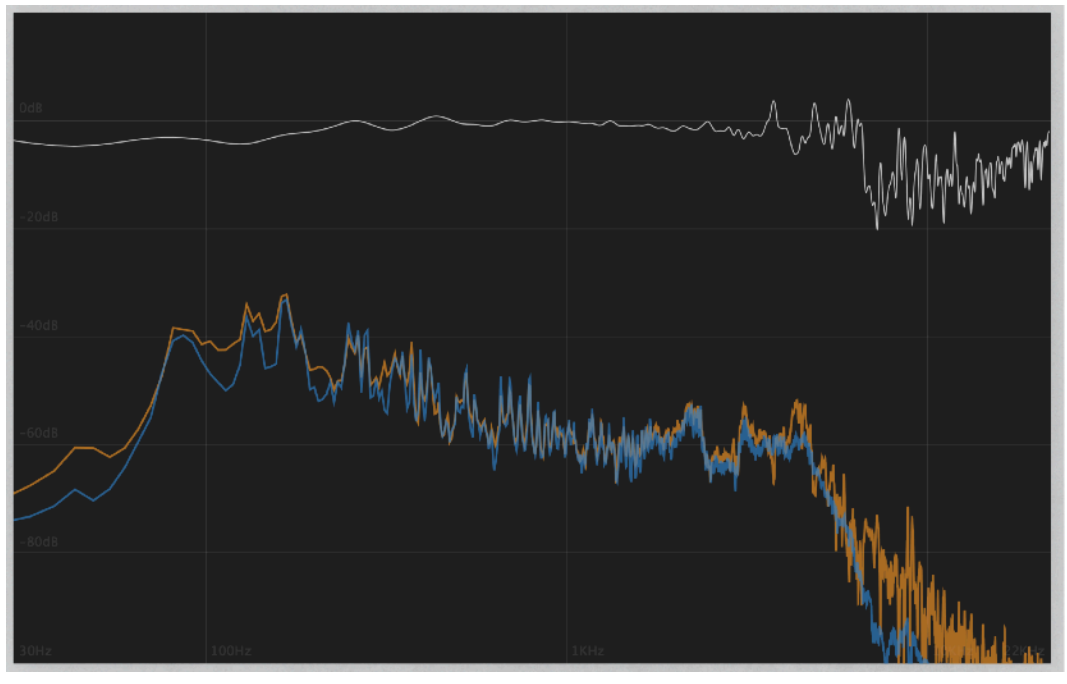
This takes a little more effort and software than simply tweaking EQs, but can yield excellent results. Here's an outline recipe for doing it with REAPER and the free plugin Spectrum Thief.

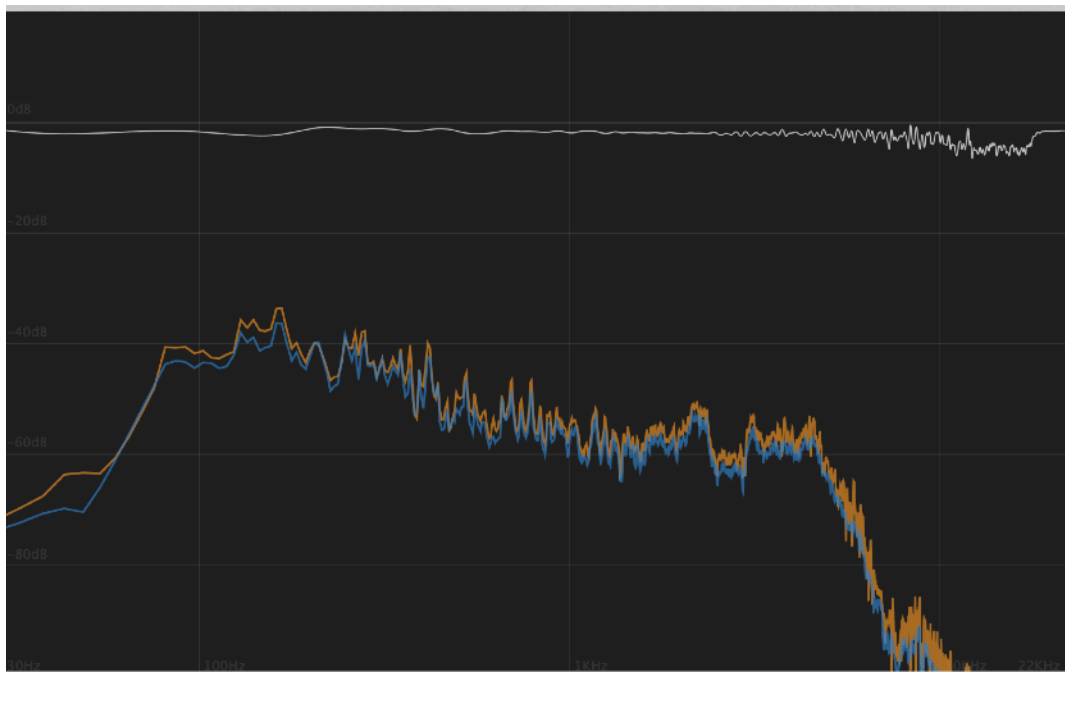
- Start a new project in REAPER, using the GP200 as interface.
- Record a minute or so your guitar, capturing the dry track.
- Re-amp this dry track through the NAM profile. Let's call the result **Track T**.
- Import the NAM into a spare snaptone slot, and re-amp the dry track through that on the GP200. We'll call that **Track S**.

- Play the two tracks together, soloing each in turn and adjusting the levels until they are matched, using your ears or an RMS/LUFS meter.
- Put an instance of Spectrum Thief on the master bus.
- Record Track T into Spectrum Thief as “Target”
- Record Track S into Spectrum Thief as “Source”
- Now turn the Smooth knob up and turn the IR resolution down to 1024 (since that’s what the GP200 uses). Export the IR.
- Import the IR into a spare CAB slot, and place this directly after the AMP block.

To give you an idea of how well this can work, here are two plots: one showing the frequency response errors between original NAM and the raw Snaptone import, and the second showing the errors after the fine-tuned IR is added.

As you can see, the high-end sizzle is almost completely tamed and the low end emphasis is controlled (the limited 1024 sample IR resolution prevents perfect correction)





One thing to watch out for is that you may hit the bug that makes imported EQ IRs very quiet: fortunately the output Volume control of the Snaptone loader gives you enough level to compensate.

Using a custom IR to tailor a NAM head capture

You can even adapt this technique for head-only NAMs, if you stick to using one or two cab IRs. Just create Track T by re-amping the dry track through the original NAM *and* your favourite cab IR, so that you end up creating a special IR that combines the frequency response of the CAB and the fine tunings for the snaptone.

Acknowledgements for this appendix

Thanks to the Redditors of these two posts who made me realise this was possible:

https://www.reddit.com/r/guitarpedals/comments/1mqtgdb/valeton_snaptone_gp5_gp200_scks_and_i_discovered/

https://www.reddit.com/r/ValetonGP2OO/comments/1pc7o76/i_sorta_fixed_the_nam_training_on_the_valeton/]

Also thanks to Željko Sremčev and Niq Scott for information on installation paths in Windows, and to Fernando Twr for some sample dry guitar files to test high-gain NAMs.

Thanks to Jacinto Alvarez for the trick to tailor NAM head captures with a combined EQ and cabinet IR.

Thanks to Jesus Serra for his work analysing the Hotone Soundclone format and where this aligns with/differs from the Snaptone format.

Thanks to Thomas Bronwyn Phillips for his work probing the actual Snaptone code on the GP200.

APPENDIX 5: ELECTRIC & ACOUSTIC SWITCHING RIG

This section is a more detailed and worked-through application of the hacks given in the FX Loop chapter, to create a setup that can be useful for guitarists that have to swap between two instruments throughout a set.

A not uncommon setup is to have a solid-body electric and an electro-acoustic guitar, sometimes even to swap during a song. Assuming the latter already has a low-impedance output (is active) or can be connected via a buffering pedal, then you can use the FX RETURN jack as a second input and so leave both guitars connected all the time, saving the need to plug and unplug or buy an additional switcher pedal. This has already been mentioned earlier.

It's possible to take this further, so that you have two separate outputs - one for the electric and one for the acoustic. This means that you can have them feeding two different amps, or the electric going to a real tube amp while the acoustic goes direct to PA.

CONNECTIONS

1. Connect the solid-body electric guitar to the main INPUT jack,
2. Connect the electro-acoustic guitar to the FX RETURN jack, using a buffer pedal if the guitar has a passive piezo pickup.
3. Connect the FX SEND jack to the amplification for the solid-body.
4. Connect the main output (unbalanced or XLR, as you prefer) to the amplification for the acoustic.

PATCH SETUP

Having made the connections, the trick is to set up patches so that the unused input and outputs are effectively muted.

Patches for the electric solidbody

In these patches, you move the FX Send point right to end of the signal chain, after VOL (or whatever block you have last). Set FX Return level to 0. Set patch volume to 0 (yes, really). Use FX Send level as your patch volume.

Explanation: because FX return is 0, no signal from your acoustic is arriving. Because Patch Volume is 0, nothing is being sent from the main outputs. But the signal at the end of the chain is still emerging from the FS SEND jack.

Patches for the acoustic

In these patches, move the FX Return point right to the start of the signal chain, and set FX Return level to 100. Set FX Send level to 0 and set FX Loop mode to Serial.

Explanation: Setting the loop mode to serial stops the main input travelling through the signal path, muting the electric. Setting FX Send to 0 stops any signal going to the FX SEND jack and so to the electric amplifier.

OVERALL RESULT

If you set things up correctly, then you can swap guitars easily and naturally mid-performance as follows:

1. Turn down volume on guitar, take it off, put on stand.

2. Select new patch (for the other guitar)

3. Put on new guitar, turn up volume.

No need to adjust amp settings or unplug leads.

If you have decided to route both outputs to the PA, then your sound guy has independent control over your electric and acoustic levels, which will please them.

CAVEATS

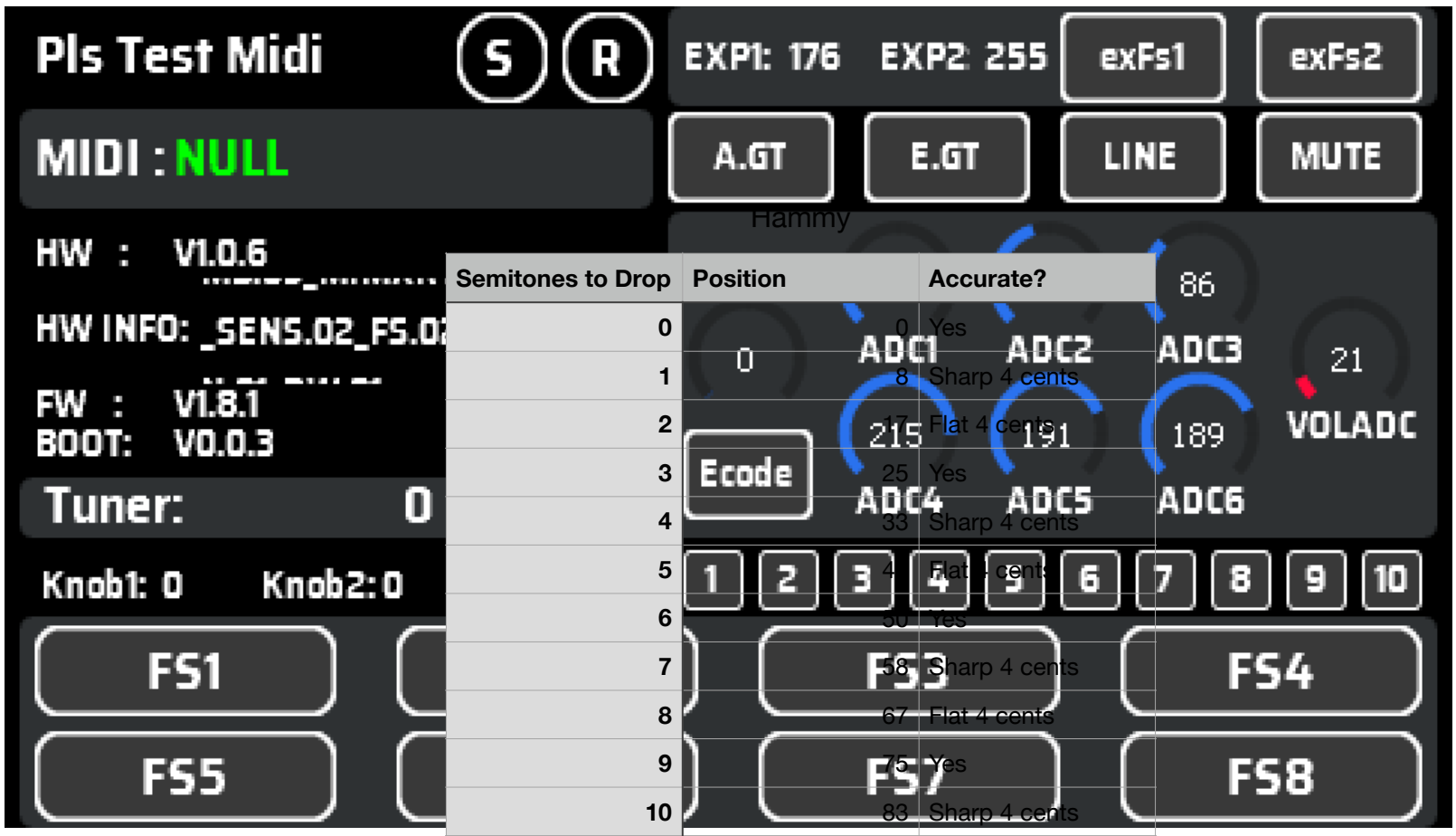
- You can’t use the FX loop as an FX loop (obviously)

- The electric has to be mono.

APPENDIX 6: THE HIDDEN SELF-TEST SCREEN

Huge thanks for this discovery to Thomas Bronwyn Phillips, with whose permission this is reproduced. Check his GP200 blog at https://gp200-reversing.hashnode.dev/

If you hold down the BACK and SAVE buttons while powering on the unit, you get this hidden self-test screen:



Semitones to Drop	Position	Accurate?
0	0	Yes
1	8	Sharp 4 cents
2	17	Flat 4 cents
3	25	Yes
4	33	Sharp 4 cents
5	42	Flat 4 cents
6	50	Yes
7	58	Sharp 4 cents
8	67	Flat 4 cents
9	75	Yes
10	83	Sharp 4 cents
11	92	Flat 4 cents
12	100	Yes

You can twiddle all knobs and press all buttons and check they are working smoothly. The three quick adjust knobs below the screen are shown as Knob1 to Knob3. PARA is shown as “Ecode” and turning it also should adjust screen brightness. Plugging a cable into the SEND or RETURN jacks will light up the S or R indicators. Pressing the BACK, GLOBAL, DRUMS and SAVE buttons will line the A.GT through to MUTE indicators and also switch the input impedance just like the Input Mode setting in global settings.

To exit the screen either press BACK and SAVE together again, or just turn the unit off and on again.

APPENDIX 7: DOWN-TUNING TABLE FOR HAMMY

This table shows the values to use for Position when using the Hammy module (with range set to -1 Octave) as a down-tuning block early in the signal chain.

The “Accurate?” column indicates whether the output of the Hammy will be sharp or flat relative to equal temperament. For example: if your guitar is tuned to E standard and you use a Position value of 8 to simulate tuning to Eb standard, the end result will be slightly sharp. Whereas if you use 25 to simulate tuning to C# standard, it will be right in tune (whether it sounds good is another matter).

4 cents sharp or flat may or may not be perceptible to you or the audience, depending on many factors such as:

- How in tune your guitar actually is;
- How in tune the rest of the band is;
- Genre;
- Type of chords being played;
- Level of inebriation of audience.